

Python script for MS3.08: Introduction to Computer Controlled Experiments

Version 1.1 - Updated 7 October 2025 by BFS

Nick Brooks

The experiments are conducted by **Devan Hu** and **Angze Li**.

Aim of the Experiment

The aim of this experiment is **to measure and analyse the current–voltage (I–V) characteristics of light-emitting diodes (LEDs) using a computer-controlled Arduino-based data acquisition system programmed in Python, in order to investigate how electronic structure and material properties influence charge transport behaviour.**

Specifically, the experiment aims to achieve:

1. Compare the I–V responses of LEDs and a standard resistor to demonstrate how the presence of a p–n junction and carrier injection processes lead to Shockley diode behaviour, in contrast to the Ohmic response of a resistor.
2. Investigate how semiconductor material properties determine the turn on voltage of LEDs.

Extension

1. Investigate how temperature influences the I–V characteristics and turn-on voltages of LEDs by comparing performance at room temperature and under liquid-nitrogen cooling (77 K).
2. Investigate how the differences in turn-on voltage affect current distribution and brightness in parallel-connected LEDs of different emission colours.

Through direct comparison of LEDs and a standard resistor, the experiment aimed to demonstrate the fundamental distinction between linear resistive conduction and non-linear semiconductor conduction, reinforcing the relationship between macroscopic electrical measurements and microscopic material properties.

MS3.08 Workflow chart

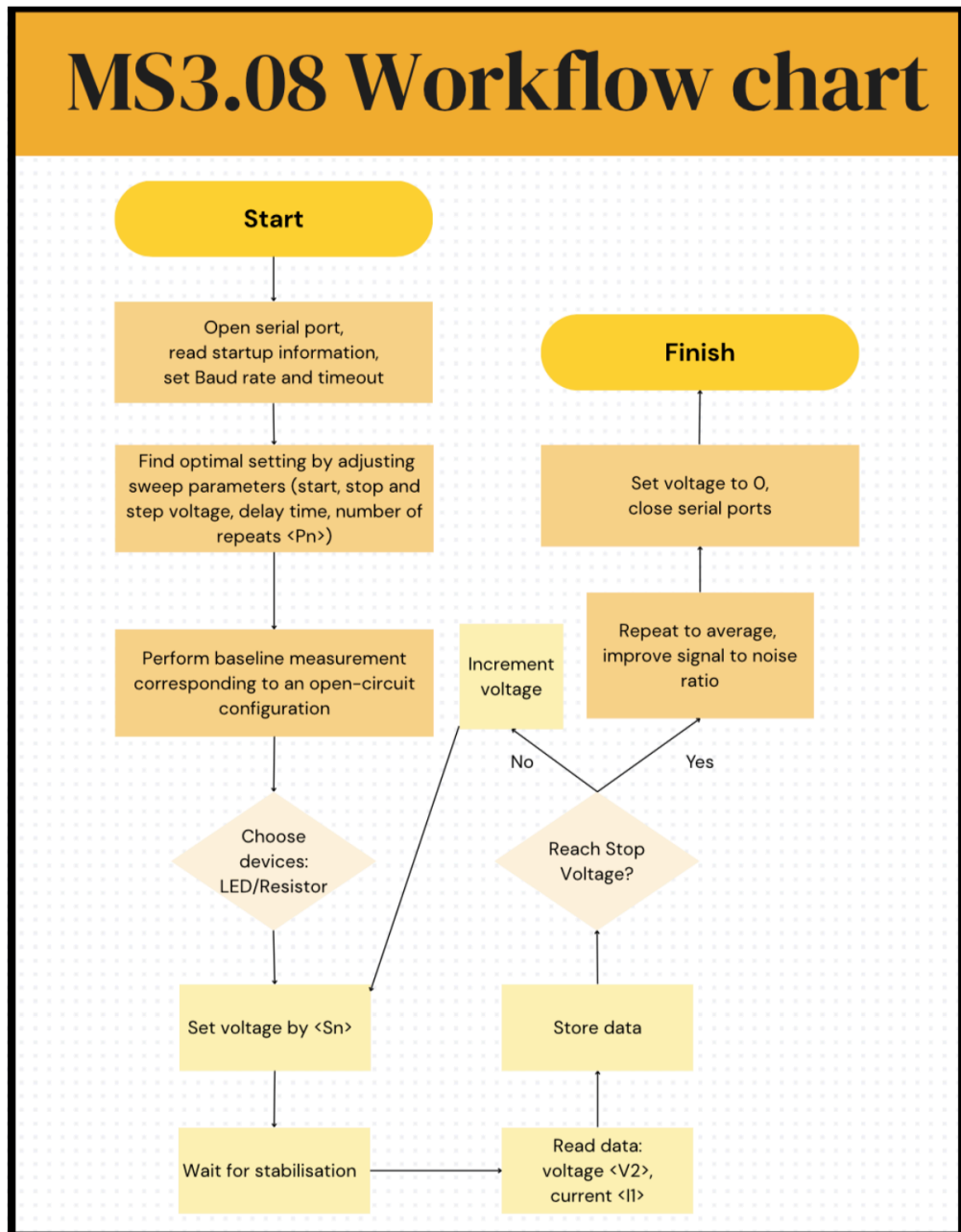


Figure 1. Computer-controlled workflow for automated I–V characterisation using the Arduino-based control and measurement system.

Instructions

This notebook provides an outline for developing a program that will allow you to investigate the conductivity of simple electronic components. It communicates with an Arduino that is configured to provide as a voltage source, as well as read the potential across the test component and current flowing through it. A full description of the experiment and the way that you communicate with the Arduino is available on Blackboard. Before you begin, you will need to install a module that lets Python communicate through a serial connection with other pieces of equipment. You only need to do this once to get everything set up.

Before you start, you need to install a Python module that allows you to work with serial connections to your computer, you only need to do this once and it will stay installed in your Python distribution.

Run the cell below and wait for the module to download and install.

```
In [66]: pip install pyserial
```

Requirement already satisfied: pyserial in /Users/devan/anaconda3/lib/python3.11/site-packages (3.5)

Note: you may need to restart the kernel to use updated packages.

If you receive a 'syntax error' when running to command above, please run the same command in a terminal window. If you are using Windows, you can launch a terminal window by typing CMD into the main Windows search bar (this is usually towards the left of the main toolbar at the bottom of your screen) and running the CMD program. If you are using MacOS or Linux, you can open the 'Terminal' program from your application launcher.

Next you need to point the program that you'll be writing towards the libraries that it will need to run. The libraries that you need are the 'serial' library that contains the commands for serial communication, some parts of serial tools so you can find the name of the port that your instrument is connected to, 'time' for allowing your program to wait and the standard tools that you have used in the past for data manipulation, curve plotting and fitting.

Run the cell below to import the appropriate libraries.

```
In [1]: # note: all functions used in the following section are imported below.
import os, csv, time
import serial
import serial.tools.list_ports
import time
%pylab inline
from scipy.optimize import curve_fit
import numpy as np
import matplotlib.pyplot as plt
import csv
from datetime import datetime
```

%pylab is deprecated, use %matplotlib inline and import the required libraries. Populating the interactive namespace from numpy and matplotlib

So that you can tell your program where to send and receive information, you need to find the name of the port that you have connected your instrument to. The port is a virtual serial port that runs over the USB connection (effectively the serial commands are sent over USB and then decoded in the instrument).

Start by making sure that your Arduino is **disconnected** from your computer, run the cell below and make a note of the port names that show up (there may be no ports shown at this point).

Next **connect** your instrument to a USB port on your computer, wait for a moment to check that any required drivers are installed (on Windows you may see a message saying that your device is being prepared and then one to say that it is ready for use), and then run the cell below again to get an updated list of the available serial ports. The name of the extra port that shows up with your Arduino connected is the one that you should use in the subsequent code. Note that the port naming scheme is different depending on your operating systems:

Windows. If you are using Windows, your Arduino will show up on a port named 'COM' and then a number (e.g. COM7).

MacOS or Linux. If you are using MacOS or Linux, your Arduino port will be named similar to '/dev/cu.usbmodem' and then a number.

```
In [21]: # note: this code lists the available serial ports to confirm that the device is connected
ports = serial.tools.list_ports.comports()
```

```
for p in ports:
    print(p.device)
print (len(ports), 'ports found')
```

```
/dev/cu.URT0
/dev/cu.Bluetooth-Incoming-Port
2 ports found
```

To open the serial connection between your computer and Arduino, you need to specify the parameters that the Arduino uses to communicate. You will need to specify the port name, a baud rate (the number of bits per second) of 9600, and a timeout time of 2 seconds. You can use the following code outline to open a serial connection that we're going to name 'ser':

```
ser = serial.Serial('port name', baudrate=x, timeout=x)````
```

For example (note that you will need to change the port name to match the one found above):

```
``python
ser = serial.Serial('/dev/cu.usbmodem144301', baudrate=9600, timeout=2)````
```

```
In [3]: # note:
# '/dev/cu.usbmodem14201' : specifies the serial port connected to the device
# baudrate=9600           : sets the communication speed between Python and the device
# timeout=2 s             : limits how long the program waits for a response

ser = serial.Serial('/dev/cu.usbmodem14201', baudrate=9600, timeout=2)
```

When the serial connection is opened, the Arduino will reset and it will send three lines of information back to the computer. You can read each line into a variable called `b` using the code:

```
b = ser.readline()````
```

The text `is read in as` a list of 8-bit characters which you can print out directly, but it may be useful to convert it to a string. You can use the following code to convert the character list to a string named `readstring`:

```
``python
readstring = b.decode("utf-8")````
```

Finally, you can print each line `as it is` read using:

```
``python
print(readstring)````
```

You may find it useful to run these commands `in` a ```for``` loop that repeats 3 times rather than just repeating them 3 times explicitly.

```
In [4]: # note: this reads and prints the first three lines received from the serial device

for i in range(3):
    b = ser.readline()
    readstring = b.decode("utf-8")
    readstring
    print(readstring)
```


Next, you should try to send some commands to the Arduino and then read back the response. You can send a command using the code: `ser.write(bytestowrite)`, but remember from above that the information is sent and received as a list of 8-bit characters which is subtly different from a python string. To get a command ready to send, you need to define a string to send, for example using:

```
writestring = '<S0>'`
```

Then this string needs to be encoded as a list of 8-bit ASCII characters using:

```
`python
```

```
bytestowrite = writestring.encode()`
```

After the command has been sent, the Arduino will receive it, process **is** and respond **with** one line which you should read **as** you did **in** the cell above (but remember there will only be one line now rather than three).

```
In [7]: writestring = '<S0.00>' # note: send a command string that instructs the device to
bytestowrite = writestring.encode() # note: converts the command from a Python string
ser.write(bytestowrite) # note: sends the encoded command to the device via the serial
```

```
Out[7]: 7
```

```
In [8]: response = ser.readline() # note: this reads one complete line of data sent back by
print(response.decode('utf-8'))
```

```
S 0.000
```

Now try to set up a loop that sets the applied voltage in small steps between 0 and 5 volts and at each step reads the current running through the LED, and the voltage across it. You will need to store each reading in a list so you can plot them later. You should think about if it is appropriate to insert short delays in the program (for instance to allow the set voltage to stabilise before reading your data), you can do this using the command:

```
time.sleep(0.1)`
```

Where the number **in** brackets **is** the number of seconds to wait (**100** ms **in** the example above).

Plot out your data and think about how you could extend the experiment, for example, can you measure multiple LEDs and compare their response, how does the response of an LED compare to a standard resistor?

When you are finished, you should close the serial connection. Only one program at a time can send information to a particular serial port, so by closing the connection, you free it up so that other programs could communicate with your instrument. Make sure you open the connection again when you want to start working again.

You can close the connection using the following command:

```
ser.close()`
```

```
In [9]: ser.close()
```

0. Functions

All functions required for this LED/Resistor I–V experiments (Experiments 1–5) are defined in this cell. No experimental measurements are executed in this cell. Execution is performed explicitly in subsequent cells.

```
In [7]: # =====
# 0) OUTPUT ROOTS
# =====
FIG_ROOT = "/Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to co
CSV_ROOT = "/Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to co

# =====
# 1) TIMESTAMP + PATH HELPERS
# =====
def make_timestamp():
    return datetime.now().strftime("%Y%m%d_%H%M%S")

def safe_tag(x, ndp=3): # note: 0.01 -> "0p01", 0.3 -> "0p3"
    return f"{x:.{ndp}f}".rstrip("0").rstrip(".").replace(".", "p")

def ensure_dir(path): # note: this creates a directory if it does not already exist
    os.makedirs(path, exist_ok=True)
    return path

def exp_dirs(exp_name):
    """
    note: this creates and returns dedicated figure and data directories for a give
    """
    fig_dir = ensure_dir(os.path.join(FIG_ROOT, exp_name))
    csv_dir = ensure_dir(os.path.join(CSV_ROOT, exp_name))
    return fig_dir, csv_dir

# =====
# 2) CONSISTENT PLOT STYLING
# =====
def style_axes(ax):
    for spine in ax.spines.values():
        spine.set_linewidth(1.8)
    ax.tick_params(axis='both', width=1.8, length=6)
    for t in ax.get_xticklabels() + ax.get_yticklabels():
        t.set_fontweight('bold')
        t.set_fontsize(14)

def style_legend(leg):
    frame = leg.get_frame()
    frame.set_facecolor('white')
    frame.set_edgecolor('black')
    frame.set_linewidth(1.2)
    frame.set_alpha(0.85)

# =====
# 3) SERIAL PARSING + ACQUISITION
# =====
def parse_reply(line, cmd_name):
    """
    note: ensures reliable parsing of voltage or current values returned by the Arc
```

```

"""
parts = line.decode(errors="replace").strip().split()
if len(parts) < 2:
    raise RuntimeError(f"Bad {cmd_name} reply: {parts} (raw={line!r})")
return float(parts[1])

def acquire_iv_sweep(
    ser,
    v_start=0.0,
    v_stop=5.0,
    v_step=0.1,
    delay=0.1,
    Pn=200,
    eps=1e-12
):
    """
    note: provides a single, centralised routine for performing an automated I-V sweep
    and acquiring calibrated voltage and current measurements.
    """

    ser.write(f"<P{Pn}>".encode())
    ser.readline() # ack

    V_list, I_list = [], []
    v = v_start
    while v <= v_stop + eps:
        ser.write(f"<S{v:.2f}>".encode())
        ser.readline() # ack

        time.sleep(delay)

        ser.write(b"<V2>")
        V_meas = parse_reply(ser.readline(), "<V2>")

        ser.write(b"<I1>")
        I_meas = parse_reply(ser.readline(), "<I1>")

        V_list.append(V_meas)
        I_list.append(I_meas)
        v += v_step

    return np.array(V_list), np.array(I_list)

# =====
# 4) REPEATS + TRIMMING + AVERAGING
# =====
def trim_to_common_length(V_runs, I_runs):
    """
    note: guarantees consistent array dimensions for averaging and comparison across
    """

    n = min(len(V) for V in V_runs)
    V_out, I_out = [], []
    for V, I in zip(V_runs, I_runs):
        if len(V) > n:
            drop = len(V) - n
            V, I = V[drop:], I[drop:]
        else:
            V, I = V[:n], I[:n]
        V_out.append(V)
        I_out.append(I)
    return V_out, I_out

def collect_repeats(ser, repeats, require_enter=False, **acq_kwargs):
    """

```

```

note: automates repeated measurements and produces averaged I-V data to reduce
"""
V_runs, I_runs = [], []
for r in range(1, repeats + 1):
    if require_enter:
        input(f"Repeat {r}/{repeats}: press Enter to start...")
    V, I = acquire_iv_sweep(ser, **acq_kwargs)
    V_runs.append(V)
    I_runs.append(I)

V_runs, I_runs = trim_to_common_length(V_runs, I_runs)
V_ref = V_runs[0]
I_stack = np.vstack(I_runs)
I_avg = I_stack.mean(axis=0)
return V_ref, I_stack, I_avg

# =====
# 5) FITTING UTILITIES
# =====
def fit_linear(V, I):
    """
    note: fit  $I = mV + c$ , return (m,c,R,R2).
    assumes I is in mA => R(ohm)=1000/m.
    """
    m, c = np.polyfit(V, I, 1)
    I_fit = m * V + c
    ss_res = np.sum((I - I_fit) ** 2)
    ss_tot = np.sum((I - np.mean(I)) ** 2)
    R2 = 1 - ss_res / ss_tot if ss_tot != 0 else np.nan
    R = 1000 / m
    return (m, c, R, R2)

def fit_split_linear(V, I, V_split):
    """
    note: here is for making two linear fits: below/above V_split.
    also compute R_high and R2_high on the high region.
    """
    low = V < V_split
    high = V > V_split
    if low.sum() < 2 or high.sum() < 2:
        raise ValueError(f"Not enough points for split fit: low={low.sum()}, high={high.sum()}")

    m_low, c_low = np.polyfit(V[low], I[low], 1)
    m_high, c_high = np.polyfit(V[high], I[high], 1)

    I_high_fit = m_high * V[high] + c_high
    ss_res_high = np.sum((I[high] - I_high_fit) ** 2)
    ss_tot_high = np.sum((I[high] - np.mean(I[high])) ** 2)
    R2_high = 1 - ss_res_high / ss_tot_high if ss_tot_high != 0 else np.nan

    R_high = 1000 / m_high
    return (m_low, c_low), (m_high, c_high), R_high, R2_high

# =====
# 6) SAVING CSV
# =====
def save_repeats_csv(path, V, I_stack, I_avg=None):
    """
    note: this is for saving one CSV containing:
    Voltage, Current_run1..runN, (optional) Current_avg
    """
    I_stack = np.array(I_stack)
    with open(path, "w", newline="") as f:

```

```

w = csv.writer(f)
header = ["Voltage (V)"] + [f"Current_run{i+1} (mA)" for i in range(I_stack
if I_avg is not None:
    header += ["Current_avg (mA)"]
w.writerow(header)

for j in range(len(V)):
    row = [V[j]] + [I_stack[i, j] for i in range(I_stack.shape[0])]
    if I_avg is not None:
        row += [I_avg[j]]
    w.writerow(row)
print(f"Saved CSV → {path}")

# =====
# 7) PLOTTING BUILDING BLOCKS
# =====
def plot_open_circuit_single(ax, V, I, repeat_idx=None):
    ax.scatter(V, I, s=28, color="0.25", label="Measured data")
    ax.plot(V, I, linewidth=2.2, color="0.25")

    title = "I-V plot, nothing connected"
    if repeat_idx is not None:
        title += f" - repeat {repeat_idx}"
    ax.set_title(title, fontsize=20, fontweight="bold")

    ax.set_xlabel("Voltage / V", fontsize=22, fontweight="bold")
    ax.set_ylabel("Current / mA", fontsize=22, fontweight="bold")

    leg = ax.legend(
        prop={"size": 11, "weight": "bold"},
        loc="upper left",
        frameon=True,
        fancybox=True
    )
    style_legend(leg)

def plot_resistor_single_with_fit(ax, V, I):
    m, c, R, R2 = fit_linear(V, I)
    I_fit = m * V + c

    data_color = '0.25'
    fit_color = 'royalblue'
    ax.scatter(V, I, s=28, color=data_color, label="Resistor data")
    ax.plot(V, I_fit, linewidth=2.2, color=fit_color,
            label=f"Linear fit:  $R \approx \{R:.2f\} \Omega$ ,  $R^2 = \{R2:.5f\}")

    ax.set_xlabel("Voltage / V", fontsize=22, fontweight="bold")
    ax.set_ylabel("Current / mA", fontsize=22, fontweight="bold")

def plot_led_avg_with_splitfits(
    ax,
    V, I_avg,
    led_label,
    V_split,
    color,
    repeats,
    delay_time=None,
    fit_mode="single"
):
    (m_low, c_low), (m_high, c_high), R_high, R2_high = fit_split_linear(V, I_avg,
    ax.plot(V, I_avg, "o-", color=color, label=f"{led_label} (avg of {repeats})")$ 
```

```

mask_low = V < V_split
mask_high = V > V_split

if fit_mode == "double":
    for V_seg, m, c, label in [
        (V[mask_low], m_low, c_low, f"Linear fit (V < {V_split:.2f} V)"),
        (V[mask_high], m_high, c_high, f"Linear fit (V > {V_split:.2f} V)")
    ]:
        V_fit = np.linspace(V_seg.min(), V_seg.max(), 100)
        ax.plot(V_fit, m * V_fit + c, "--", linewidth=2.5, label=label)

elif fit_mode == "single":
    V_fit = np.linspace(V[mask_high].min(), V[mask_high].max(), 100)
    ax.plot(
        V_fit, m_high * V_fit + c_high,
        "--", linewidth=2.5,
        label=f"Linear fit (V > {V_split:.2f} V)"
    )

else:
    raise ValueError("fit_mode must be 'single' or 'double'")

ax.axvline(V_split, linestyle=":", linewidth=2.5, label="~ turn-on voltage", color="red")

ax.text(
    0.02, 0.55,
    f"{led_label}\n"
    f"Above {V_split:.2f} V:\n"
    f"slope = {m_high:.3f} mA/V\n"
    f" $R \approx \{R\_high:.3f\} \Omega$ \n"
    f" $\$R^2\$ = \{R2\_high:.4f\}$ ",
    transform=ax.transAxes,
    fontsize=12, fontweight="bold",
    bbox=dict(boxstyle="round", facecolor="white", alpha=0.8)
)

ax.set_xlabel("Voltage / V", fontsize=22, fontweight="bold")
ax.set_ylabel("Current / mA", fontsize=22, fontweight="bold")

title = f"{led_label} I-V (averaged)"
if delay_time is not None:
    title += f", delay time = {delay_time:.2f} s"
ax.set_title(title, fontsize=18, fontweight="bold")

return (m_low, c_low), (m_high, c_high), R_high, R2_high

def finalize_and_save(fig, ax, fig_path, legend=True, legend_loc="upper left", legend_style_axes(ax)):
    if legend:
        leg = ax.legend(prop={'size': legend_size, 'weight': 'bold'}, loc=legend_loc,
                        frameon=True, fancybox=True)
        style_legend(leg)
    fig.tight_layout()
    fig.savefig(fig_path, dpi=600)
    print(f"Saved figure → {fig_path}")
    plt.show()

# =====
# 8) EXPERIMENT 1.1: Green LED delay-time test (avg fig + one CSV per delay)
# =====
def run_delay_time_test(
    ser,
    delays,

```

```

repeats=5,
led_label="Green LED",
V_split=1.7,
color="green",
v_start=0.0, v_stop=5.0, v_step=0.1,
Pn=200,
require_enter_each_run=True,
fit_mode="single"
):
    fig_dir, csv_dir = exp_dirs("1_Delay_Time_Test")
    stamp = make_timestamp()

    for delay_time in delays:
        print(f"\n=== Delay time = {delay_time} s ===")

        V_ref, I_stack, I_avg = collect_repeats(
            ser, repeats,
            require_enter=require_enter_each_run,
            v_start=v_start, v_stop=v_stop, v_step=v_step,
            delay=delay_time, Pn=Pn
        )

        delay_tag = safe_tag(delay_time)
        csv_path = os.path.join(
            csv_dir,
            f"{led_label.replace(' ', '_')}_{delay}_{delay_tag}s_repeats{repeats}_{stamp}"
        )
        save_repeats_csv(csv_path, V_ref, I_stack, I_avg)

        fig, ax = plt.subplots(figsize=(8, 6))
        plot_led_avg_with_splitfits(
            ax,
            V_ref, I_avg,
            led_label=led_label,
            V_split=V_split,
            color=color,
            repeats=repeats,
            delay_time=delay_time,
            fit_mode=fit_mode
        )
        fig_path = os.path.join(
            fig_dir,
            f"{led_label.replace(' ', '_')}_{delay}_{delay_tag}s_avg_{stamp}.png"
        )
        finalize_and_save(fig, ax, fig_path, legend=True, legend_size=11)

    print("\nExperiment 1 done.")

# =====
# 9) EXPERIMENT 1.2: Green LED step-voltage test (avg fig + one CSV per step)
# =====
def run_step_voltage_test(
    ser,
    steps,
    repeats=5,
    led_label="Green LED",
    V_split=1.9,
    color="green",
    delay=0.3,
    v_start=0.0, v_stop=5.0,
    Pn=200,
    fit_mode="single",
    require_enter_each_run=True

```

```

):
    fig_dir, csv_dir = exp_dirs("2_Step_Voltage_Test")
    stamp = make_timestamp()

    for step in steps:
        print(f"\n=== Step voltage = {step} V ===")

        V_ref, I_stack, I_avg = collect_repeats(
            ser, repeats,
            require_enter=require_enter_each_run,
            v_start=v_start, v_stop=v_stop, v_step=step,
            delay=delay, Pn=Pn
        )

        step_tag = safe_tag(step)
        csv_path = os.path.join(csv_dir, f"{led_label.replace(' ', '_')}_step_{step}_
        save_repeats_csv(csv_path, V_ref, I_stack, I_avg)

        fig, ax = plt.subplots(figsize=(8, 6))
        plot_led_avg_with_splitfits(ax, V_ref, I_avg, led_label, V_split, color, re
        ax.set_title(f"{led_label} I-V (averaged), step={step} V", fontsize=18, for

        fig_path = os.path.join(fig_dir, f"{led_label.replace(' ', '_')}_step_{step}_
        finalize_and_save(fig, ax, fig_path, legend=True, legend_size=11)

    print("\nExperiment 2 done.")

# =====
# 10) EXPERIMENT 2: Open-circuit baseline (5 separate figs + 1 CSV)
# =====
def run_open_circuit_baseline(
    ser,
    repeats=5,
    v_start=0.0, v_stop=5.0, v_step=0.1,
    delay=0.1,
    Pn=200,
    fit_mode="single",
    require_enter_each_run=True
):
    fig_dir, csv_dir = exp_dirs("3_Open_Circuit_Baseline")
    stamp = make_timestamp()

    V_runs, I_runs = [], []

    for r in range(1, repeats + 1):
        if require_enter_each_run:
            input(f"Open circuit repeat {r}/{repeats}: press Enter to start...")

        V, I = acquire_iv_sweep(
            ser,
            v_start=v_start, v_stop=v_stop, v_step=v_step,
            delay=delay, Pn=Pn
        )
        V_runs.append(V); I_runs.append(I)

        fig, ax = plt.subplots(figsize=(8, 6))
        plot_open_circuit_single(ax, V, I, repeat_idx=r)
        fig_path = os.path.join(fig_dir, f"open_circuit_run{r}_{stamp}.png")
        finalize_and_save(fig, ax, fig_path, legend=False)

    V_runs, I_runs = trim_to_common_length(V_runs, I_runs)
    V_ref = V_runs[0]

```



```

I_stack = np.vstack(I_runs)
I_avg = I_stack.mean(axis=0)

csv_path = os.path.join(csv_dir, f"open_circuit_repeats{repeats}_{stamp}.csv")
save_repeats_csv(csv_path, V_ref, I_stack, I_avg)

print("\nExperiment 3 done.")

# =====
# 11) EXPERIMENT 3: Resistor (5 separate figs + 1 CSV)
# =====
def run_resistor_iv_sweep(
    ser,
    repeats=5,
    v_start=0.0, v_stop=5.0, v_step=0.1,
    delay=0.1,
    Pn=200,
    fit_mode="single",
    require_enter_each_run=True
):
    fig_dir, csv_dir = exp_dirs("4_Resistor")
    stamp = make_timestamp()

    V_runs, I_runs = [], []

    for r in range(1, repeats + 1):
        if require_enter_each_run:
            input(f"Resistor repeat {r}/{repeats}: press Enter to start...")

        V, I = acquire_iv_sweep(
            ser,
            v_start=v_start, v_stop=v_stop, v_step=v_step,
            delay=delay, Pn=Pn
        )
        V_runs.append(V); I_runs.append(I)

        fig, ax = plt.subplots(figsize=(8, 6))
        plot_resistor_single_with_fit(ax, V, I)
        ax.set_title(f"I-V plot (resistor) - run {r}", fontsize=20, fontweight="bold")

        fig_path = os.path.join(fig_dir, f"resistor_run{r}_{stamp}.png")
        finalize_and_save(fig, ax, fig_path, legend=True, legend_size=11)

    V_runs, I_runs = trim_to_common_length(V_runs, I_runs)
    V_ref = V_runs[0]
    I_stack = np.vstack(I_runs)
    I_avg = I_stack.mean(axis=0)

    csv_path = os.path.join(csv_dir, f"resistor_repeats{repeats}_{stamp}.csv")
    save_repeats_csv(csv_path, V_ref, I_stack, I_avg)

    print("\nExperiment 4 done.")

# =====
# 12) EXPERIMENT 4: 5 LED colours (each LED: 5 separate figs + 1 CSV)
# =====
def run_led_colour_set(
    ser,
    led_specs,
    repeats=5,
    v_start=0.0, v_stop=5.0, v_step=0.1,
    delay=0.1,

```

```

Pn=200,
fit_mode="single",
require_enter_each_run=True
):

fig_dir, csv_dir = exp_dirs("5_LED_Colours")
stamp = make_timestamp()

for led_label, spec in led_specs.items():
    color = spec["color"]
    V_split = spec["V_split"]

    print(f"\n=== {led_label} (V_split={V_split}) ===")

    led_fig_dir = ensure_dir(os.path.join(fig_dir, led_label.replace(" ", "_")))
    led_csv_dir = ensure_dir(os.path.join(csv_dir, led_label.replace(" ", "_")))

    V_runs, I_runs = [], []

    for r in range(1, repeats + 1):
        if require_enter_each_run:
            input(f"{led_label} repeat {r}/{repeats}: press Enter to start...")

        V, I = acquire_iv_sweep(
            ser,
            v_start=v_start, v_stop=v_stop, v_step=v_step,
            delay=delay, Pn=Pn
        )
        V_runs.append(V); I_runs.append(I)

        # note: this code is for saving figure for each run with split fits + t
        fig, ax = plt.subplots(figsize=(8, 6))
        plot_led_avg_with_splitfits(
            ax,
            V,
            I_avg=I,          # note: single run treated as "avg"
            led_label=led_label,
            V_split=V_split,
            color=color,
            repeats=1,
            delay_time=0.1,
            fit_mode=fit_mode
        )

        ax.set_title(f"I-V plot ({led_label} LED) - run {r}", fontsize=18, font

        fig_path = os.path.join(led_fig_dir, f"{led_label.replace(' ', '_')}_run
        finalize_and_save(fig, ax, fig_path, legend=True, legend_size=12)

    V_runs, I_runs = trim_to_common_length(V_runs, I_runs)
    V_ref = V_runs[0]
    I_stack = np.vstack(I_runs)
    I_avg = I_stack.mean(axis=0)

    csv_path = os.path.join(led_csv_dir, f"{led_label.replace(' ', '_')}_repeats
    save_repeats_csv(csv_path, V_ref, I_stack, I_avg)

print("\nExperiment 5 done.")

```

1. Test Variables

Aim :

The following experimental variables were modified to determine the optimal measurement conditions: the **start and stop voltage**, the **voltage step size**, and the **delay time** between successive readings.

Explanation :

To optimise the experimental parameters in a controlled and systematic manner, a single LED was selected as a reference device for all parameter exploration. The green LED was chosen due to its moderate band gap, which places its turn-on behaviour between that of the lower-band-gap red and yellow LEDs and the higher-band-gap blue LED. This intermediate operating regime makes the green LED representative of the overall LED set, allowing measurement parameters to be optimised without biasing the procedure towards extreme operating conditions.

LEDs with higher band gaps operate at higher forward voltages, where the current–voltage (I–V) relationship is more strongly nonlinear. In this regime, the measured current becomes more sensitive to small voltage fluctuations and insufficient stabilisation time following each voltage step. Conversely, lower-band-gap LEDs exhibit earlier turn-on and more gradual current increases. Using an intermediate-band-gap LED therefore provides a balanced test case for exploring system-level parameters such as voltage range, step size, and delay time.

Once the optimal experimental parameters were determined using the green LED, these parameters were fixed and applied consistently across all LED colours to ensure measurement consistency and comparability.

1.0. Start and End Voltage

Explanation :

As shown in various I–V plots below, the modification of start and end voltage is deemed unnecessary. This is due to:

1. The resistance of LEDs before turn-on voltage can be effectively seen as *infinite*, so start voltages do not affect the overall performance of LEDs;
2. Across different LEDs, the currents at ~2.5 V are already quite high and seems are automatically cut off by the microcontroller, so there is no point changing this value either

1.1. Delay Time

Explanation :

To determine the optimal delay time between successive measurements, a dedicated function, `run_delay_time_test`, was defined to perform a complete I–V sweep for the green LED while varying only the delay time. The user sets a range of delay time in advance, so when the function is called, it sets the stabilisation period between applying each voltage step and recording the voltage and current. Linear fits are applied to the measured data, and the slope of the high-voltage region is used to calculate an effective resistance as a measure of measurement stability.

Results and conclusion :

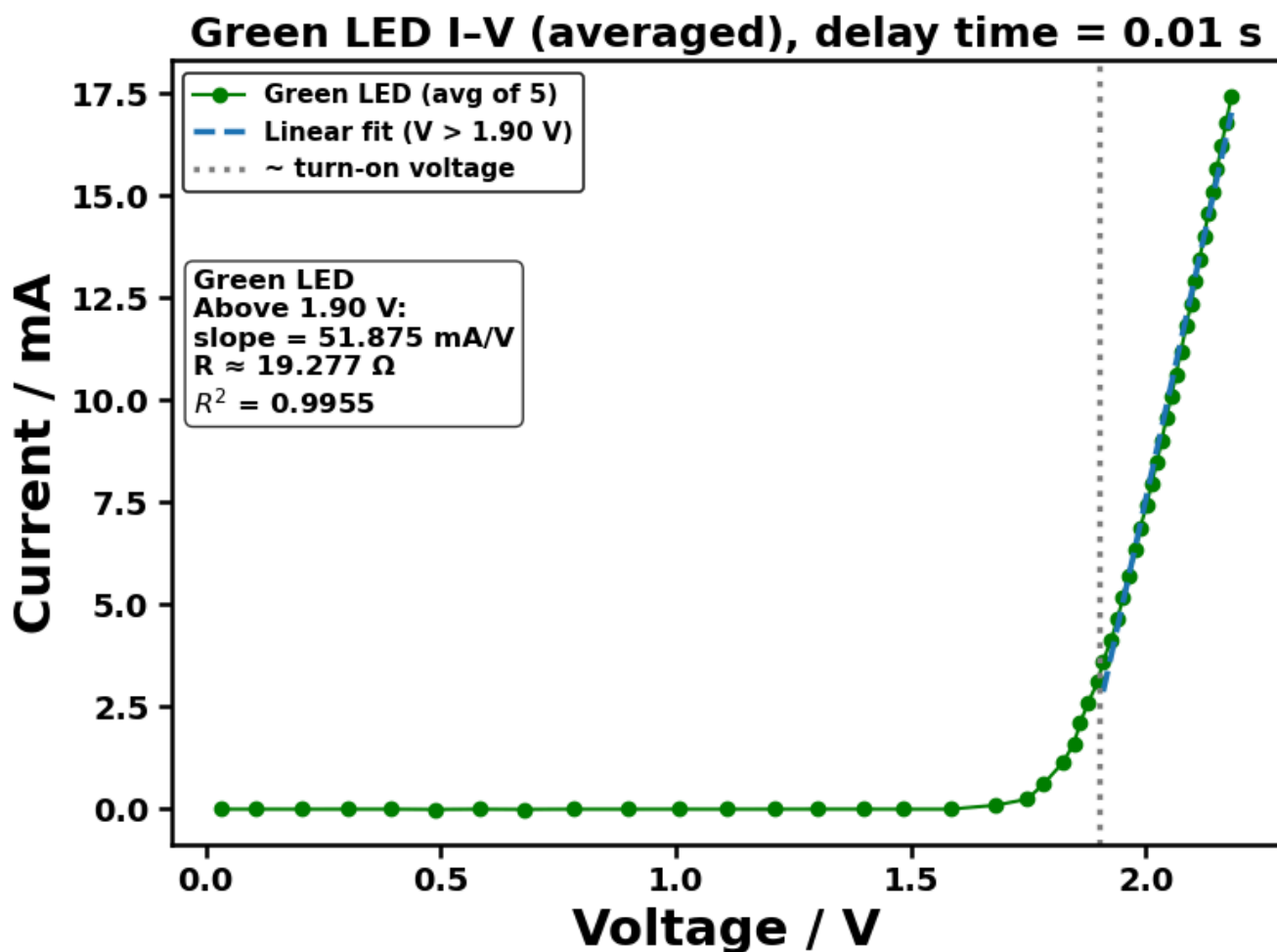
The function was repeatedly executed with delay times of 0.01 s, 0.05 s, 0.1 s, 0.15 s, 0.2 s, and 0.3 s. Short delay times produced unstable resistance values, indicating insufficient stabilisation, while longer delay times led to convergence of the extracted resistance. No significant improvement was observed beyond 0.1 s. Therefore, a delay time of 0.1 s was selected as the optimal balance between accuracy and experimental efficiency and was used for all subsequent measurements.

```
In [127... run_delay_time_test(
    ser=ser,
    delays=(0.01, 0.05, 0.10, 0.15, 0.20, 0.30),
    repeats=5,
    led_label="Green LED",
    V_split=1.9,
    color="green",
    v_step=0.1,
    Pn=200,
    require_enter_each_run=False,
    fit_mode="single" # note: this can be changed to "double-fit" for both low- a
)
```

=== Delay time = 0.01 s ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/1_Delay_Time_Test/Green_LED_delay_0p01s_repeats5_20260122_193254.csv

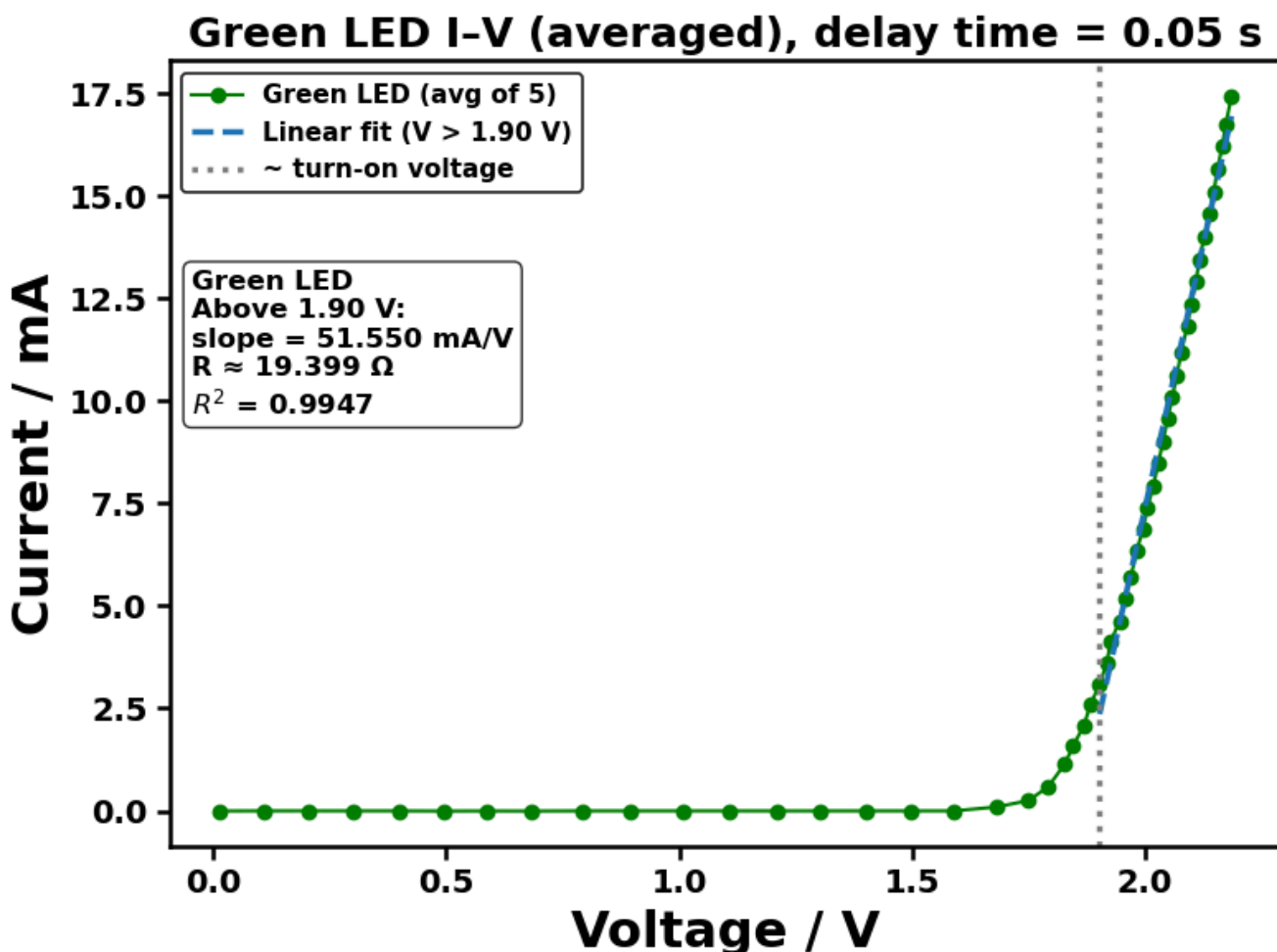
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/1_Delay_Time_Test/Green_LED_delay_0p01s_avg_20260122_193254.png



=== Delay time = 0.05 s ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/1_Delay_Time_Test/Green_LED_delay_0p05s_repeats5_20260122_193254.csv

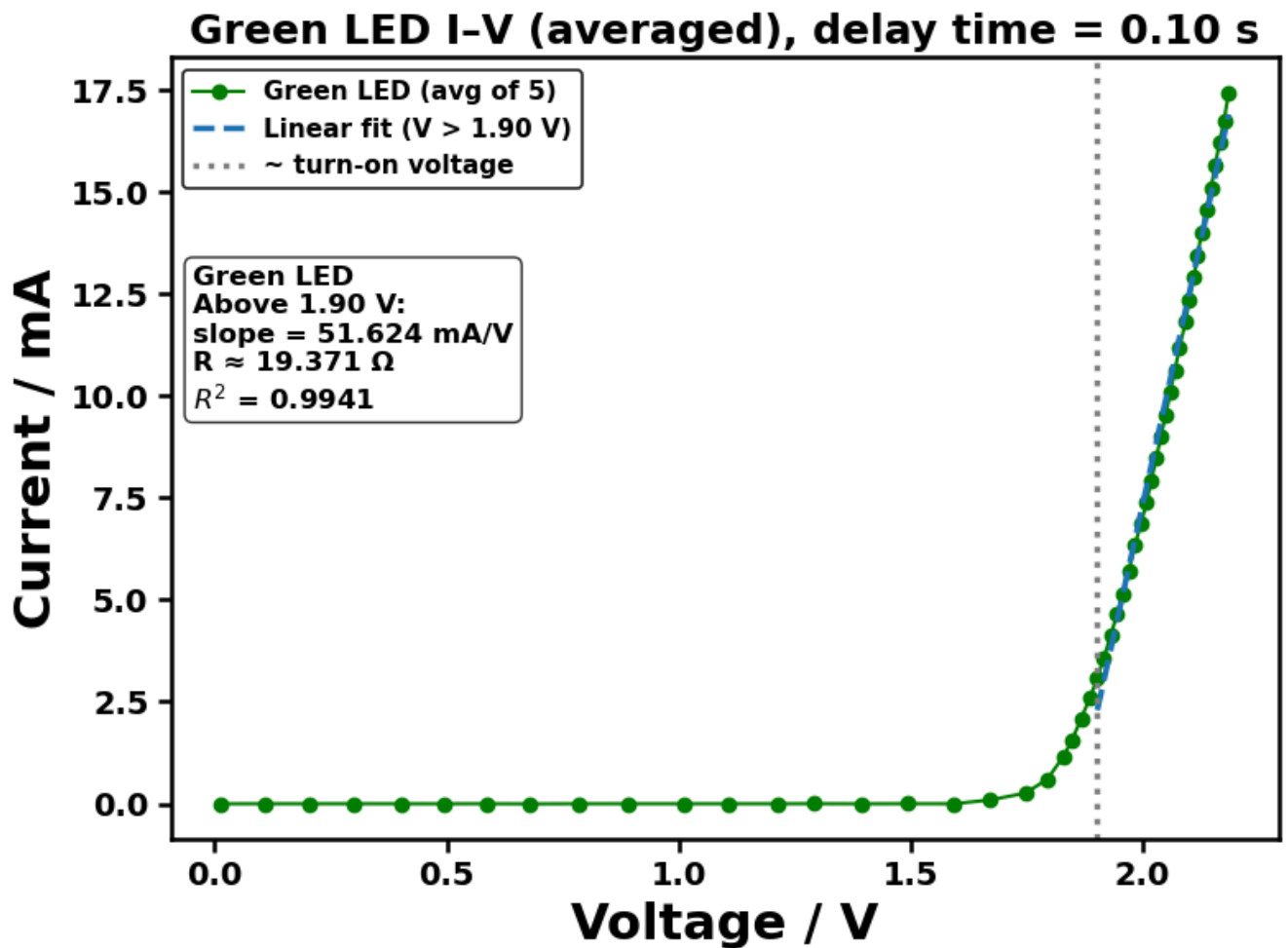
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/1_Delay_Time_Test/Green_LED_delay_0p05s_avg_20260122_193254.png



=== Delay time = 0.1 s ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/1_Delay_Time_Test/Green_LED_delay_0p1s_repeats5_20260122_193254.csv

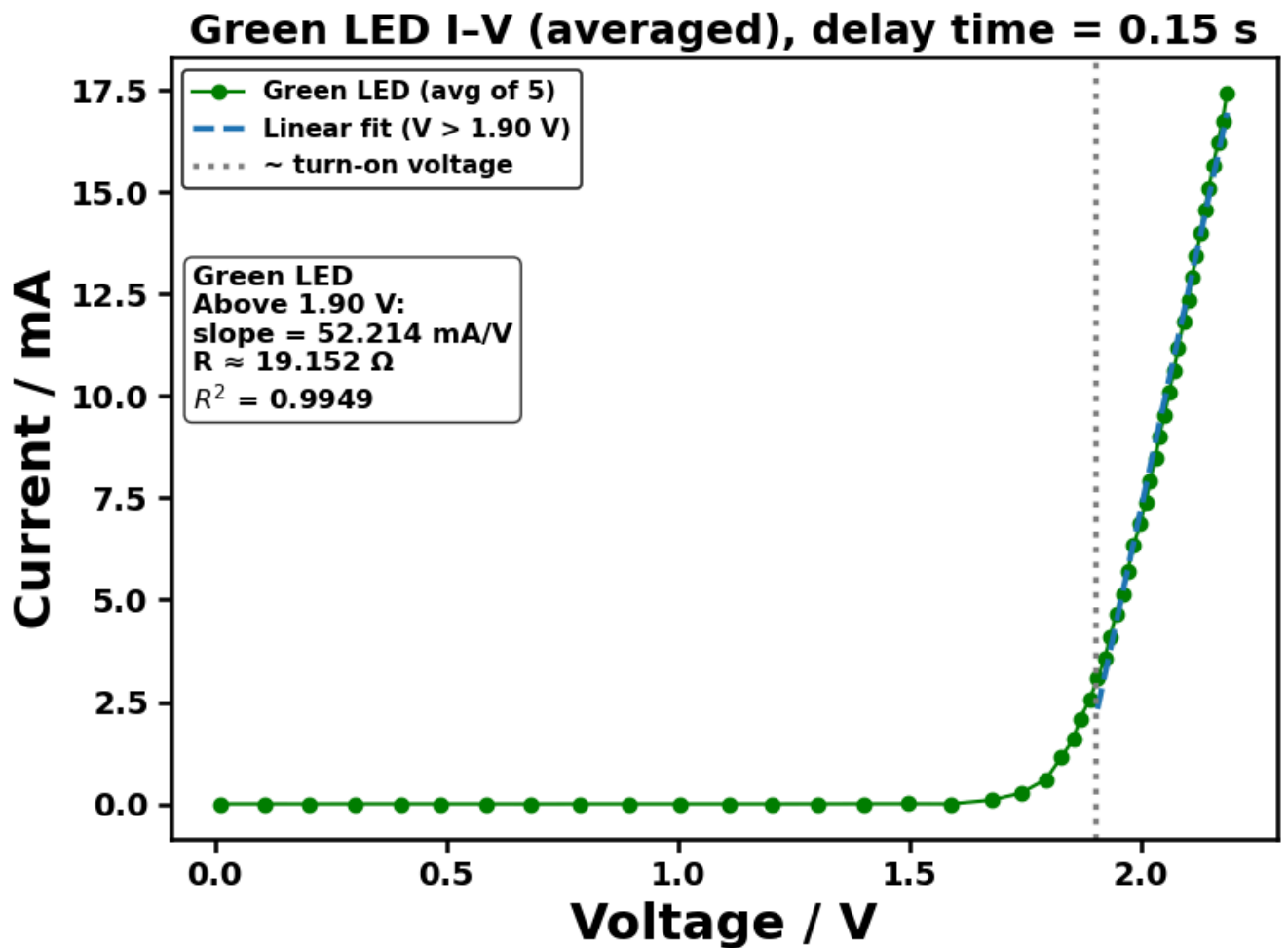
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/1_Delay_Time_Test/Green_LED_delay_0p1s_avg_20260122_193254.png



=== Delay time = 0.15 s ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/1_Delay_Time_Test/Green_LED_delay_0p15s_repeats5_20260122_193254.csv

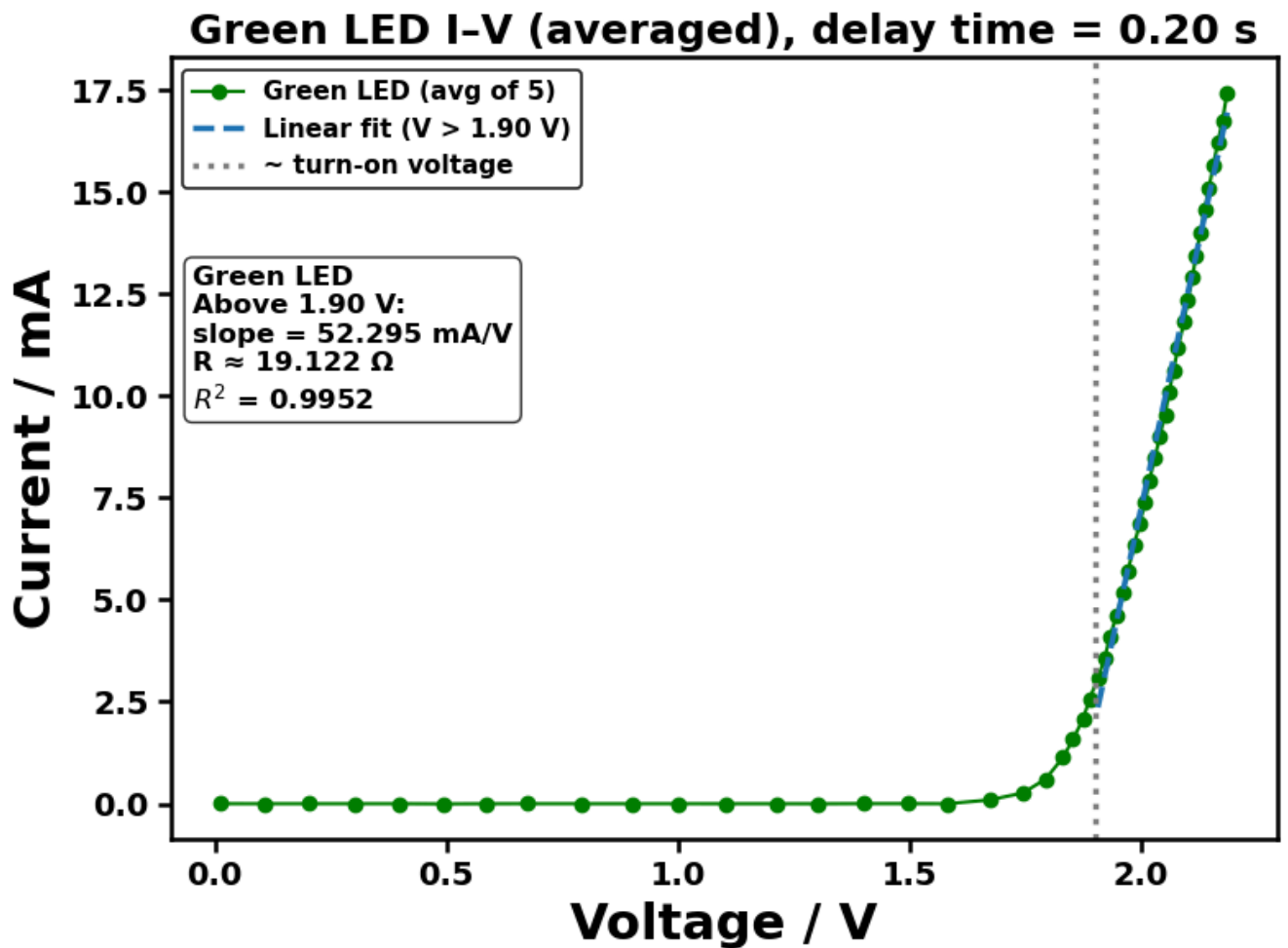
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/1_Delay_Time_Test/Green_LED_delay_0p15s_avg_20260122_193254.png



=== Delay time = 0.2 s ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/1_Delay_Time_Test/Green_LED_delay_0p2s_repeats5_20260122_193254.csv

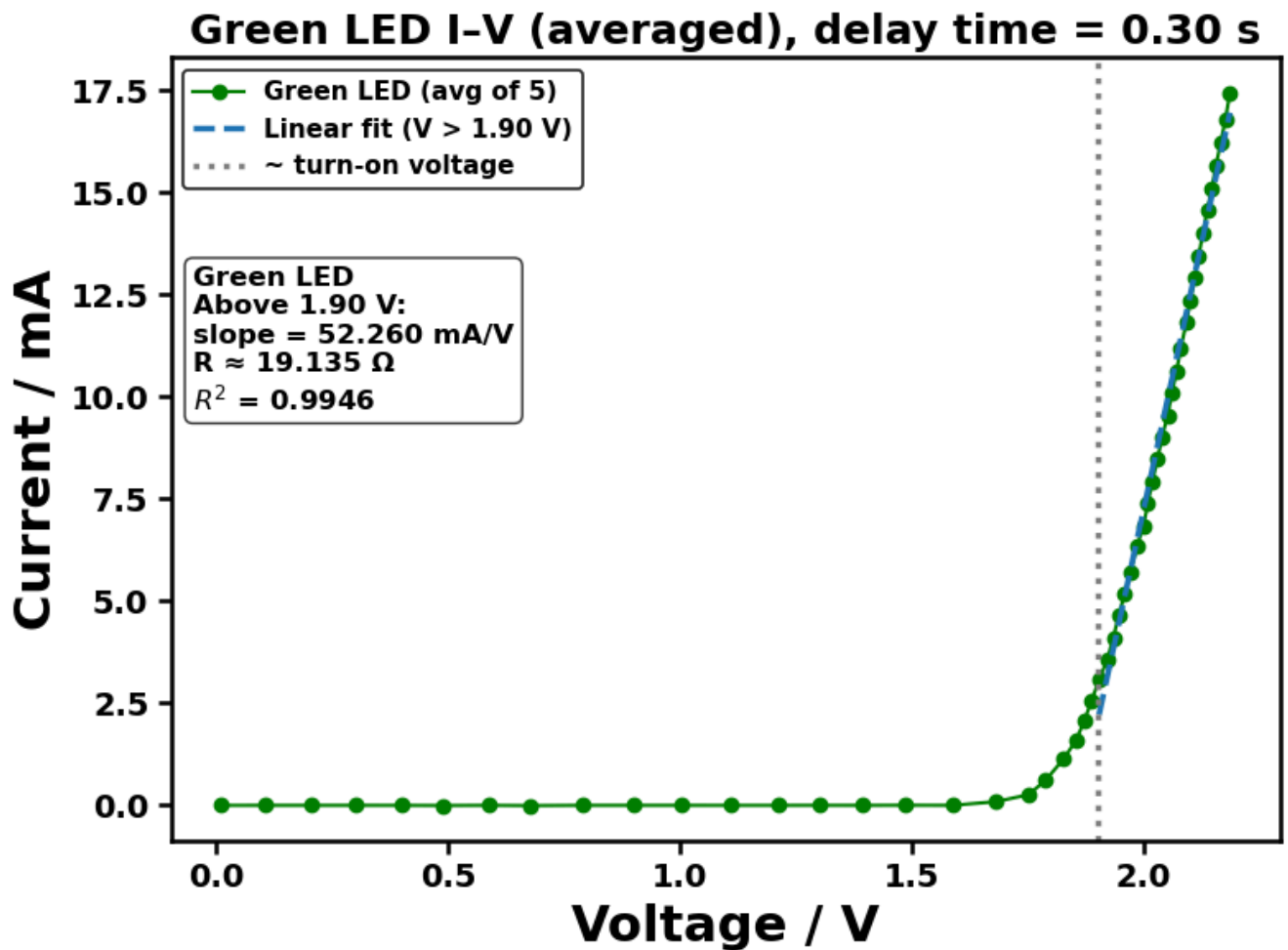
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/1_Delay_Time_Test/Green_LED_delay_0p2s_avg_20260122_193254.png



=== Delay time = 0.3 s ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/1_Delay_Time_Test/Green_LED_delay_0p3s_repeats5_20260122_193254.csv

Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/1_Delay_Time_Test/Green_LED_delay_0p3s_avg_20260122_193254.png



Experiment 1 done.

1.2. Step Voltage

Explanation :

To investigate the effect of voltage step size on the measured I-V characteristics, another function, `run_step_voltage_test`, was defined in which the step voltage is provided as a user input while all other experimental parameters, including the stabilisation delay, are held constant. Each execution of the function performs a complete I-V sweep over a fixed voltage range and applies linear fits to the high-voltage regions of the data.

Step voltages that were too large resulted in insufficient data density for reliable linear fitting, while excessively small step sizes increased experimental duration without improving the accuracy. An intermediate step voltage was therefore selected to ensure reliable linear fitting while maintaining experimental efficiency.

Results and conclusion :

We tested 0.05 V, 0.1 V, 0.2 V, 0.5 V and 1 V and found that 0.1 V is the optimal balance between efficiency and accuracy.

```
In [143... run_step_voltage_test(
    ser,
    steps=(0.05, 0.10, 0.20, 0.50, 1.00),
    repeats=5,
    led_label="Green LED",
    V_split=1.9,
    color="green",
```

```

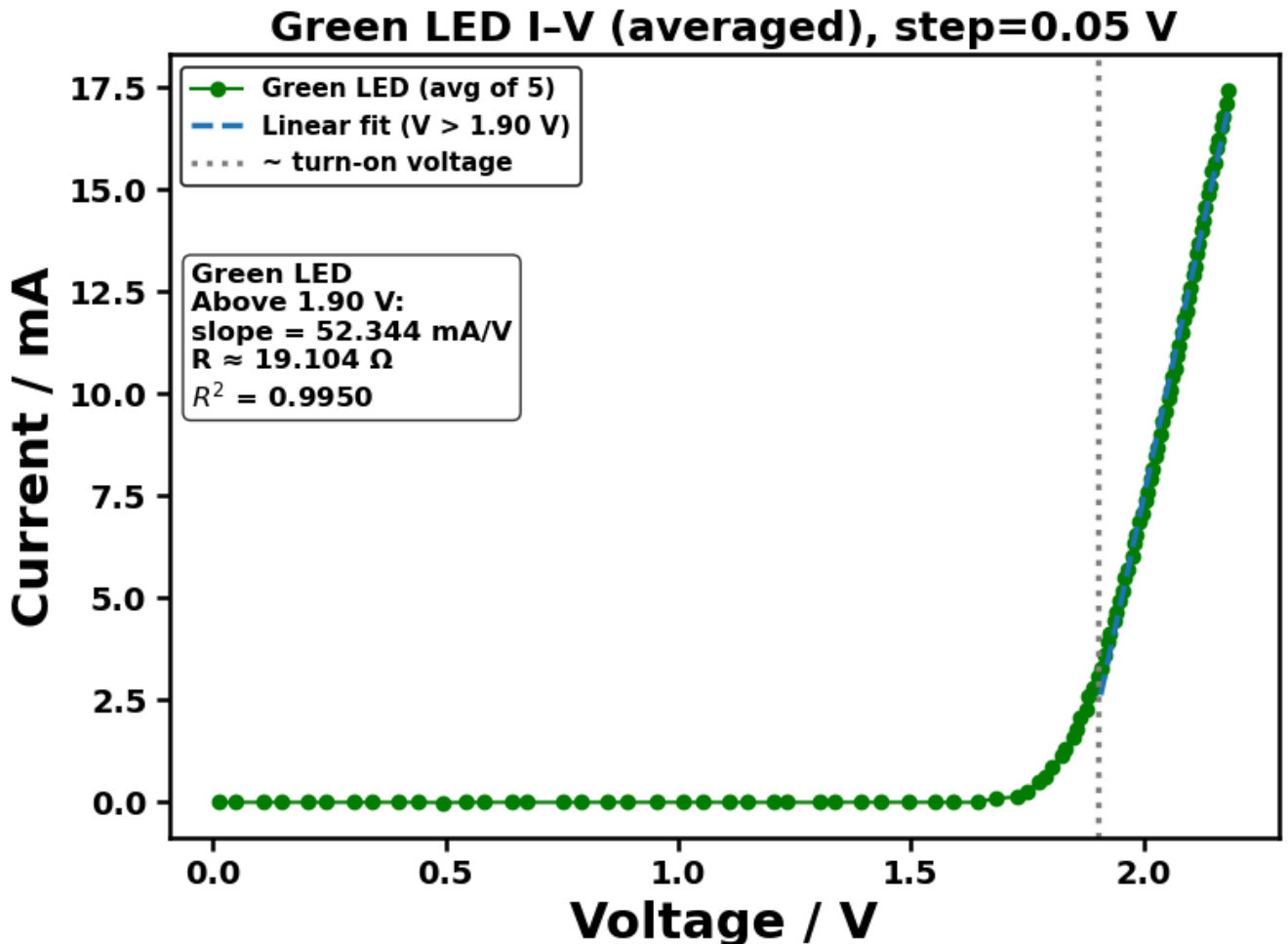
delay=0.1,
Pn=200,
fit_mode="single",
require_enter_each_run=False
)

```

=== Step voltage = 0.05 V ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/2_Step_Voltage_Test/Green_LED_step_0p05V_repeats5_20260122_195047.csv

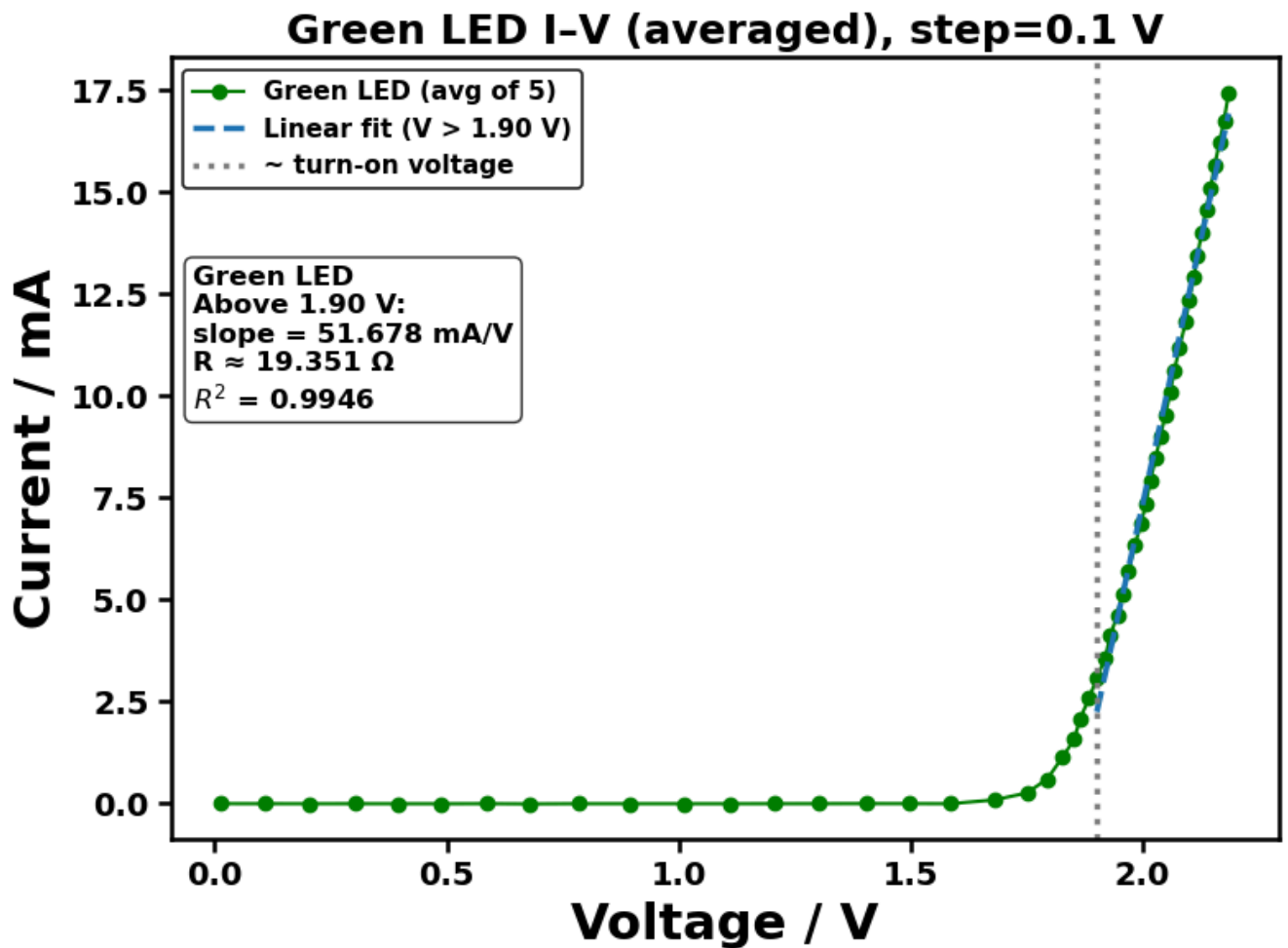
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/2_Step_Voltage_Test/Green_LED_step_0p05V_avg_20260122_195047.png



=== Step voltage = 0.1 V ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/2_Step_Voltage_Test/Green_LED_step_0p1V_repeats5_20260122_195047.csv

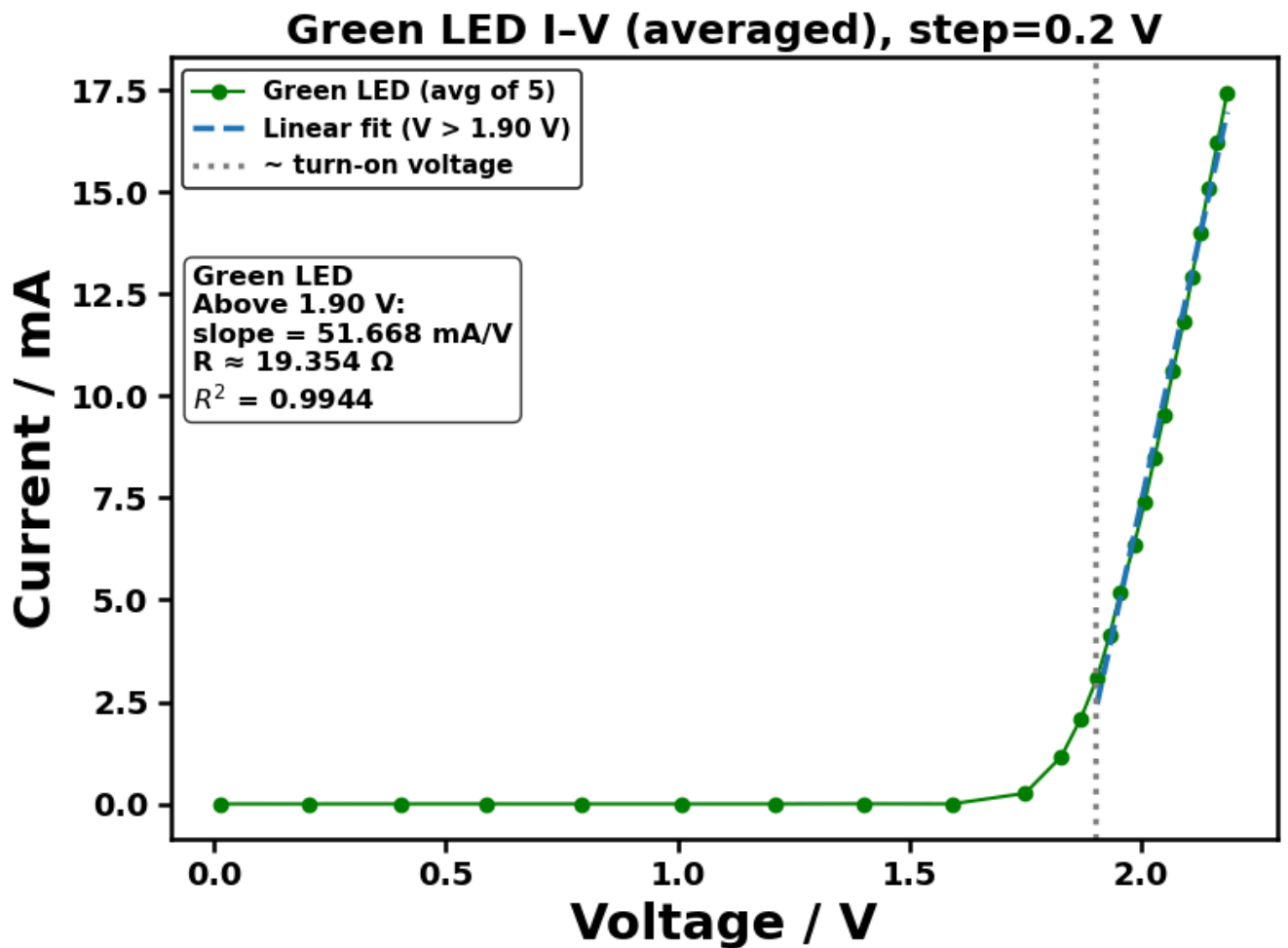
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/2_Step_Voltage_Test/Green_LED_step_0p1V_avg_20260122_195047.png



=== Step voltage = 0.2 V ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/2_Step_Voltage_Test/Green_LED_step_0p2V_repeats5_20260122_195047.csv

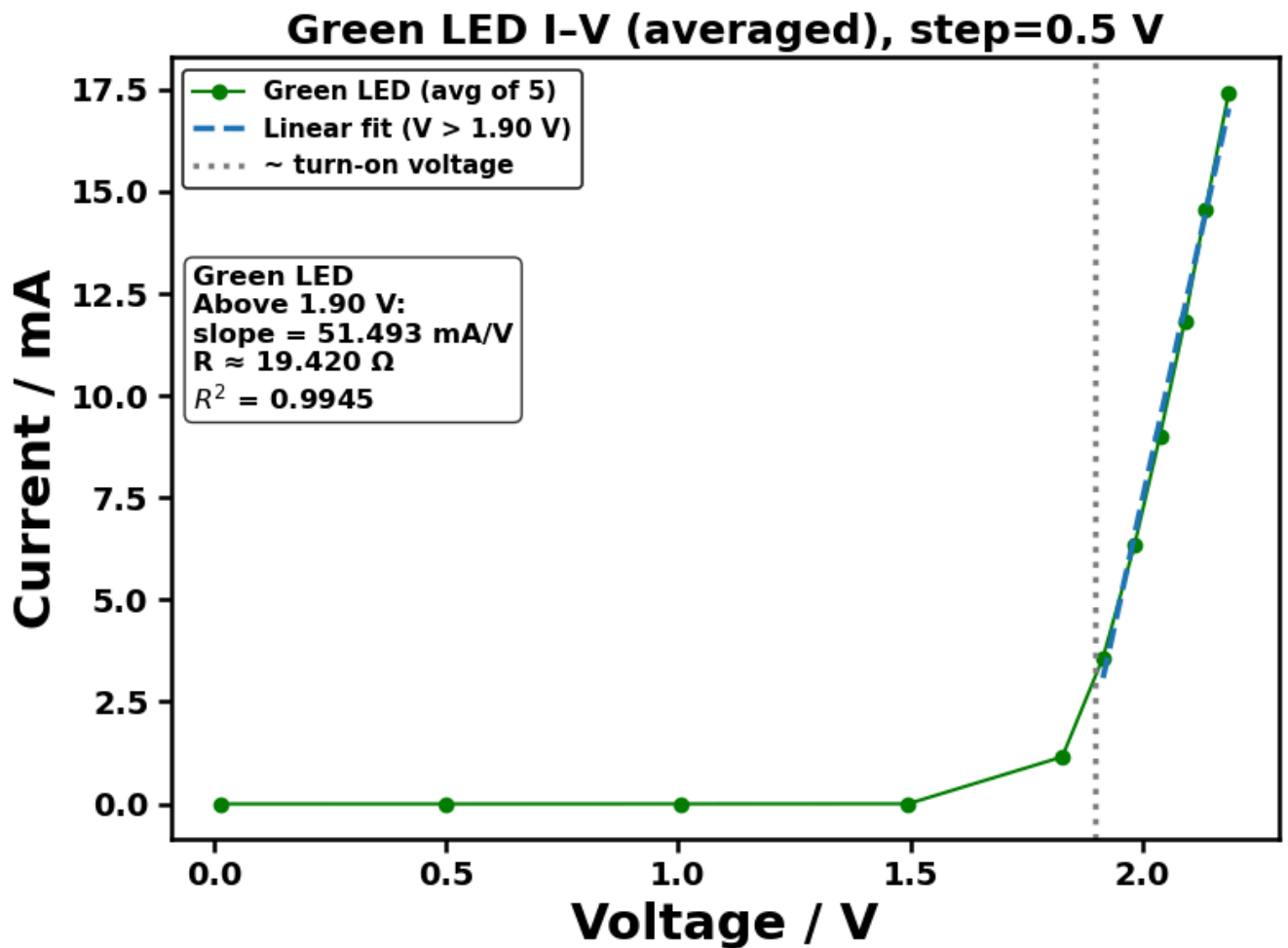
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/2_Step_Voltage_Test/Green_LED_step_0p2V_avg_20260122_195047.png



=== Step voltage = 0.5 V ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/2_Step_Voltage_Test/Green_LED_step_0p5V_repeats5_20260122_195047.csv

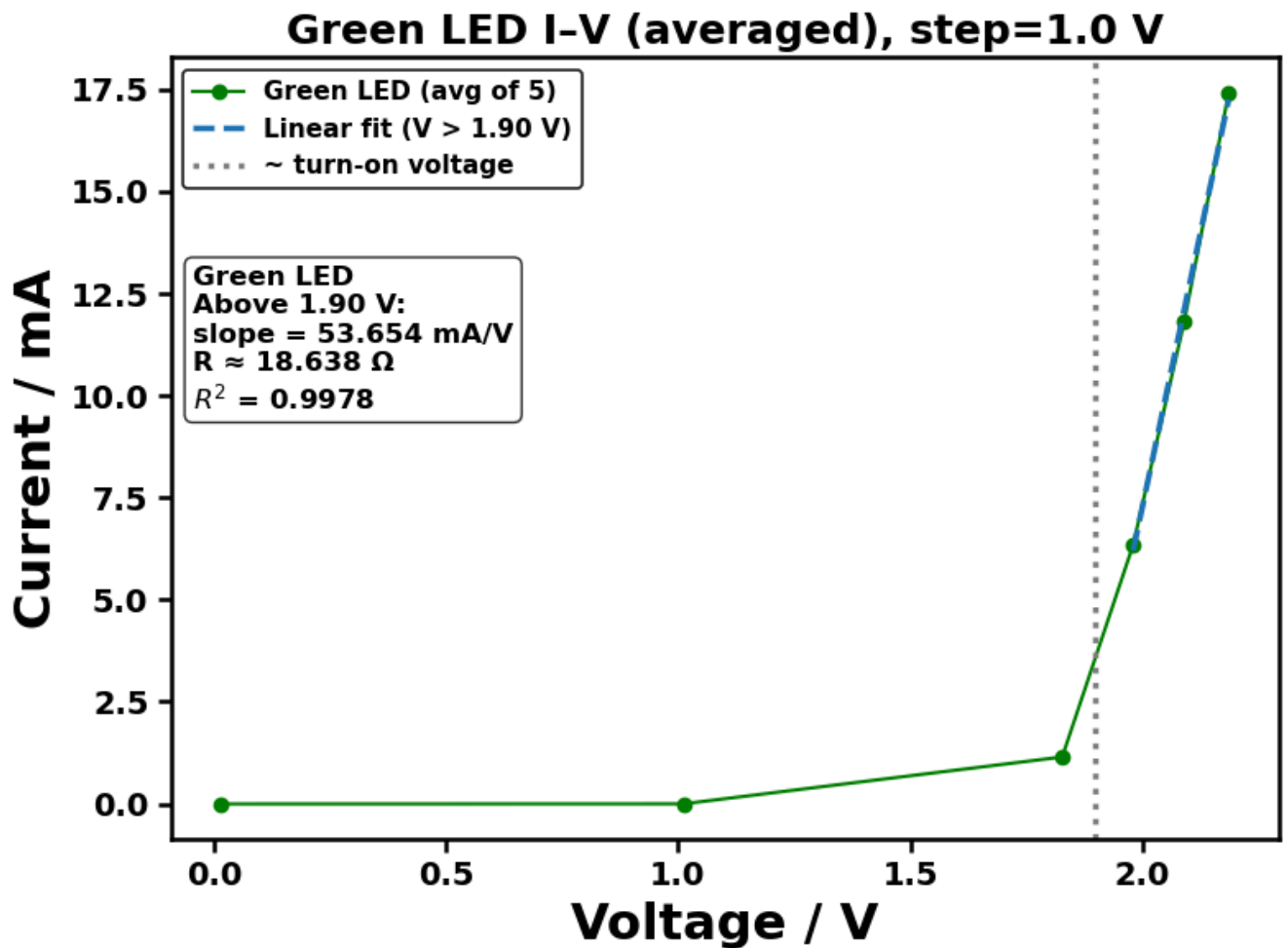
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/2_Step_Voltage_Test/Green_LED_step_0p5V_avg_20260122_195047.png



=== Step voltage = 1.0 V ===

Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/2_Step_Voltage_Test/Green_LED_step_1V_repeats5_20260122_195047.csv

Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/2_Step_Voltage_Test/Green_LED_step_1V_avg_20260122_195047.png



Experiment 2 done.

2. I-V Characteristics

2.1. Nothing Connected

Explanation :

Baseline measurement was performed with no LED or standard resistor connected, corresponding to an **open-circuit** configuration.

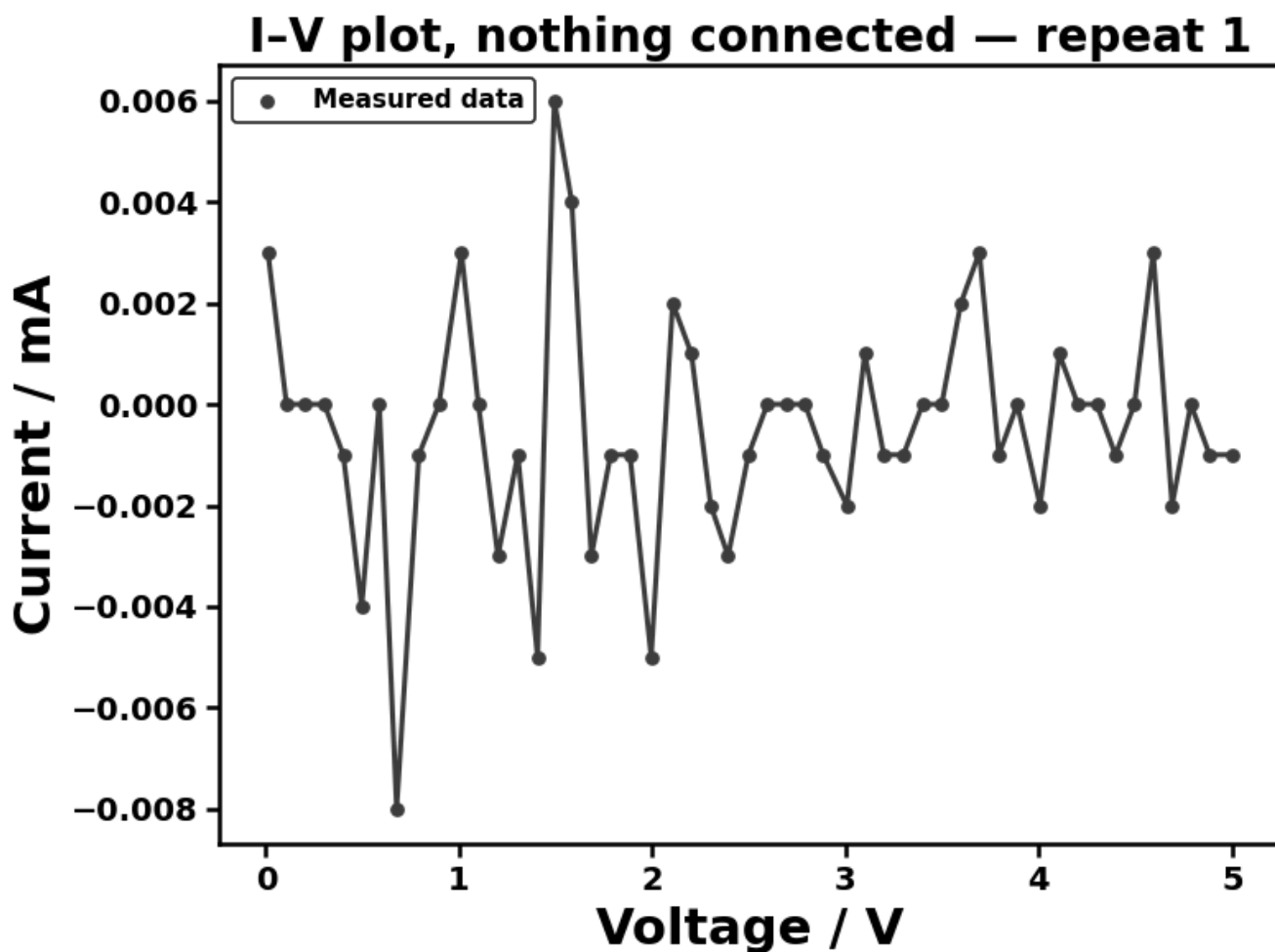
The current is calculated as:

$$I = (A0 - A1)/R$$

For an ideal open circuit, $A0 \approx A1$, and therefore $I \approx 0$. Repeated measurements under identical conditions produced different readings, indicating systematic errors from the reading of the currents.

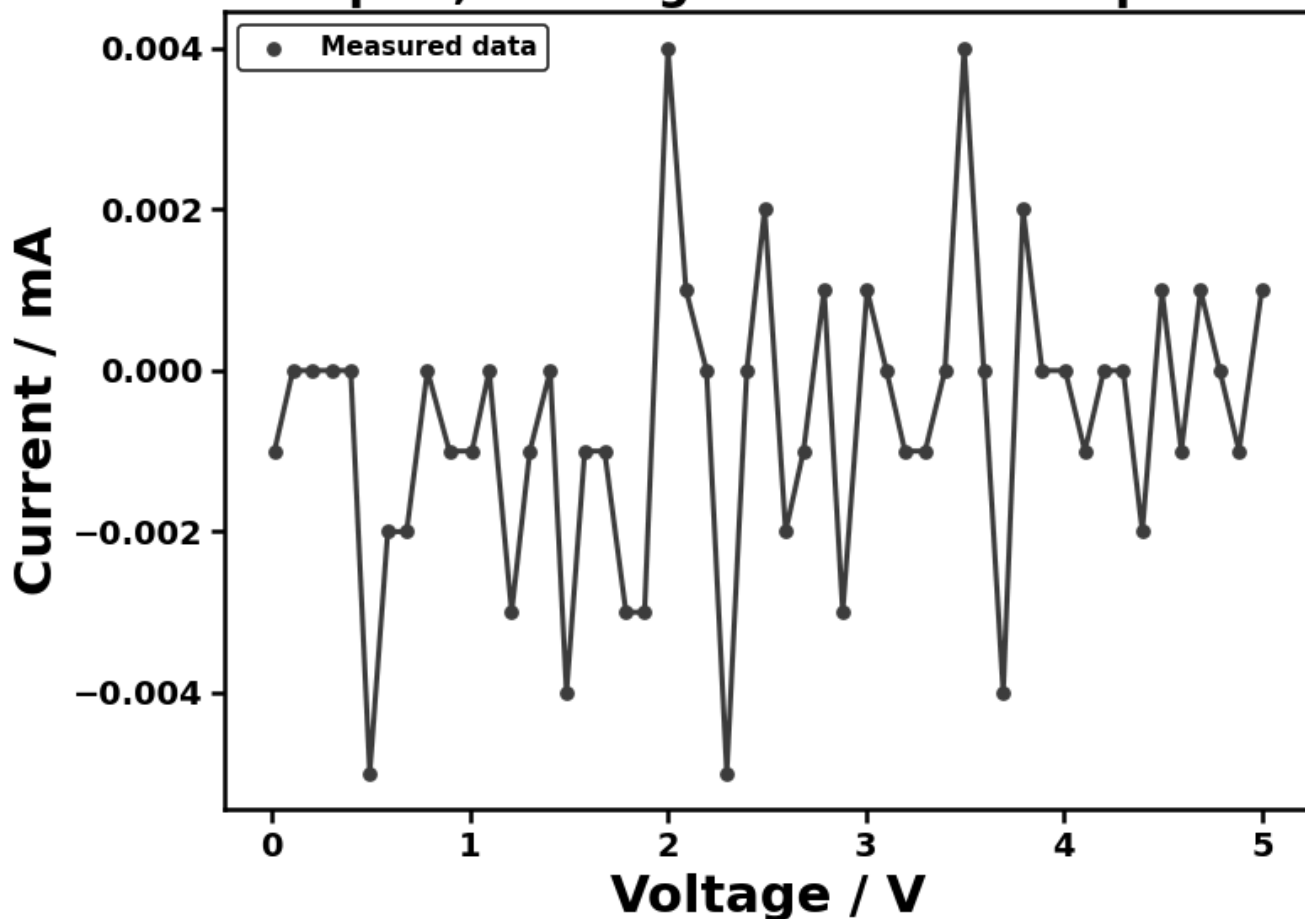
```
In [173.. run_open_circuit_baseline(
    ser,
    repeats=5,
    v_step=0.1,
    delay=0.1,
    Pn=200,
    fit_mode="single",
    require_enter_each_run=False
)
```

Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/3_Open_Circuit_Baseline/open_circuit_run1_20260122_201328.png



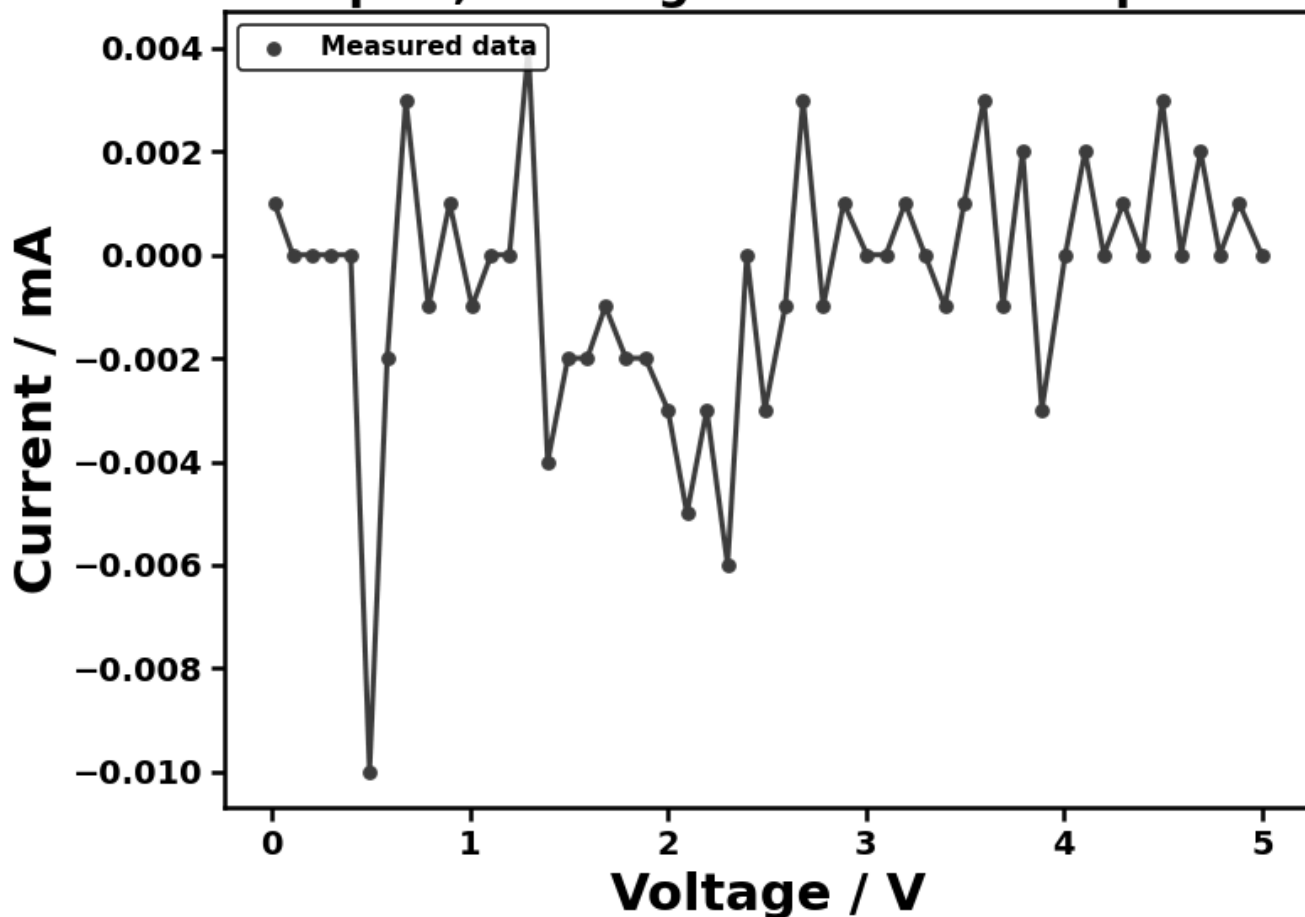
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/3_Open_Circuit_Baseline/open_circuit_run2_20260122_201328.png

I-V plot, nothing connected — repeat 2

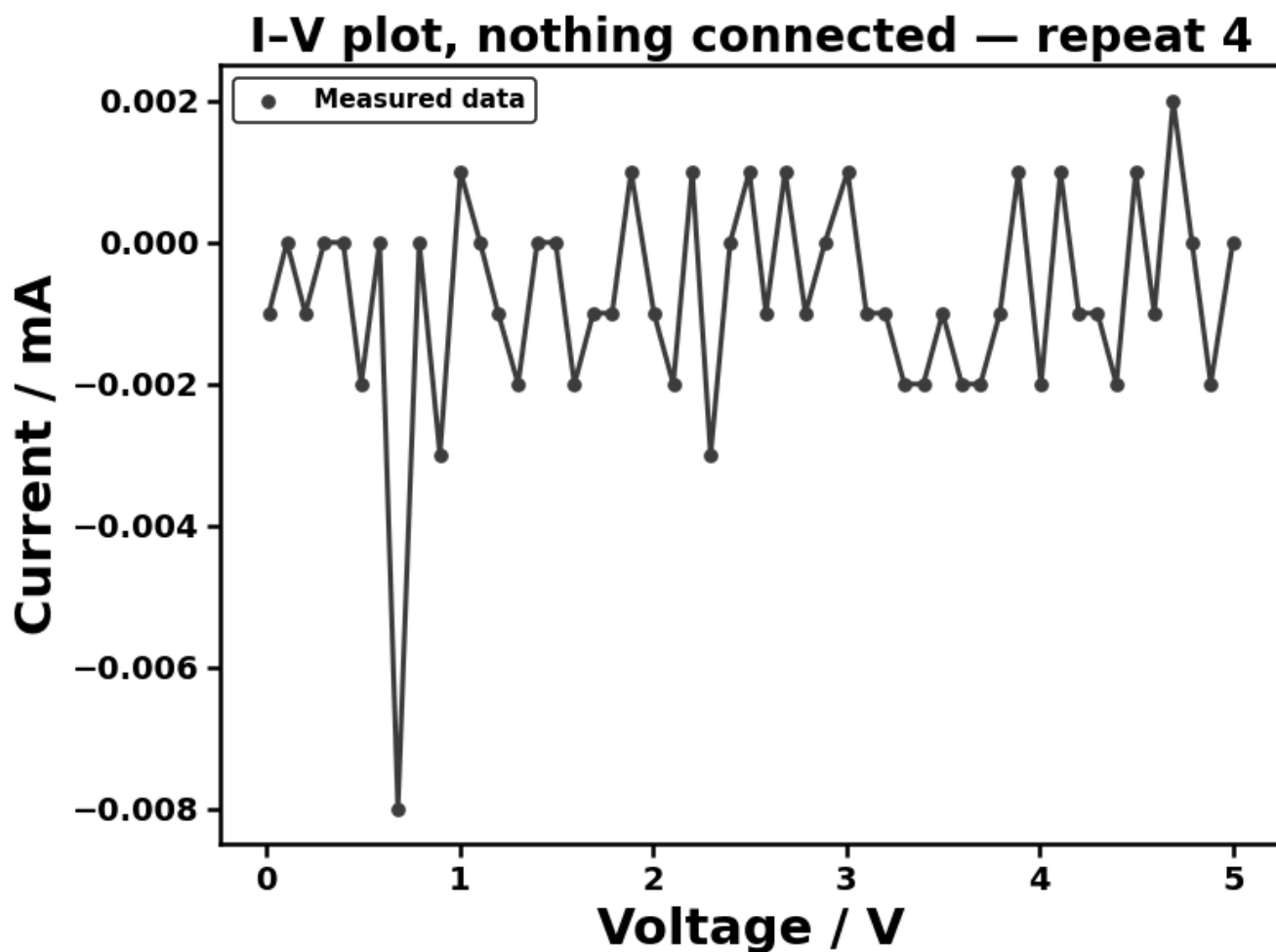


Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/3_Open_Circuit_Baseline/open_circuit_run3_20260122_201328.png

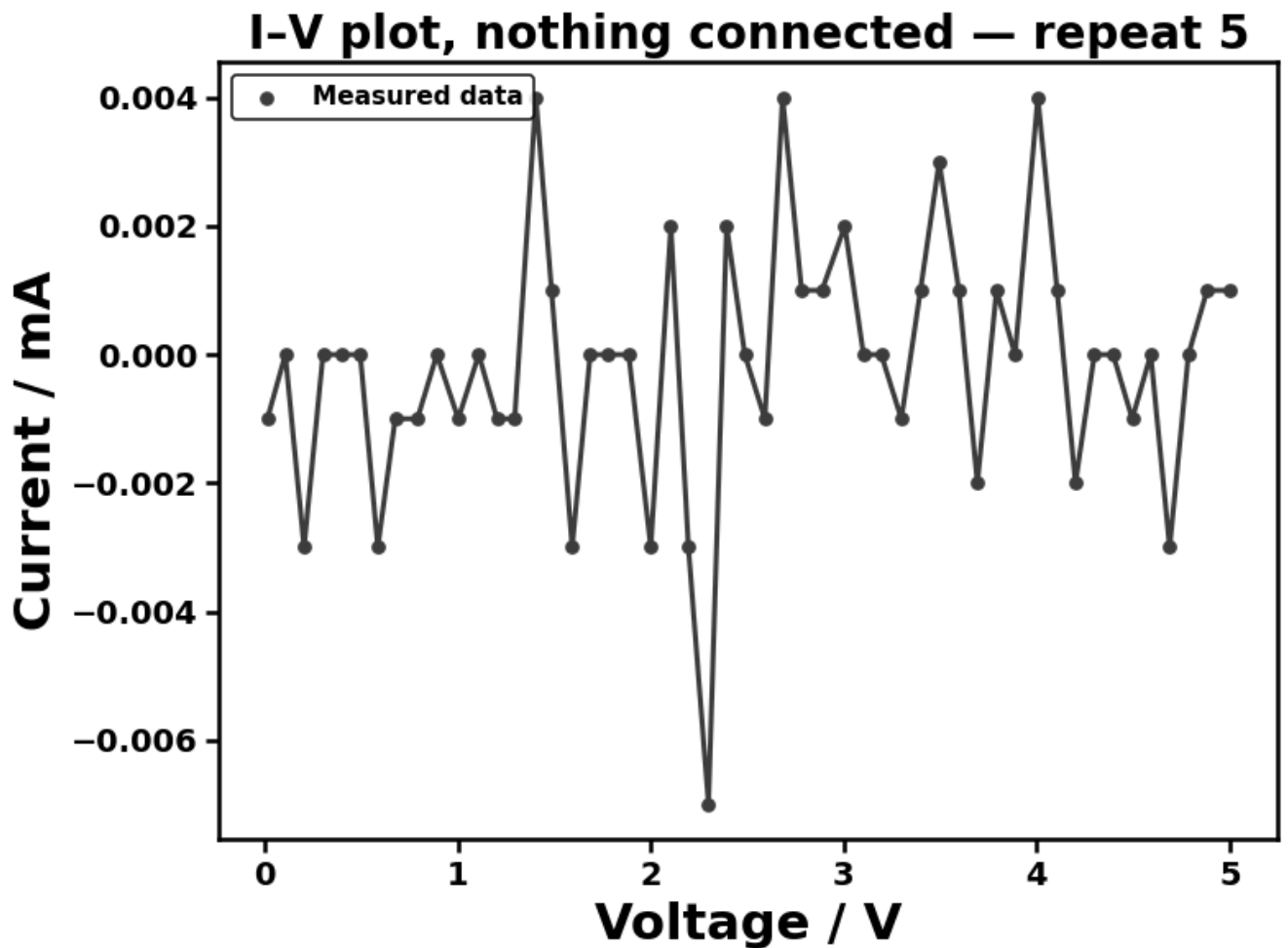
I-V plot, nothing connected — repeat 3



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/3_Open_Circuit_Baseline/open_circuit_run4_20260122_201328.png



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/3_Open_Circuit_Baseline/open_circuit_run5_20260122_201328.png



Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/3_Open_Circuit_Baseline/open_circuit_repeats5_20260122_201328.csv

Experiment 3 done.

2.2. Standard Resistor

Aim :

To measure and characterise the I-V relationship of a standard resistor, and to demonstrate ohmic-like conduction as a reference case for comparison with LEDs.

Explanation :

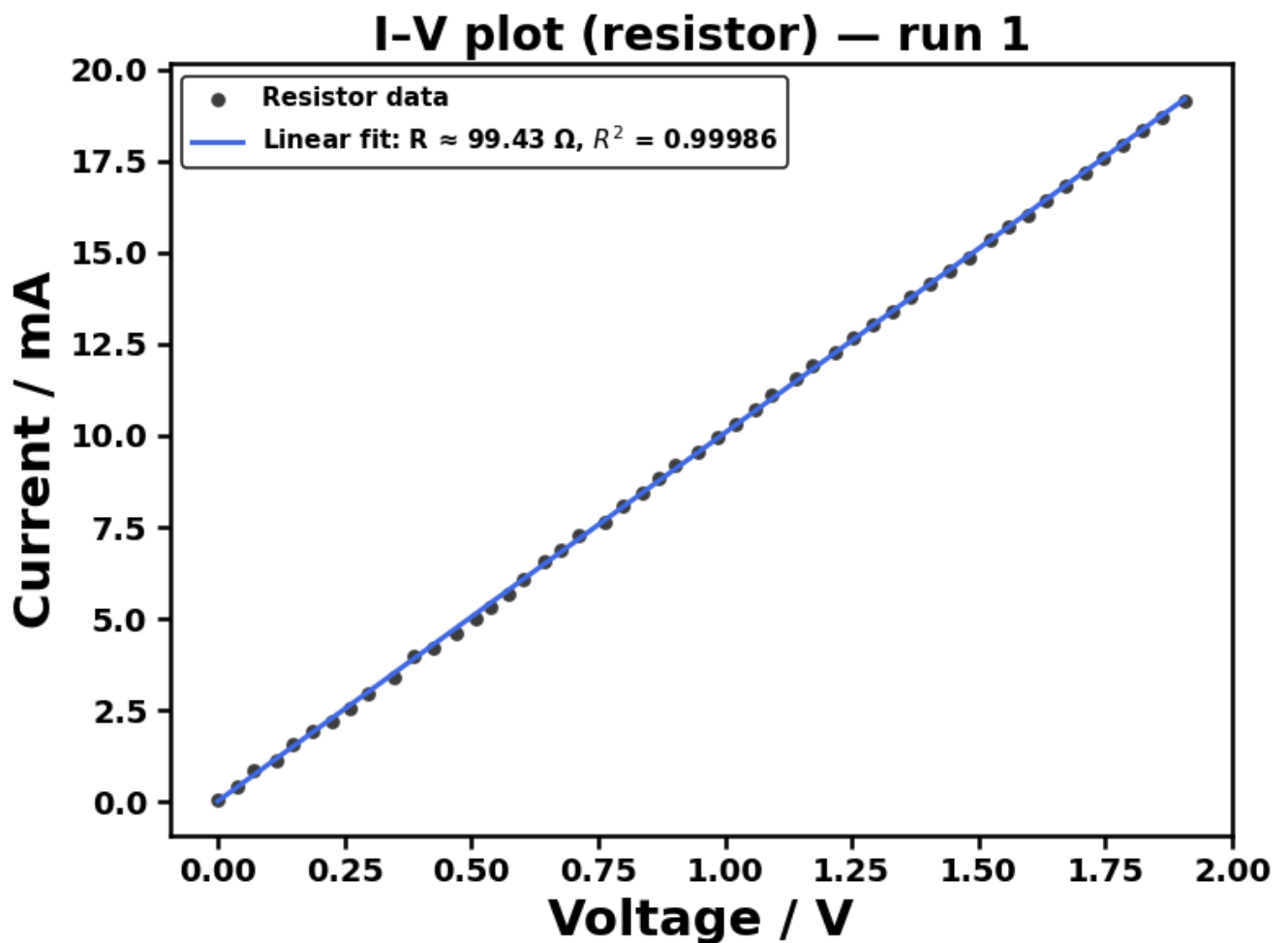
The I-V data for the standard resistor were collected using the computer-controlled acquisition system. The raw data were saved as a csv file for subsequent data processing, averaging, and plotting.

Each measurement was repeated five times at every applied voltage to reduce the influence of random experimental error and improve statistical reliability. The resulting I-V relationship was expected to exhibit a linear dependence, consistent with Ohm's law.

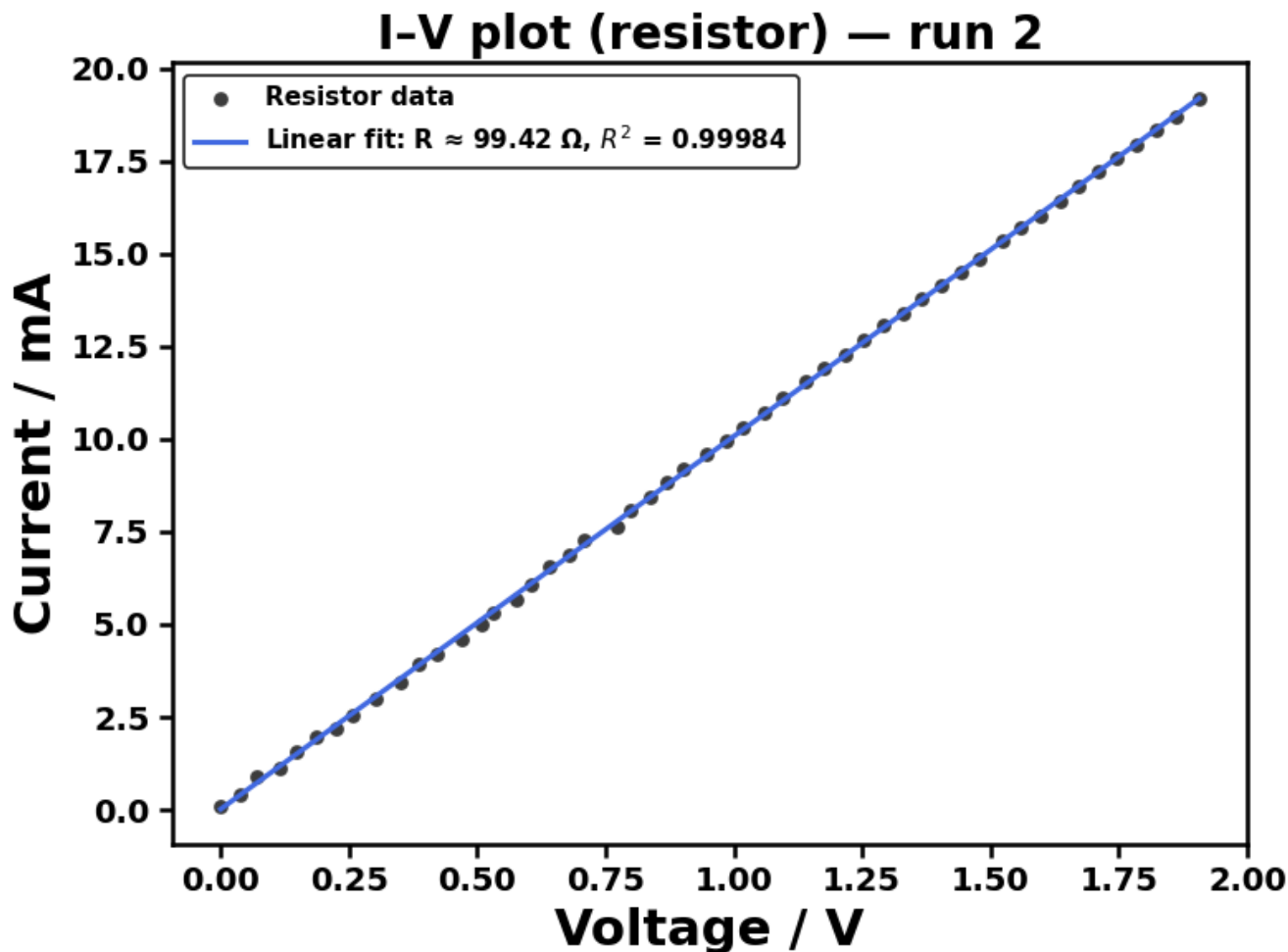
```
In [183... run_resistor_iv_sweep(
    ser,
    repeats=5,
    v_step=0.1,
    delay=0.1,
    Pn=200,
    fit_mode="single",
```

```
) require_enter_each_run=False
```

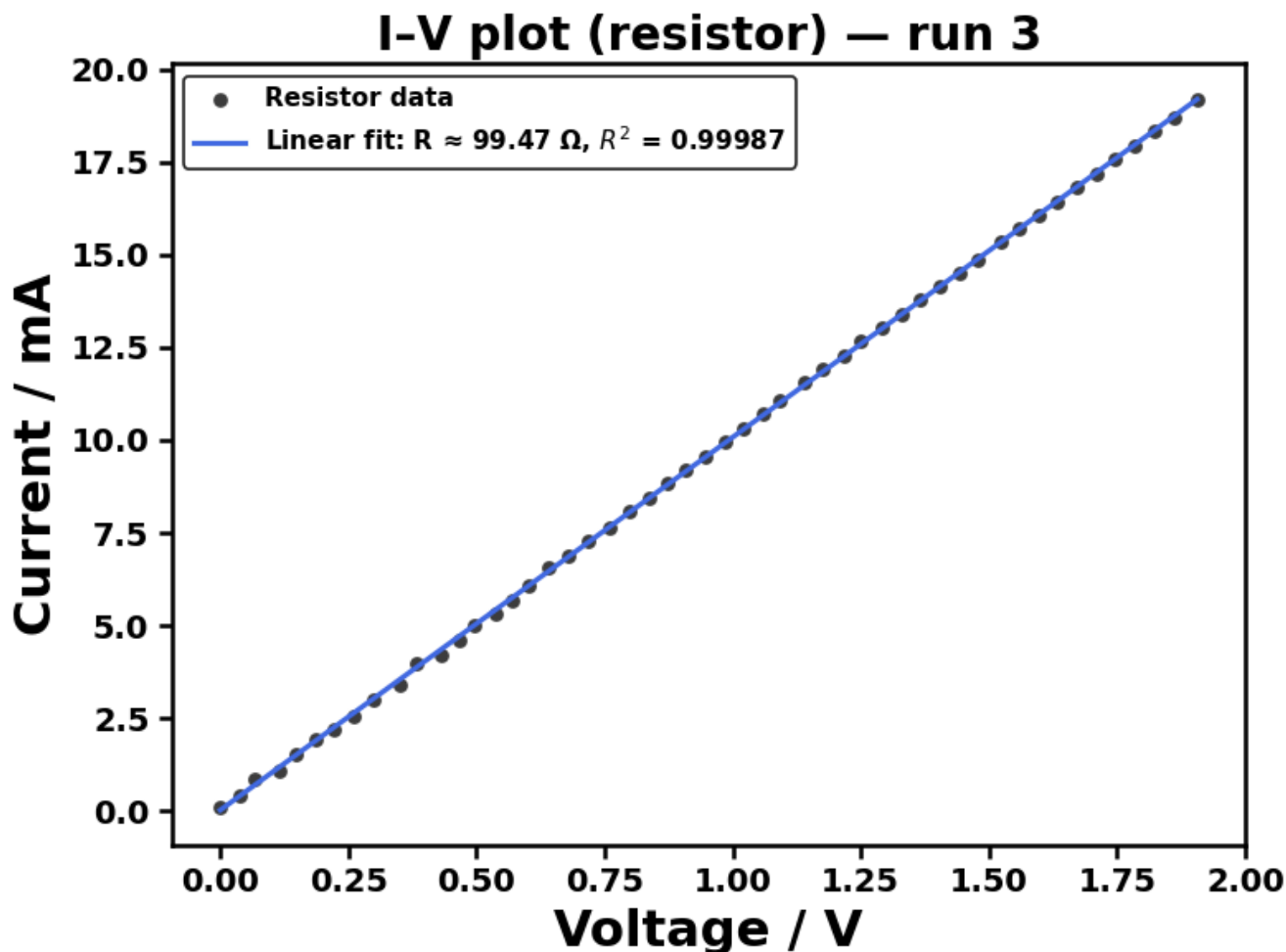
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/4_Resistor/resistor_run1_20260122_201801.png



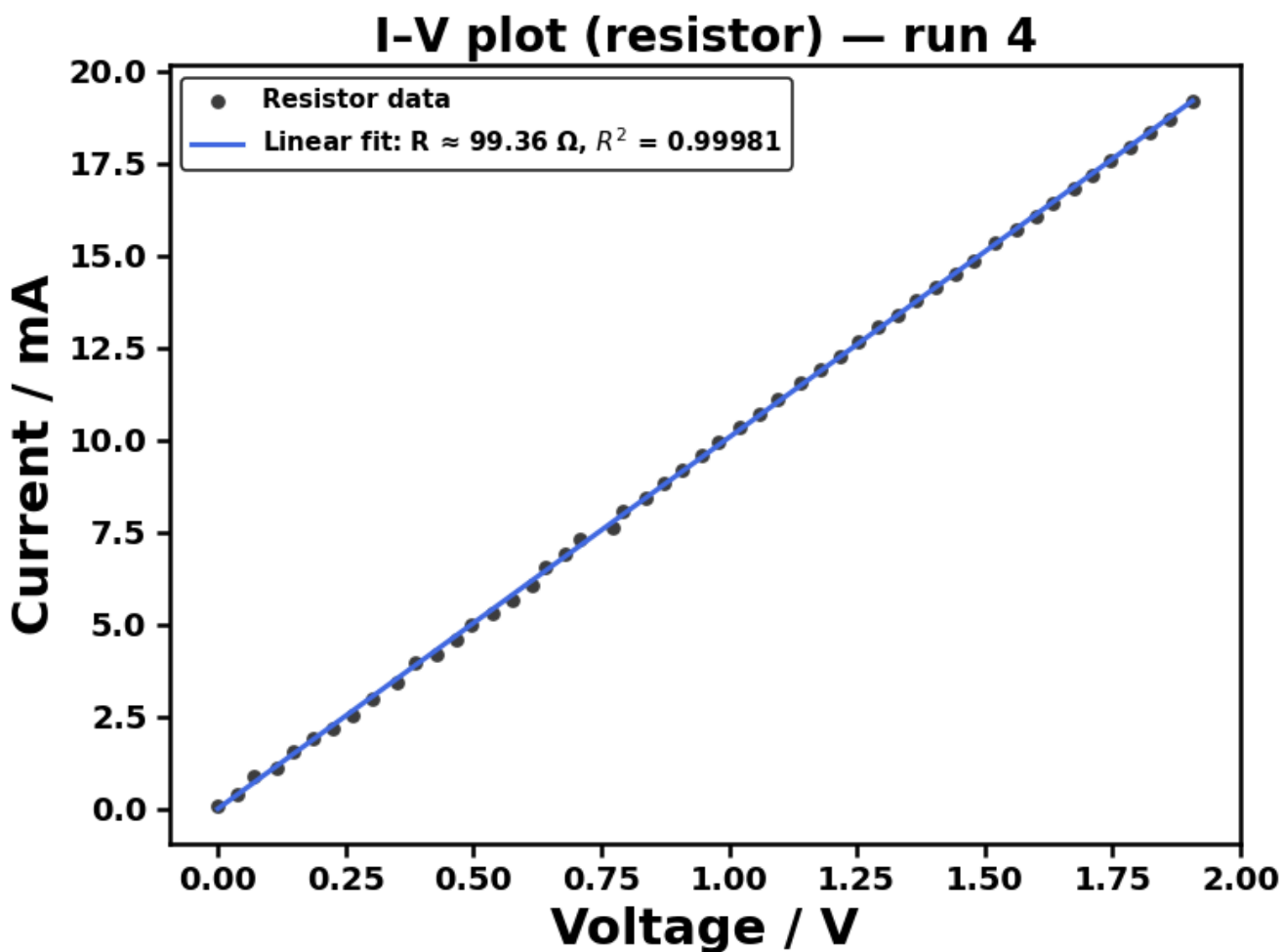
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/4_Resistor/resistor_run2_20260122_201801.png



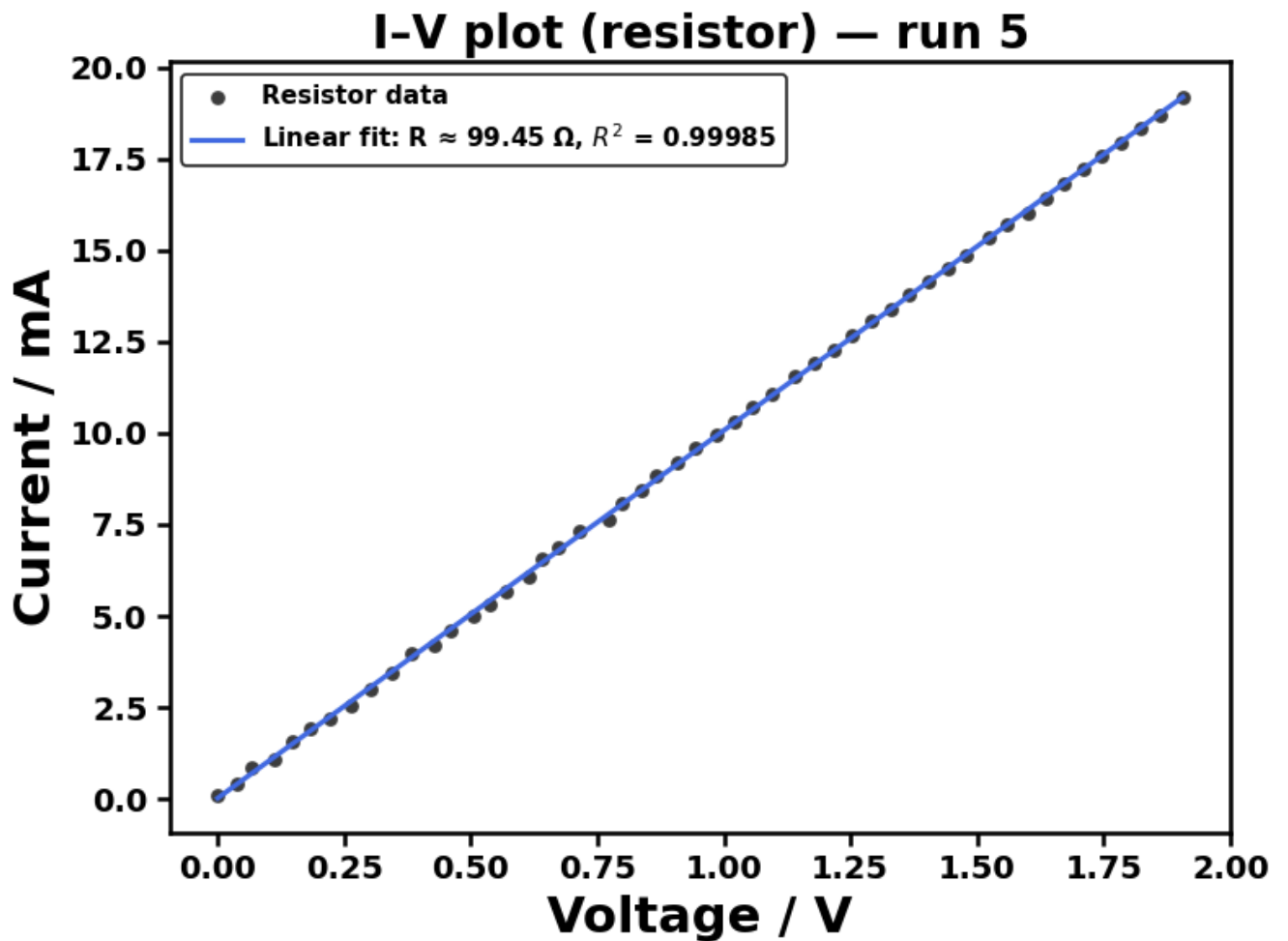
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/4_Resistor/resistor_run3_20260122_201801.png



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/4_Resistor/resistor_run4_20260122_201801.png



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/Figures/4_Resistor/resistor_run5_20260122_201801.png



Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260122/raw data_finalised/4_Resistor/resistor_repeats5_20260122_201801.csv

Experiment 4 done.

2.3. Colored LED

Aim :

To measure the I-V characteristics of LED and investigate their electronic behaviour, while comparing LEDs of different emission colours to examine the relationship between turn-on voltage and semiconductor energy band gap.

Explanation :

The applied voltage across each LED was incrementally swept and the resulting current was recorded using micro-controller with data stored in csv format for efficient processing. The LEDs were measured in the order Red, Yellow, Green, Blue and White.

The measured I-V curves exhibit characteristic behaviour, with minimal current at low voltages followed by a sharp increase beyond a specific turn-on voltage. This behaviour is governed by charge carrier injection across a p-n junction and is described by the Shockley diode equation rather than Ohm's law. Differences in turn-on voltage between LEDs of different colours, including white, were analysed and attributed to variations in semiconductor band gap energy.

```
In [23]: led_specs = {  
    "Red LED": {"color": "red", "V_split": 1.7},  
    "Yellow LED": {"color": "gold", "V_split": 1.8},
```

```

"Green LED": {"color": "green", "V_split": 1.9},
"Blue LED": {"color": "blue", "V_split": 2.6},
"White LED": {"color": "grey", "V_split": 2.65},
} # note: here, we set the code to run the experiment in batch.

```

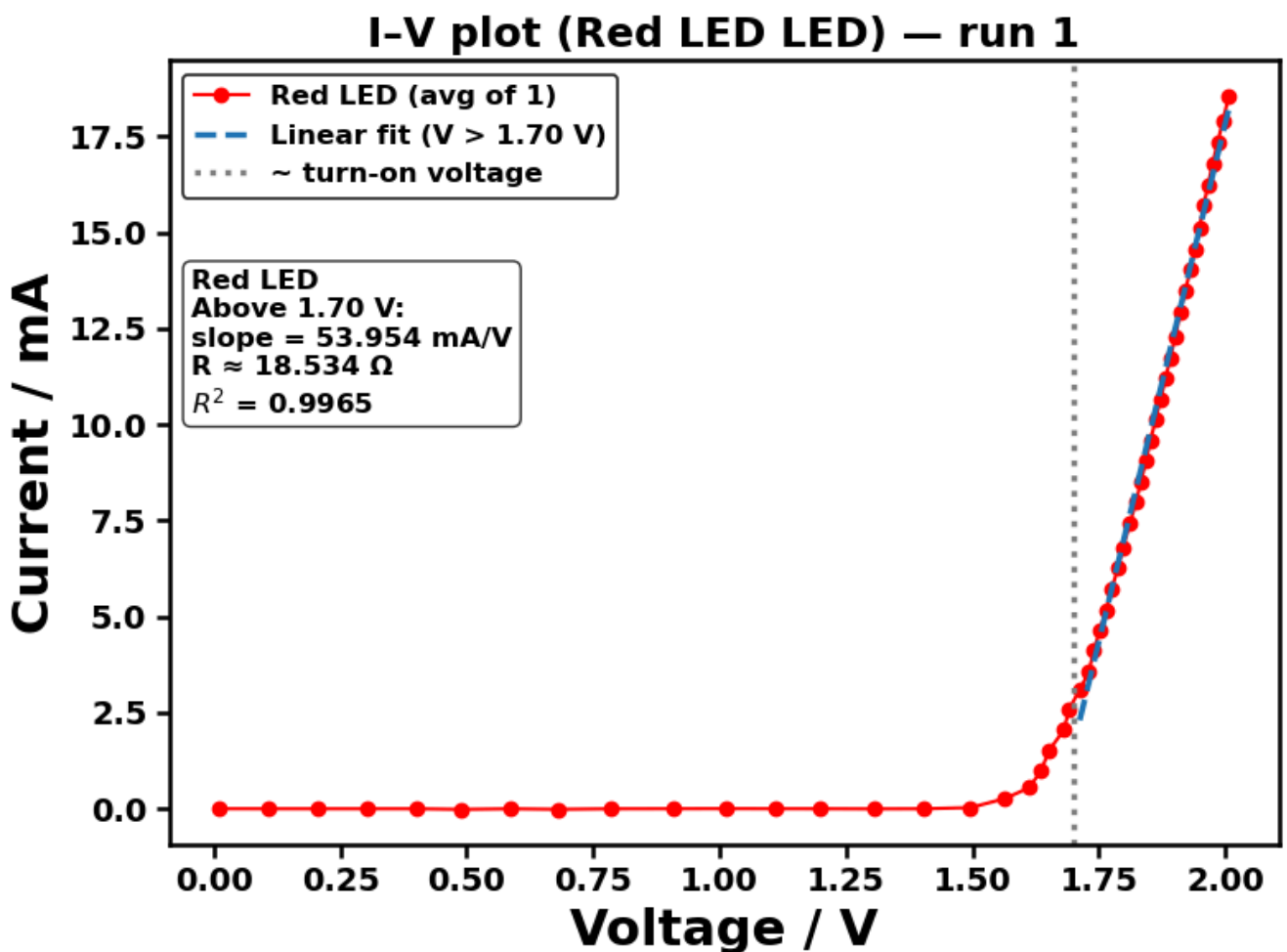
```

In [10]: run_led_colour_set(
    ser,
    led_specs=led_specs,
    repeats=5,
    v_step=0.1,
    delay=0.1,
    Pn=200,
    fit_mode="single",
    require_enter_each_run=True
)

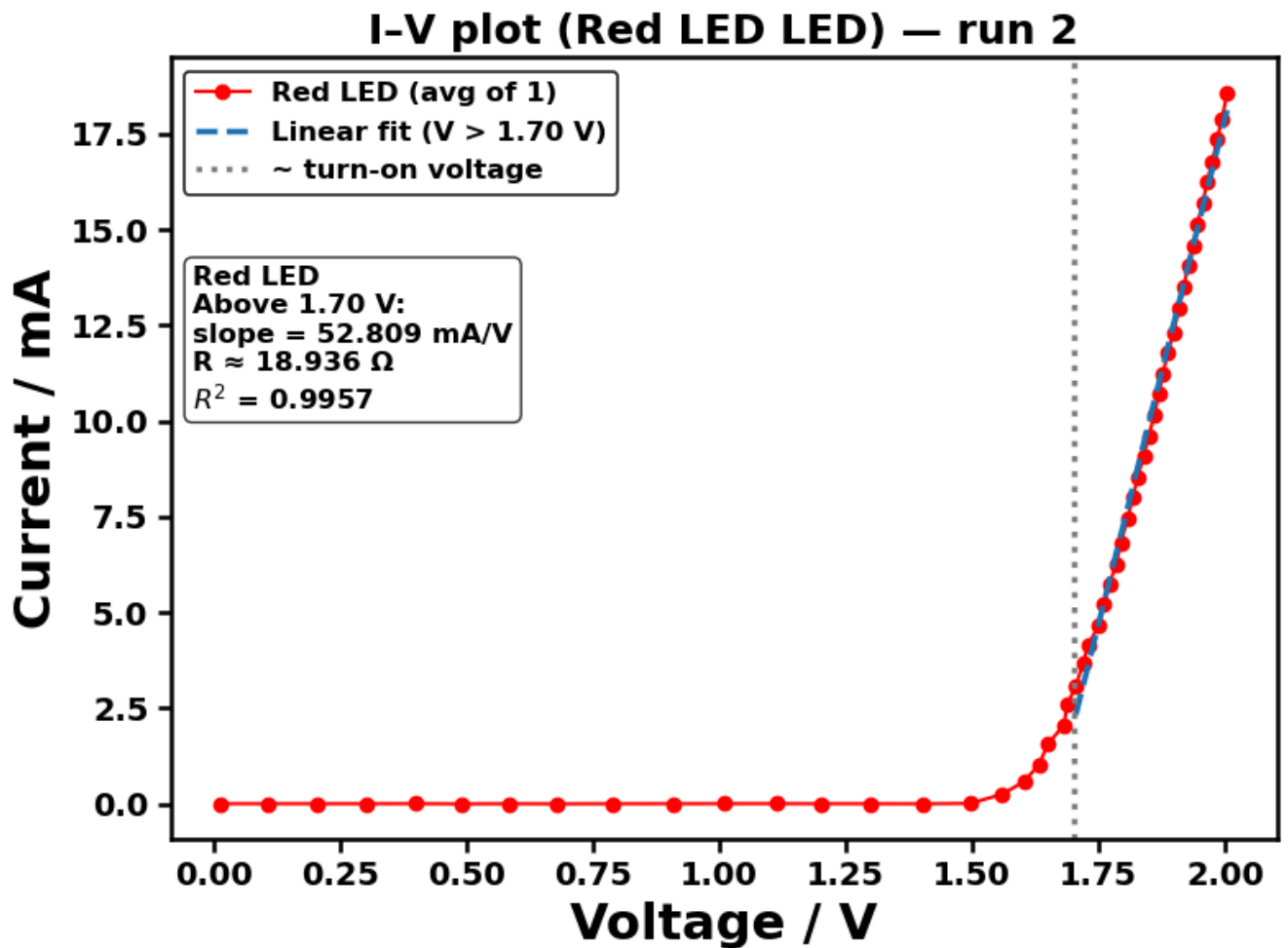
```

=== Red LED (V_split=1.7) ===

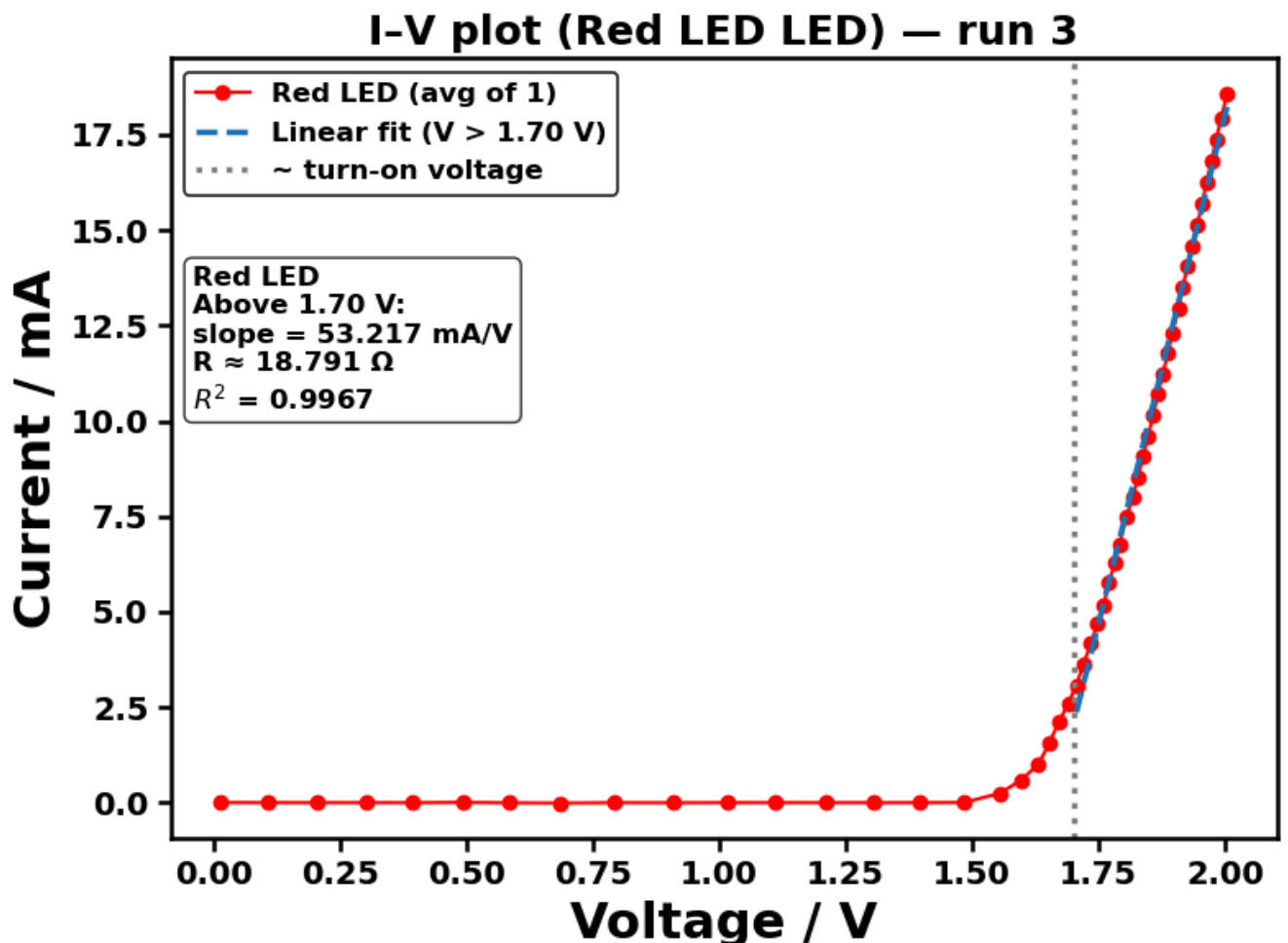
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Red_LED/Red_LED_run1_20260123_104738.png



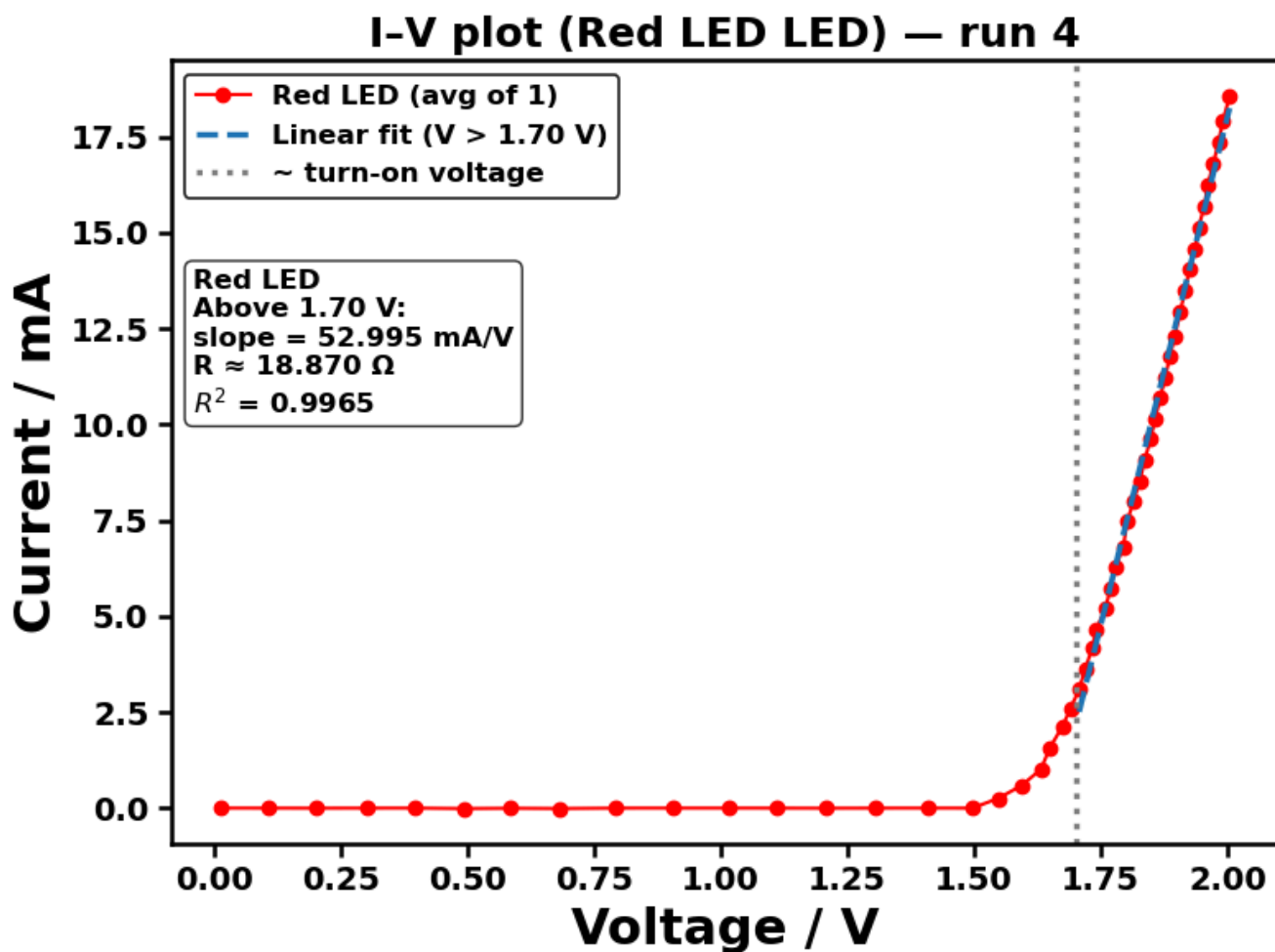
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Red_LED/Red_LED_run2_20260123_104738.png



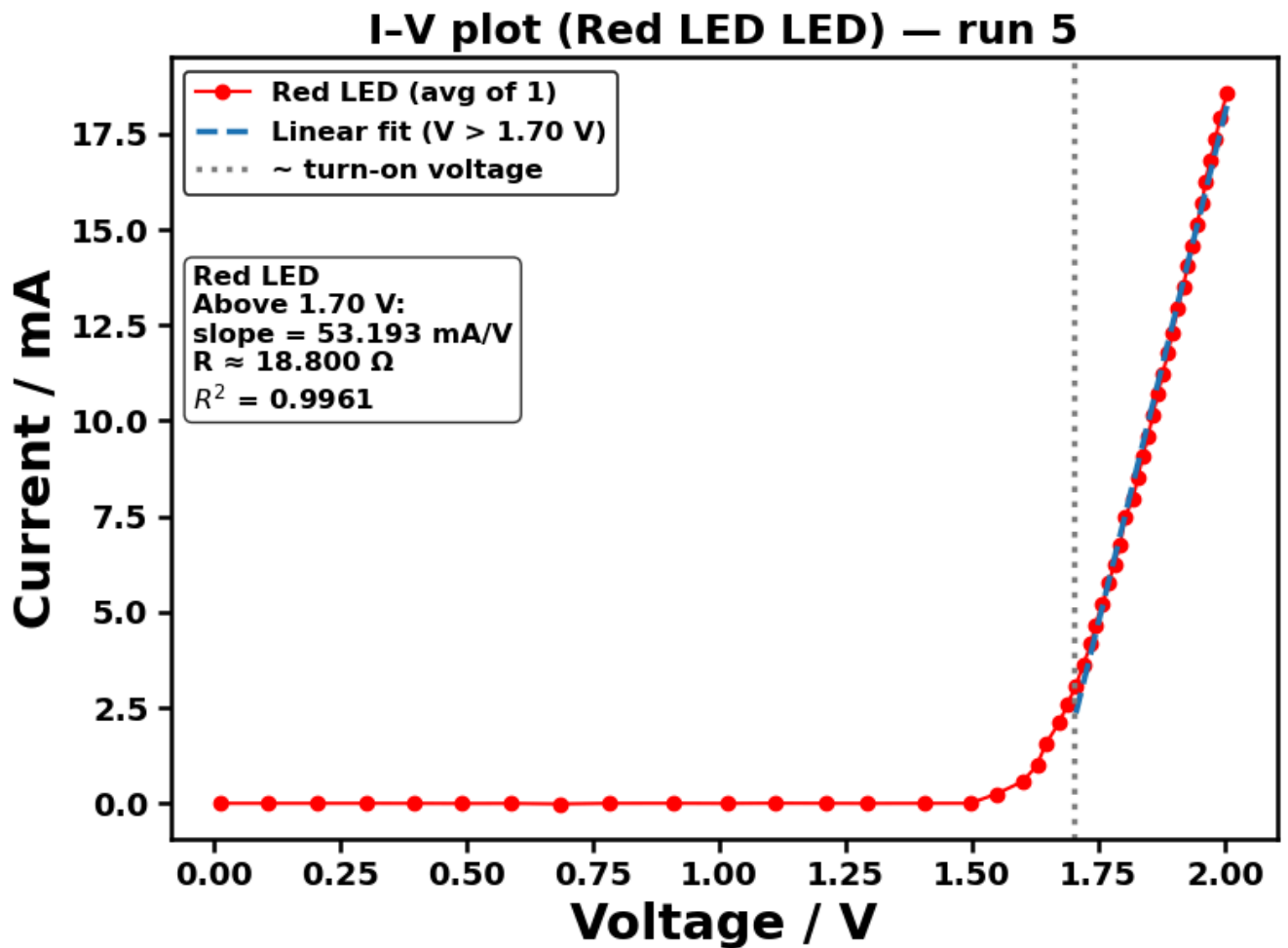
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Red_LED/Red_LED_run3_20260123_104738.png



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Red_LED/Red_LED_run4_20260123_104738.png



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Red_LED/Red_LED_run5_20260123_104738.png

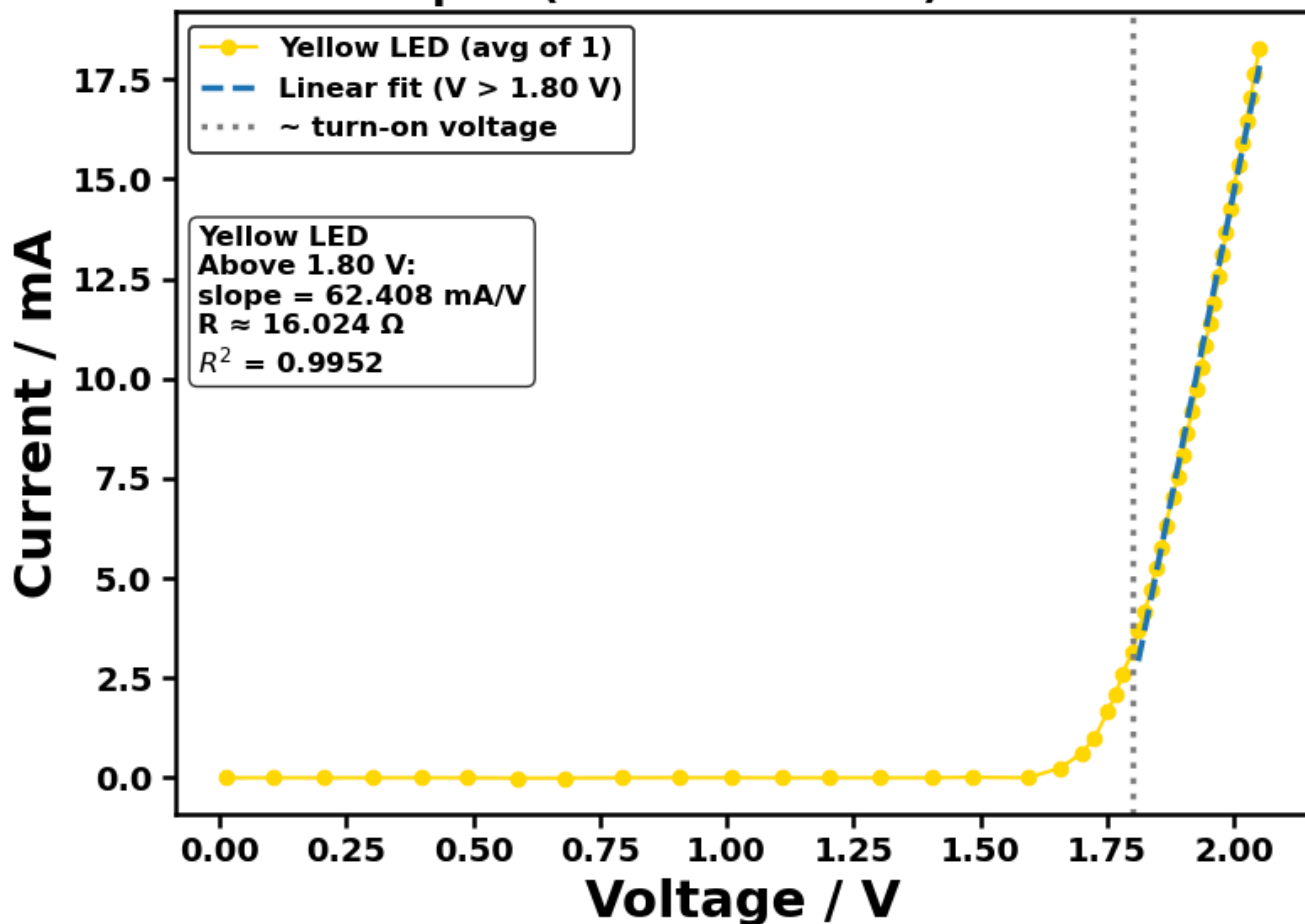


Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/raw data_finalised/5_LED_Colours/Red_LED/Red_LED_repeats5_20260123_104738.csv

=== Yellow LED ($V_{\text{split}}=1.8$) ===

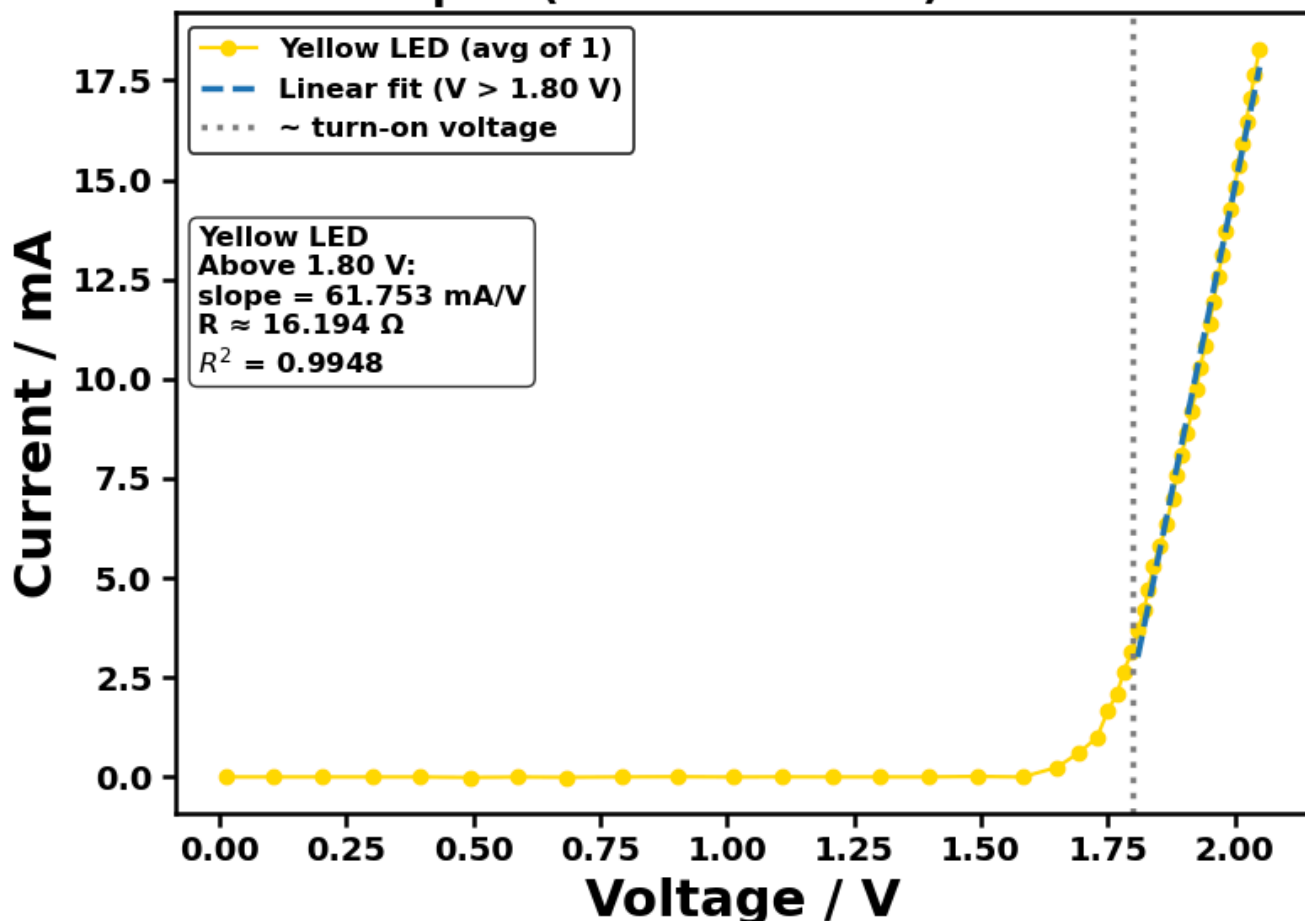
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Yellow_LED/Yellow_LED_run1_20260123_104738.png

I-V plot (Yellow LED LED) — run 1

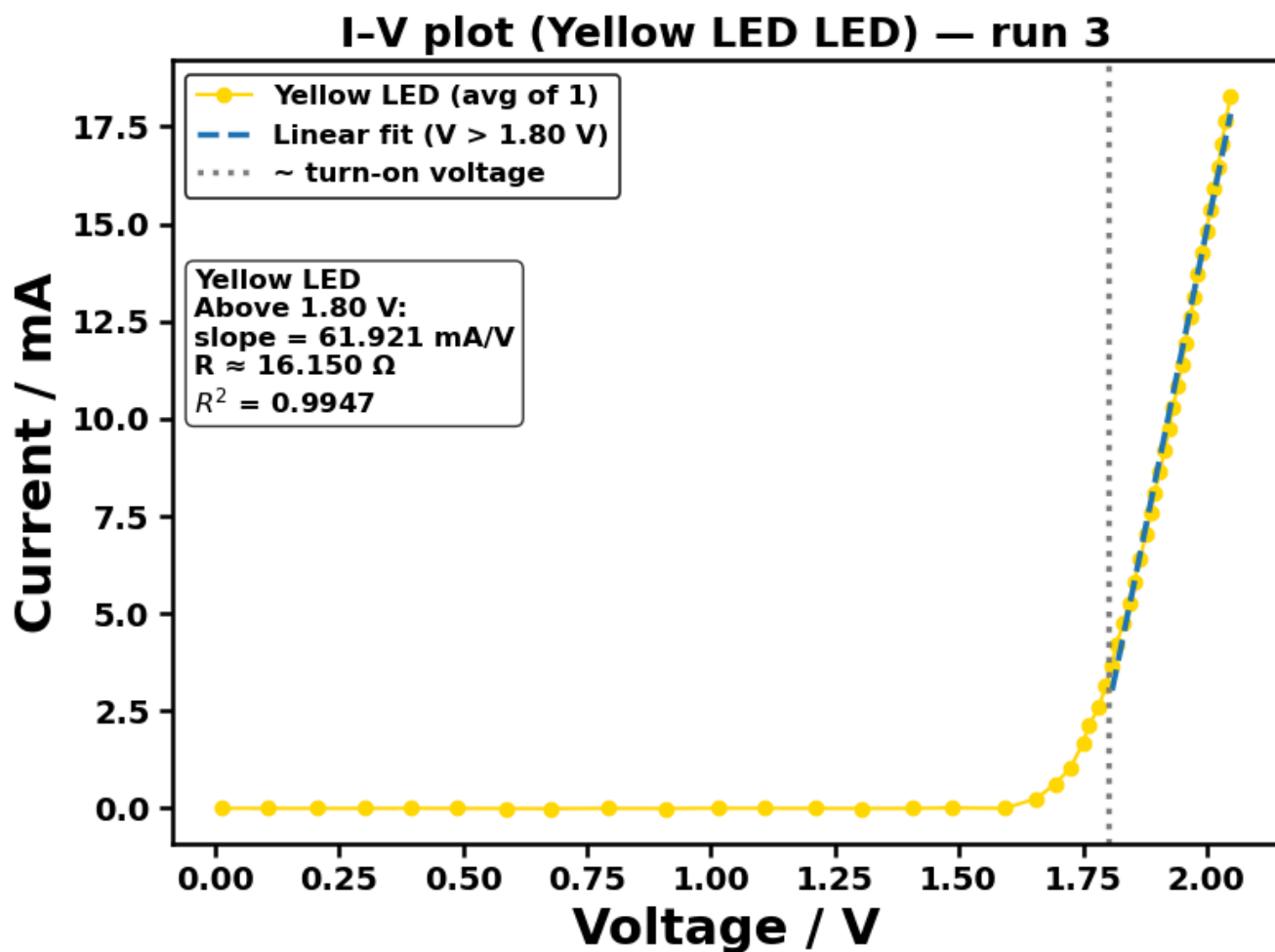


Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Yellow_LED/Yellow_LED_run2_20260123_104738.png

I-V plot (Yellow LED LED) — run 2

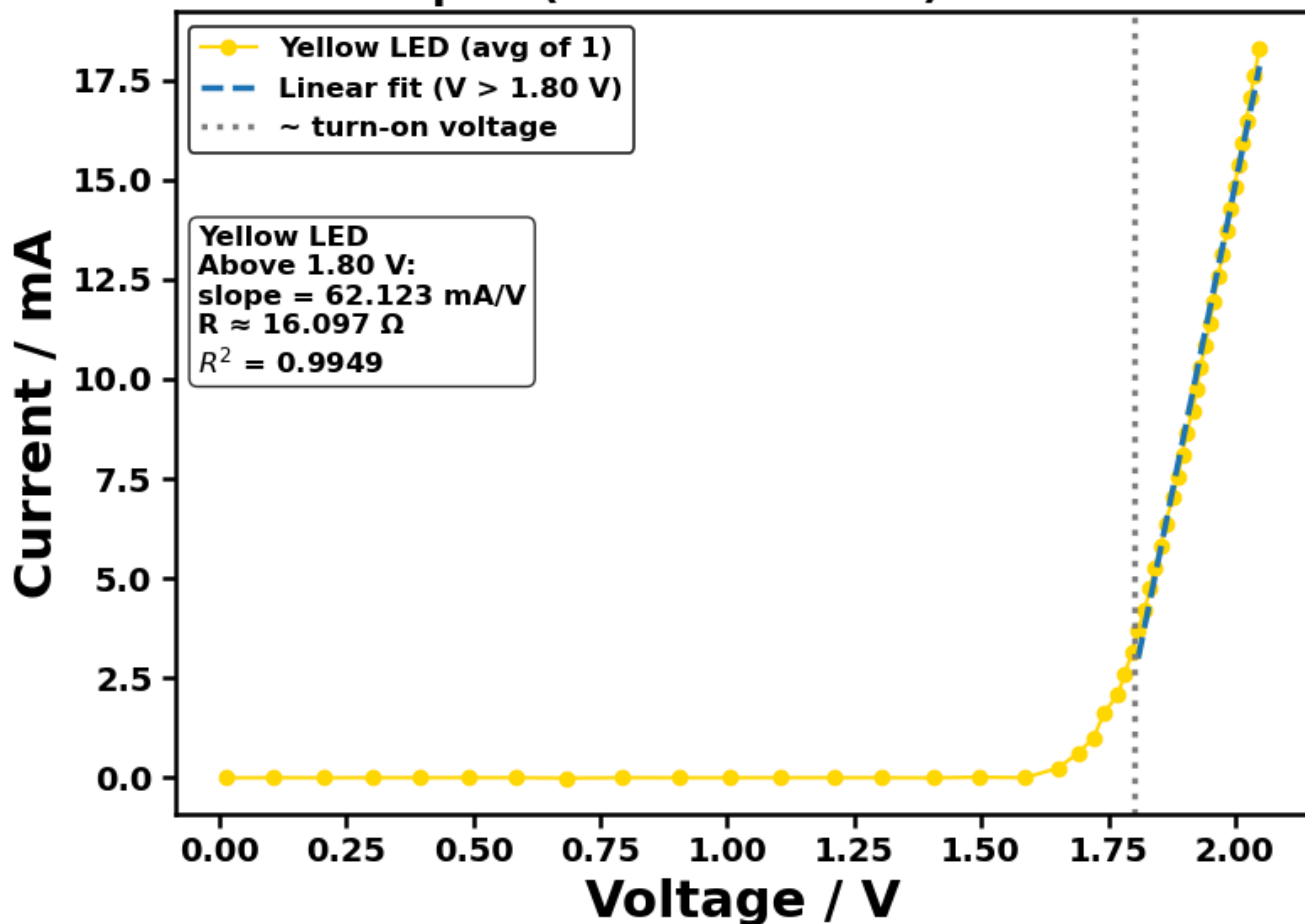


Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Yellow_LED/Yellow_LED_run3_20260123_104738.png



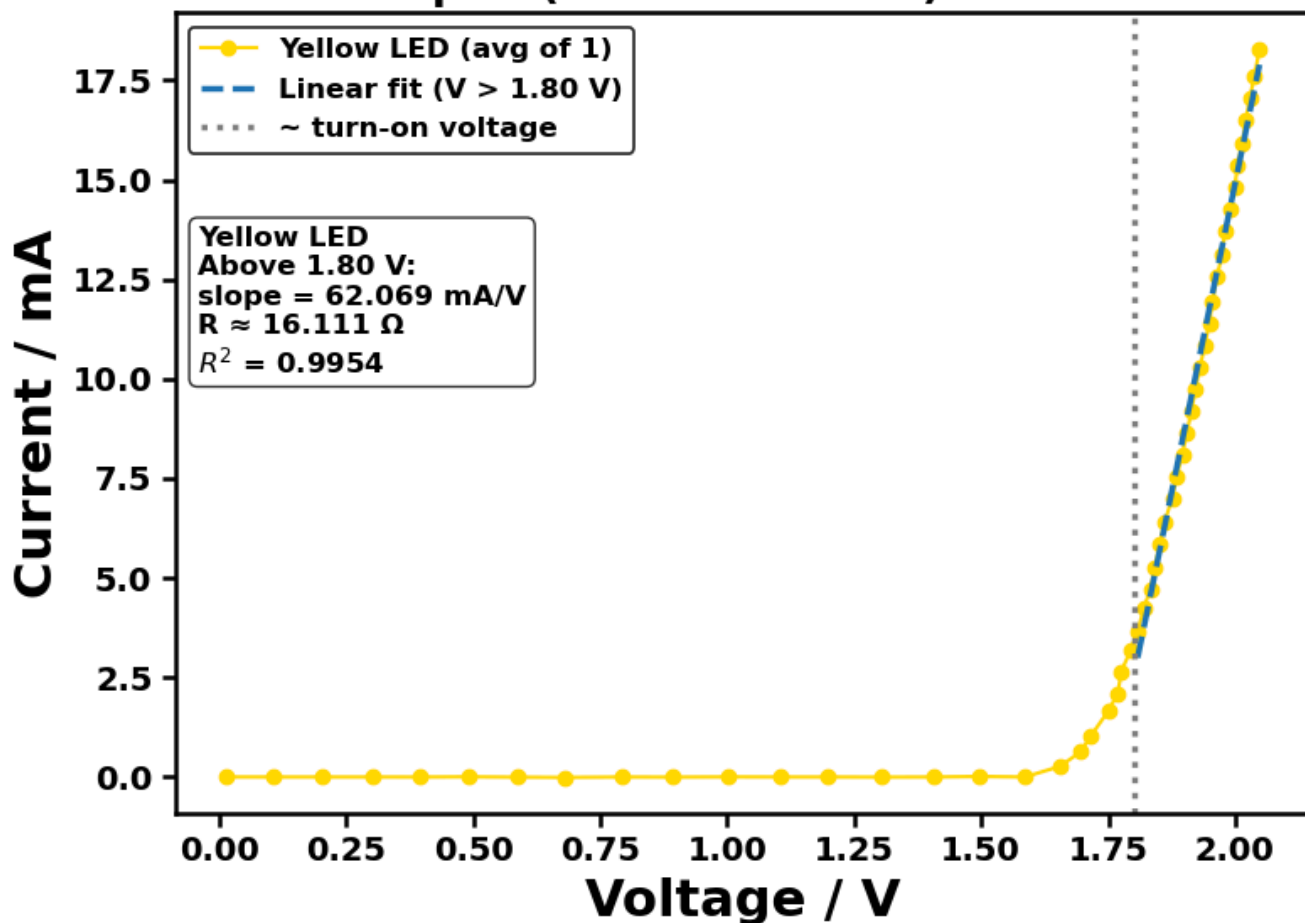
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Yellow_LED/Yellow_LED_run4_20260123_104738.png

I-V plot (Yellow LED LED) — run 4



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Yellow_LED/Yellow_LED_run5_20260123_104738.png

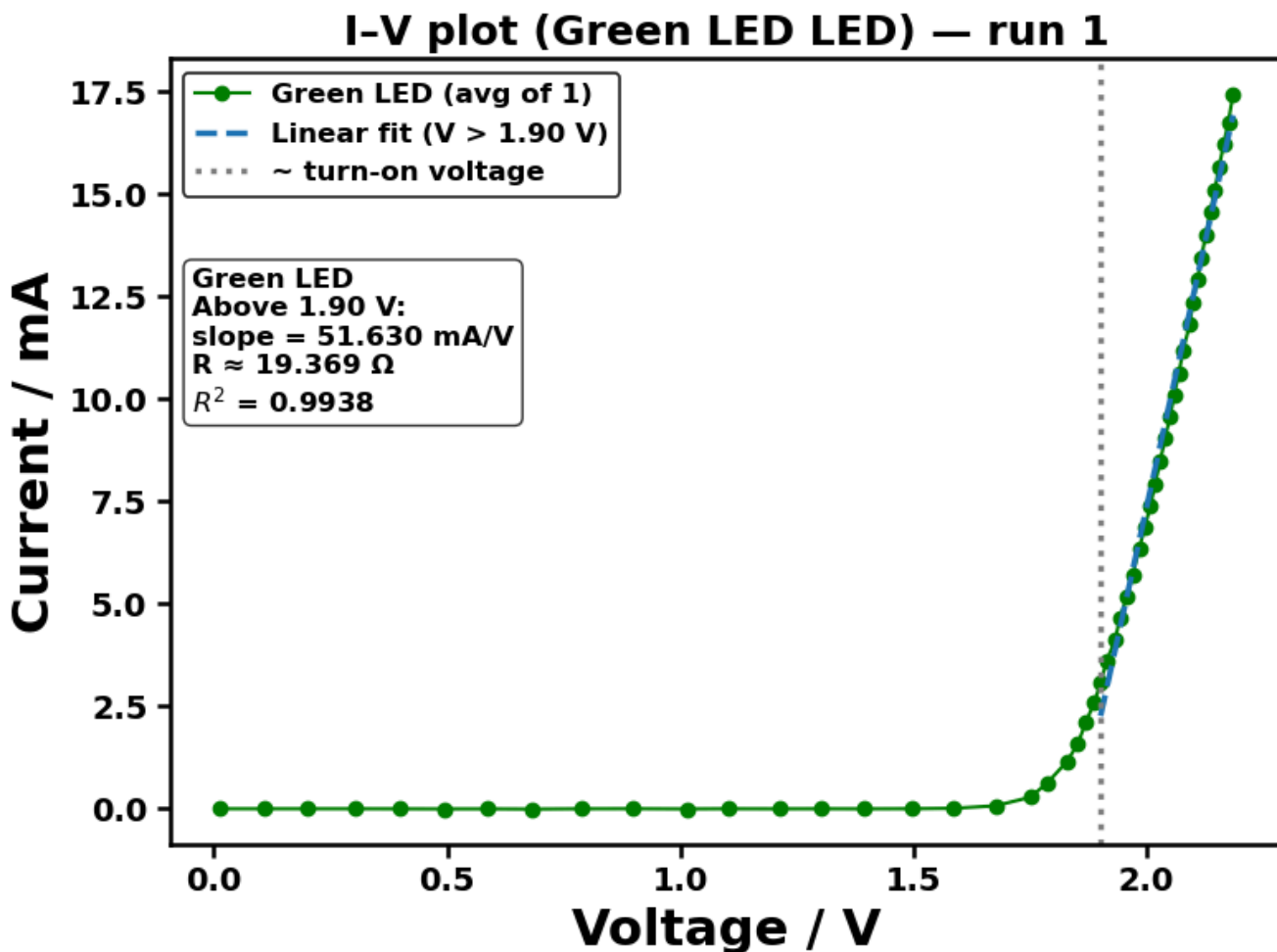
I-V plot (Yellow LED LED) — run 5



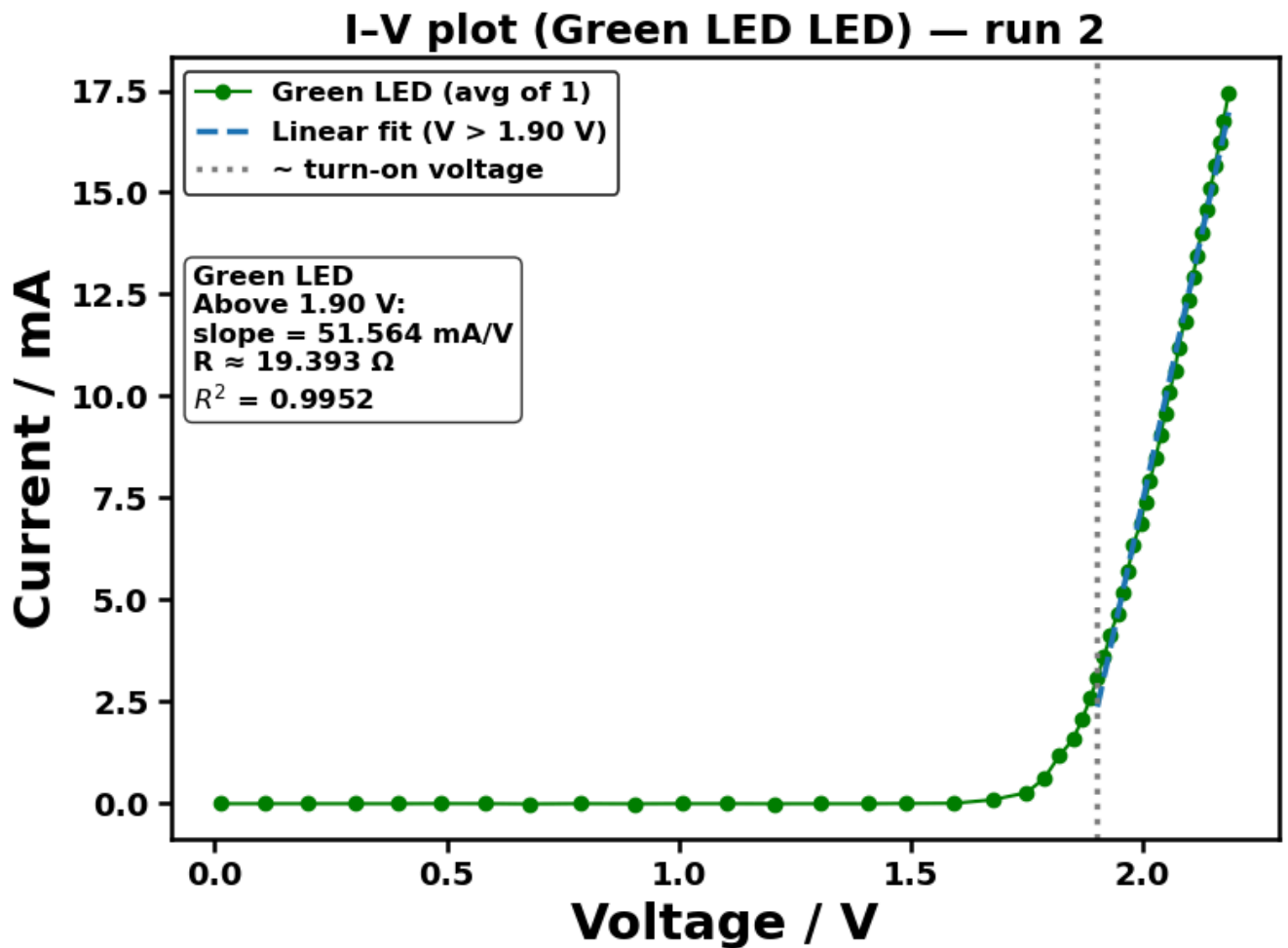
Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/raw data_finalised/5_LED_Colours/Yellow_LED/Yellow_LED_repeats5_20260123_104738.csv

=== Green LED (V_split=1.9) ===

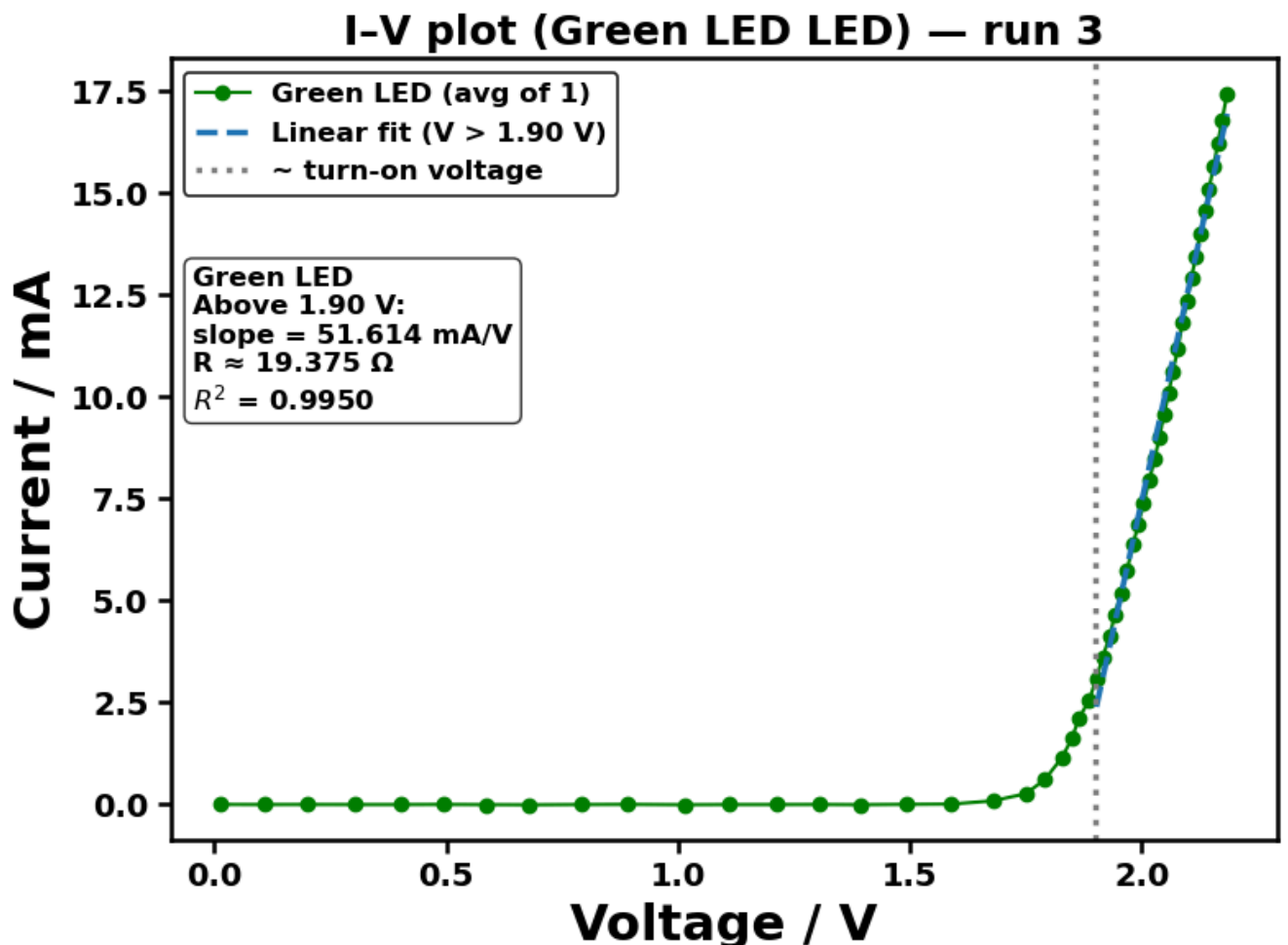
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Green_LED/Green_LED_run1_20260123_104738.png



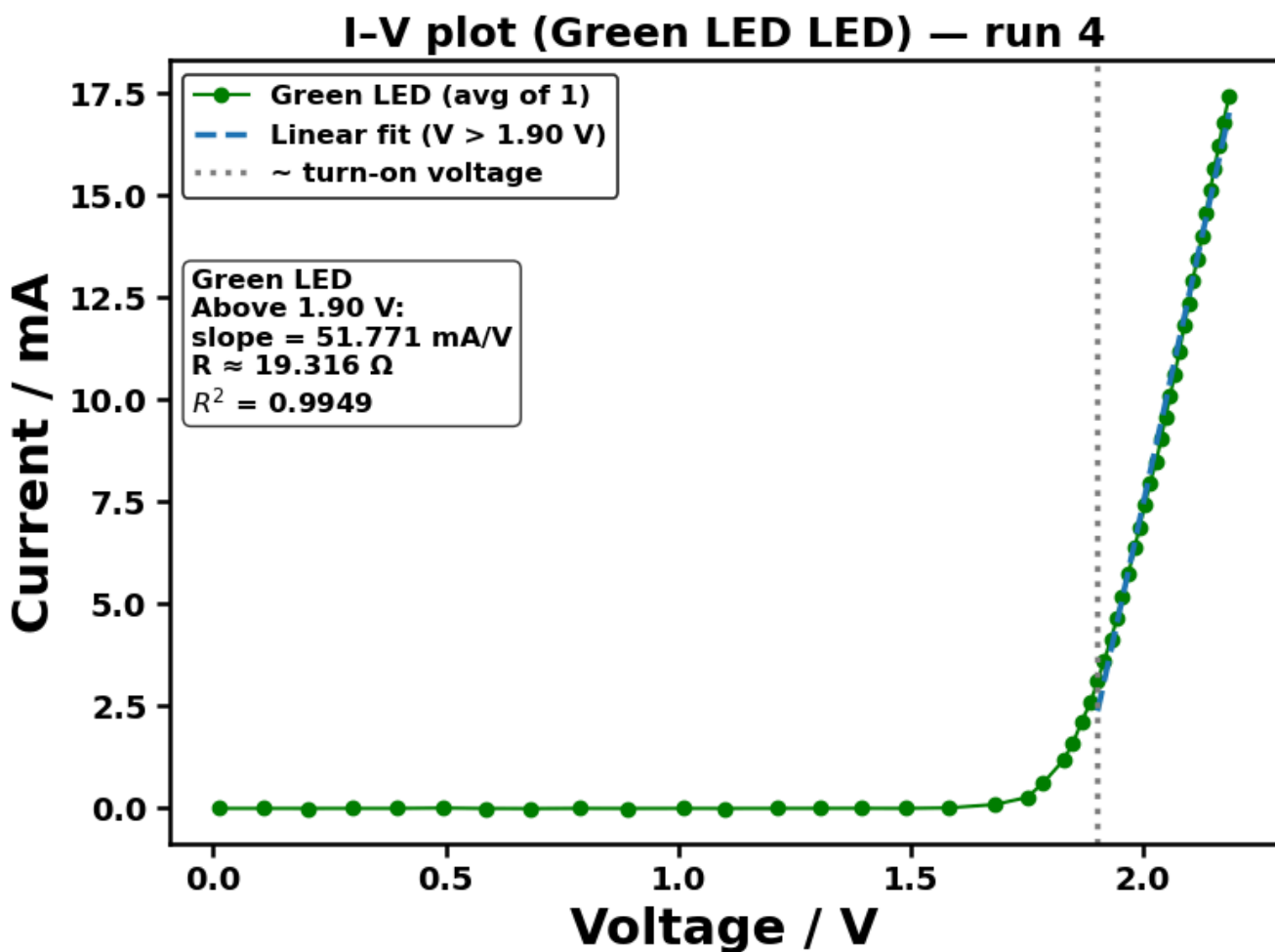
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Green_LED/Green_LED_run2_20260123_104738.png



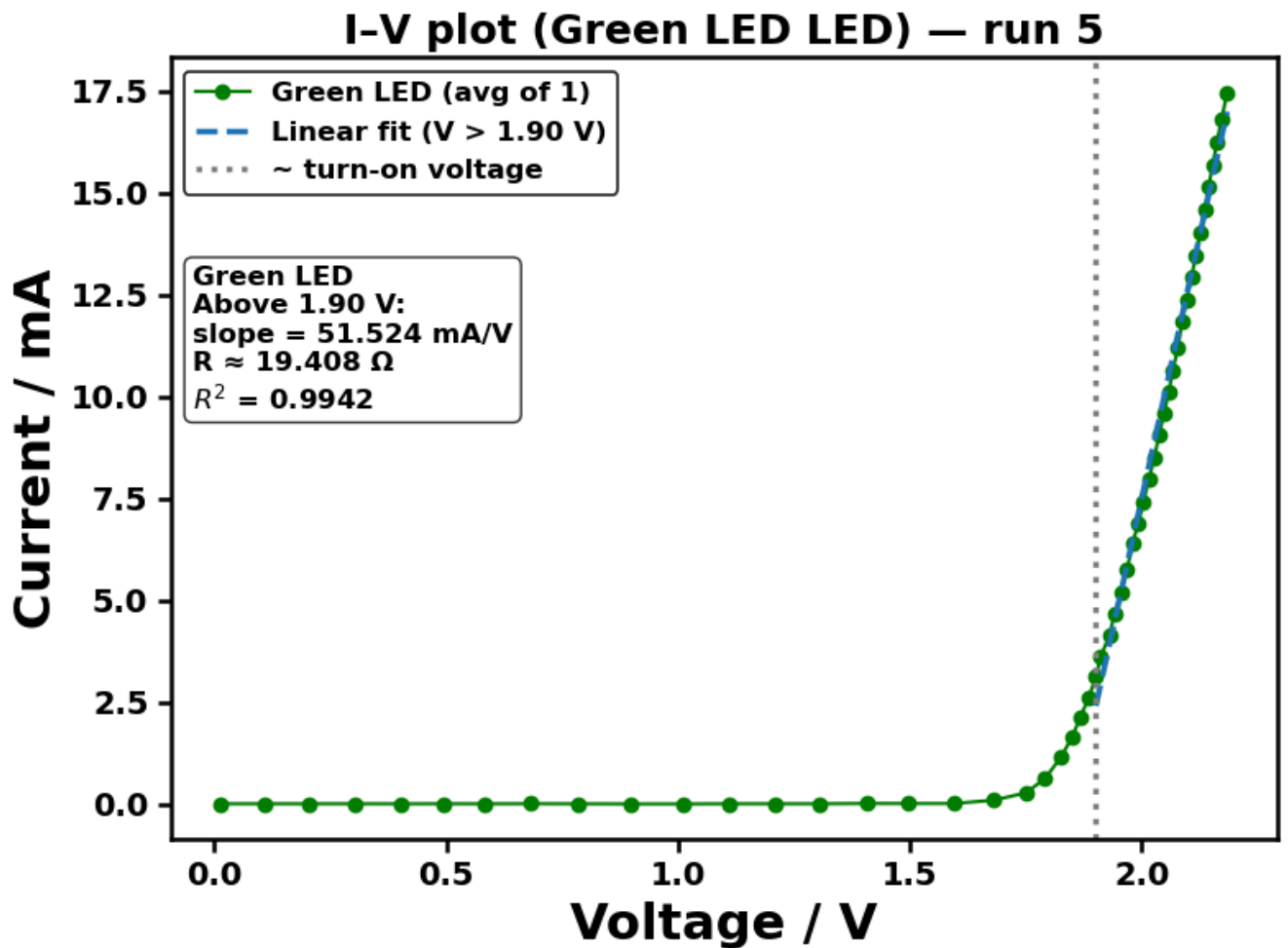
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Green_LED/Green_LED_run3_20260123_104738.png



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Green_LED/Green_LED_run4_20260123_104738.png



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Green_LED/Green_LED_run5_20260123_104738.png

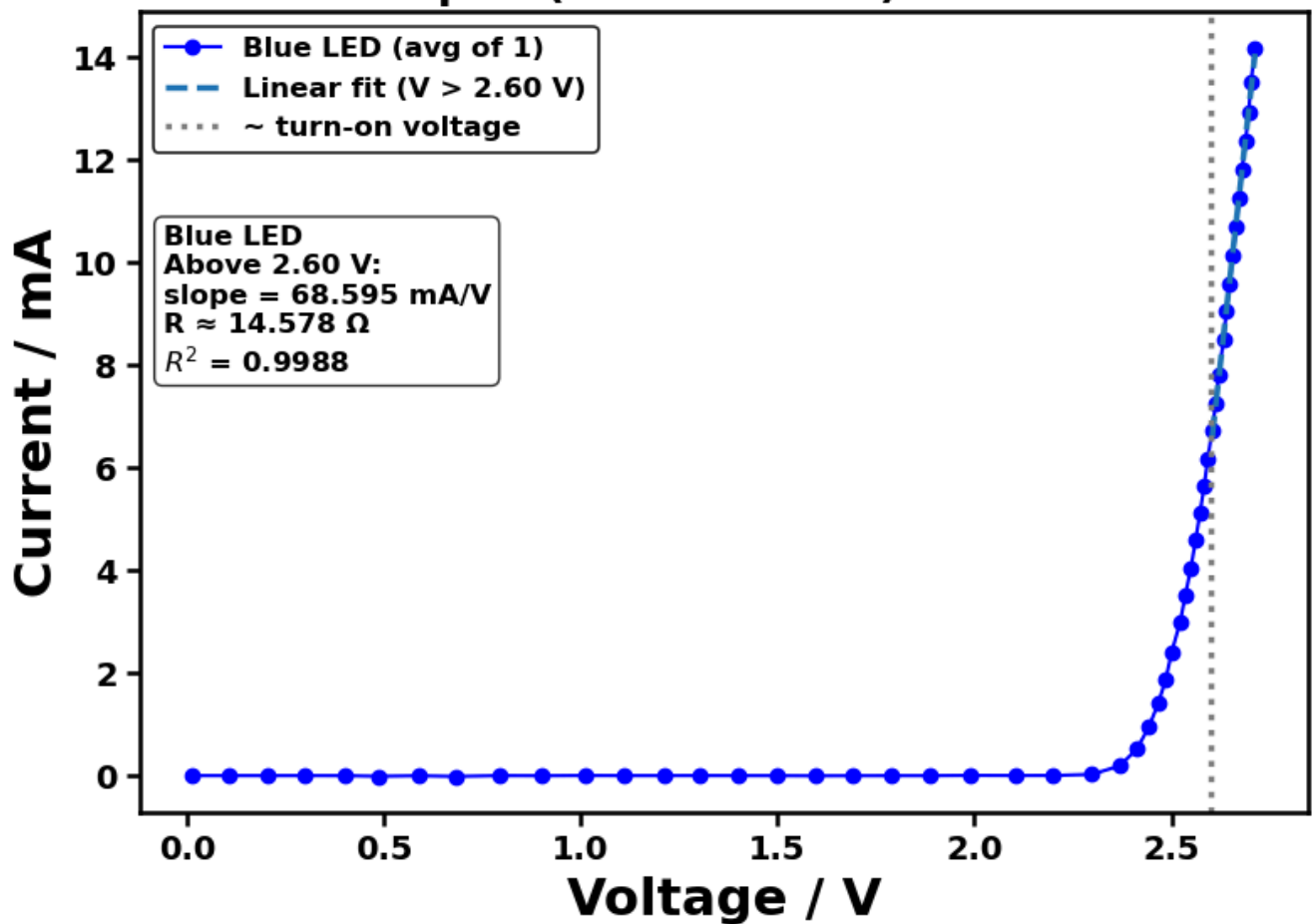


Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/raw data_finalised/5_LED_Colours/Green_LED/Green_LED_repeats5_20260123_104738.csv

=== Blue LED ($V_{\text{split}}=2.6$) ===

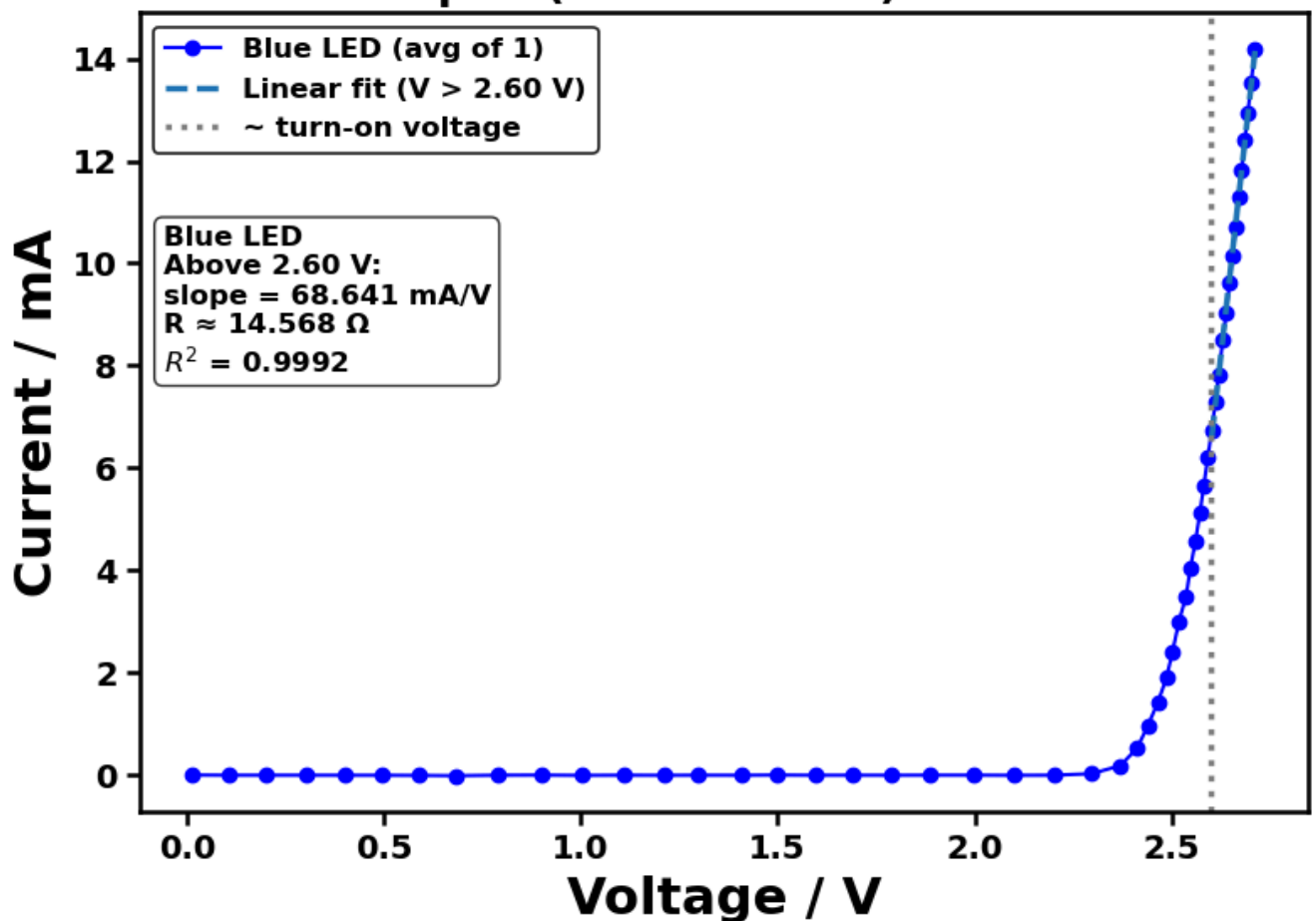
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Blue_LED/Blue_LED_run1_20260123_104738.png

I-V plot (Blue LED LED) — run 1

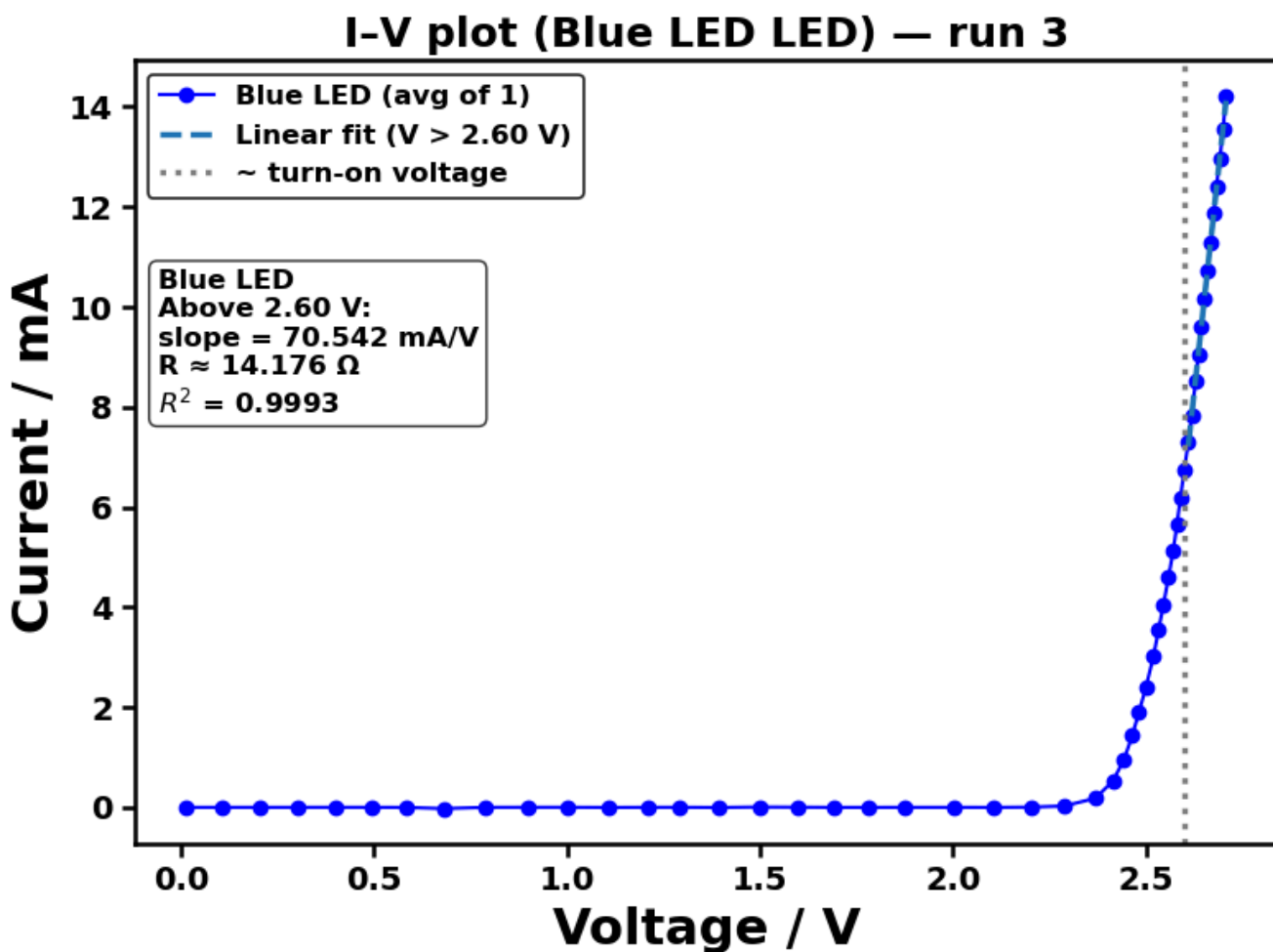


Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Blue_LED/Blue_LED_run2_20260123_104738.png

I-V plot (Blue LED LED) — run 2

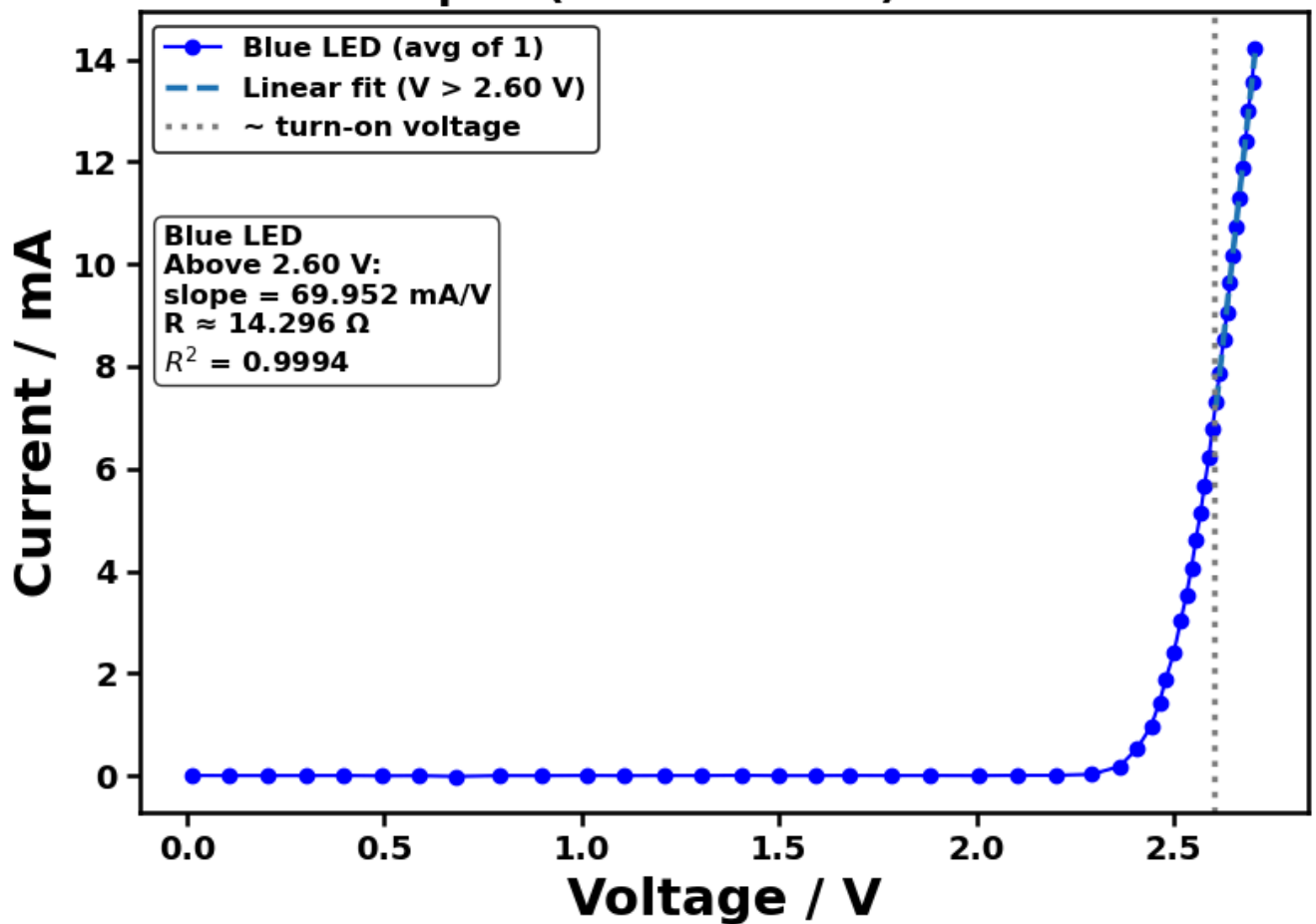


Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Blue_LED/Blue_LED_run3_20260123_104738.png



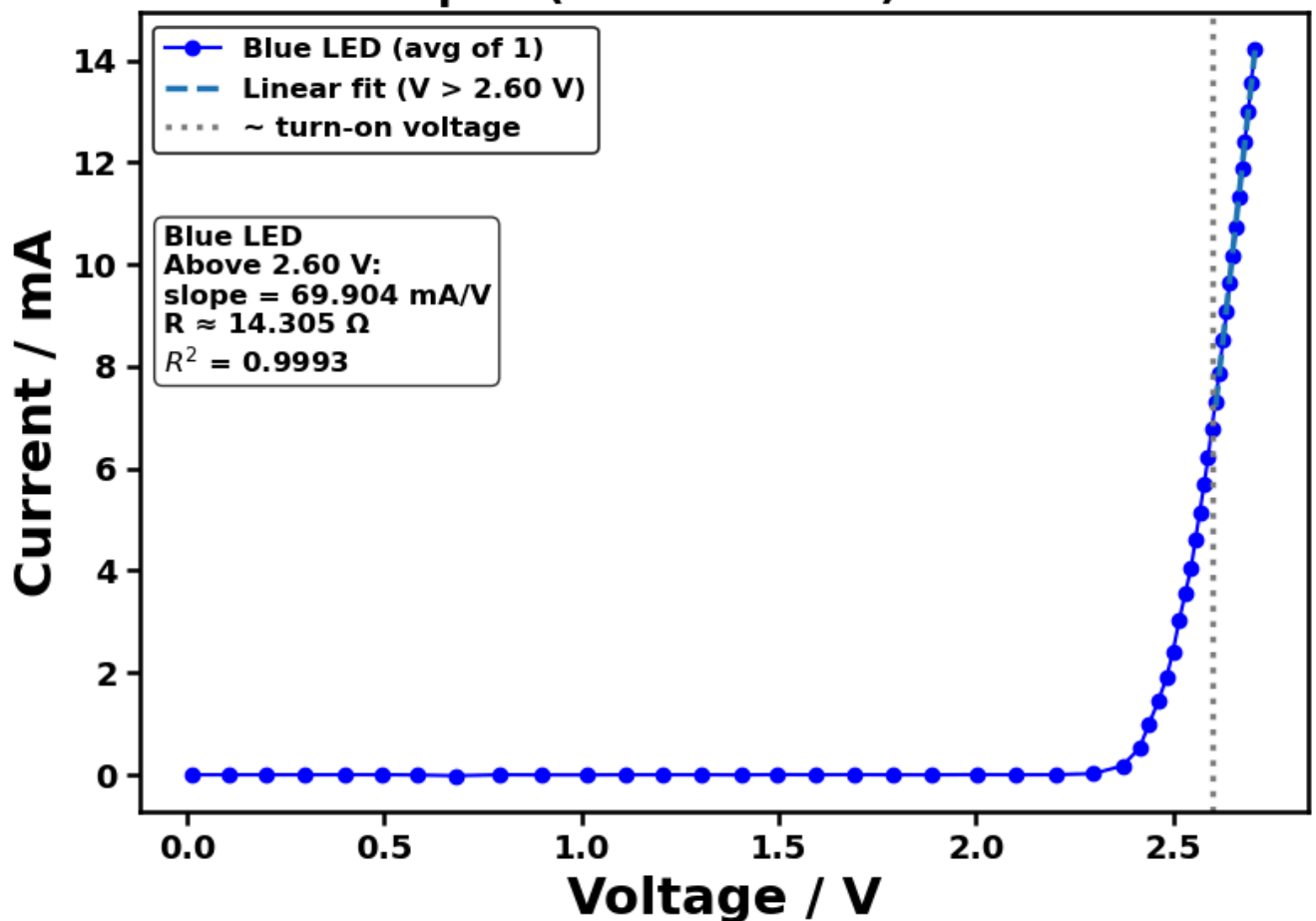
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Blue_LED/Blue_LED_run4_20260123_104738.png

I-V plot (Blue LED LED) — run 4



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/Blue_LED/Blue_LED_run5_20260123_104738.png

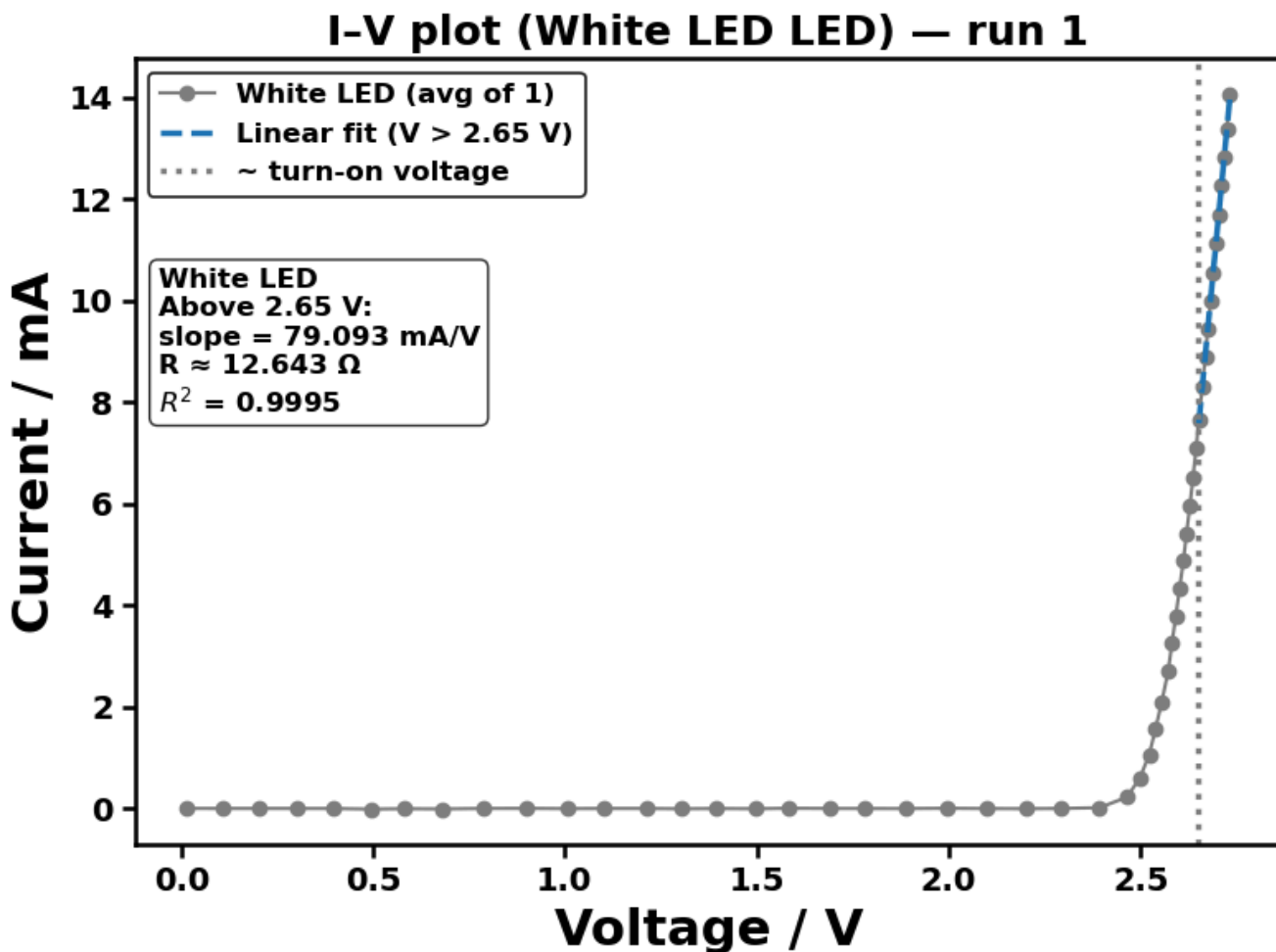
I-V plot (Blue LED LED) — run 5



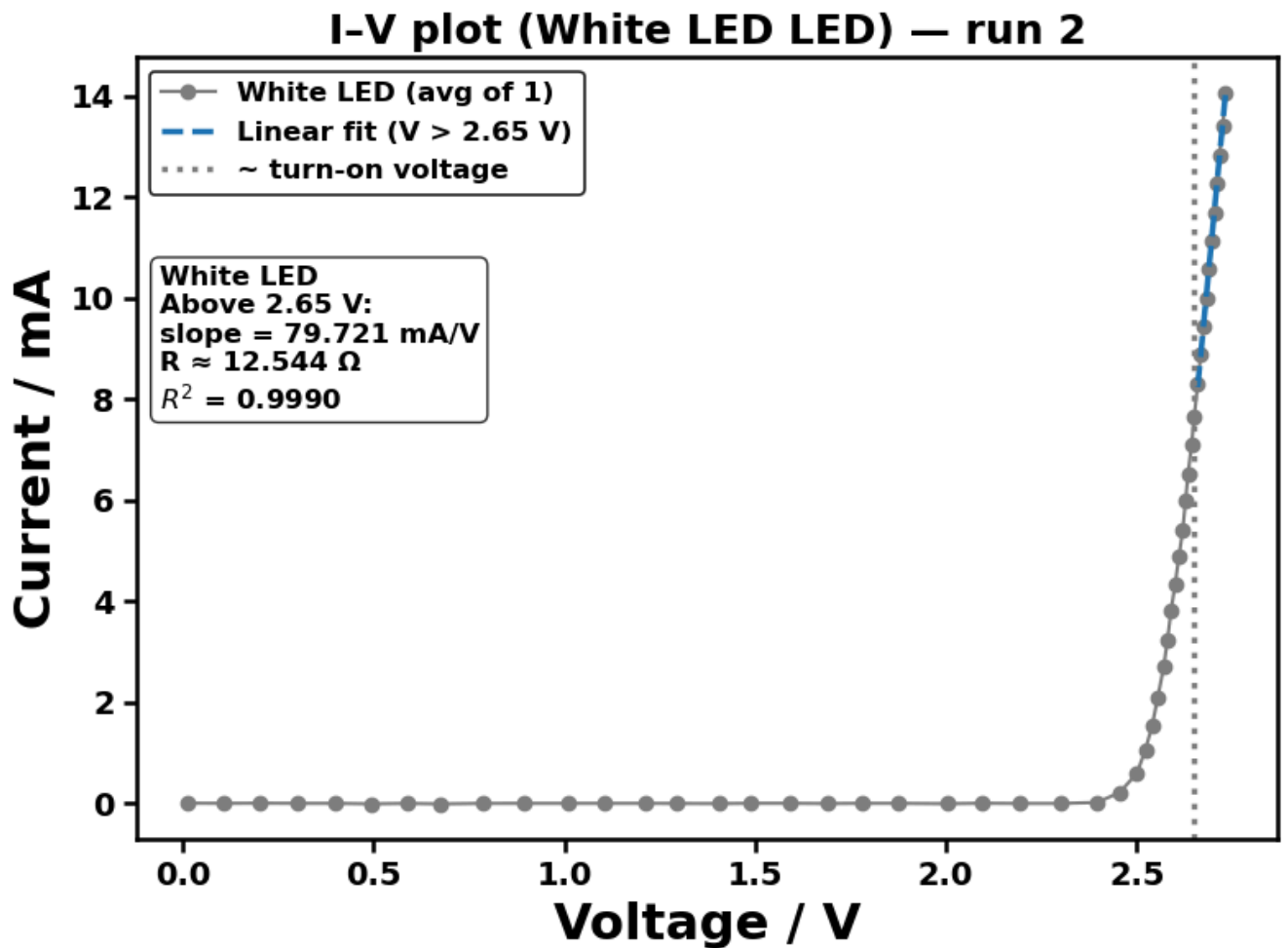
Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/raw data_finalised/5_LED_Colours/Blue_LED/Blue_LED_repeats5_20260123_104738.csv

=== White LED (V_split=2.65) ===

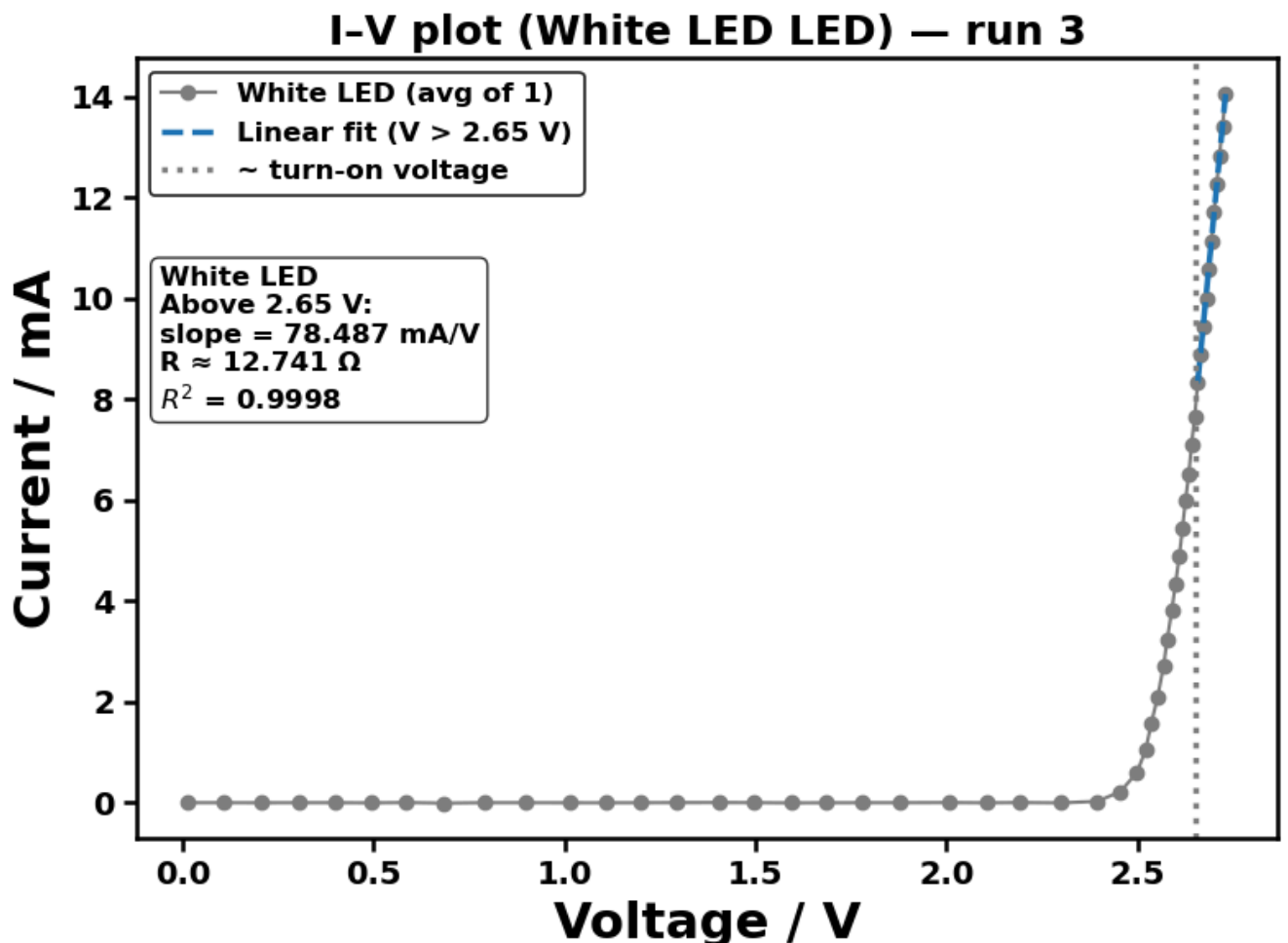
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/White_LED/White_LED_run1_20260123_104738.png



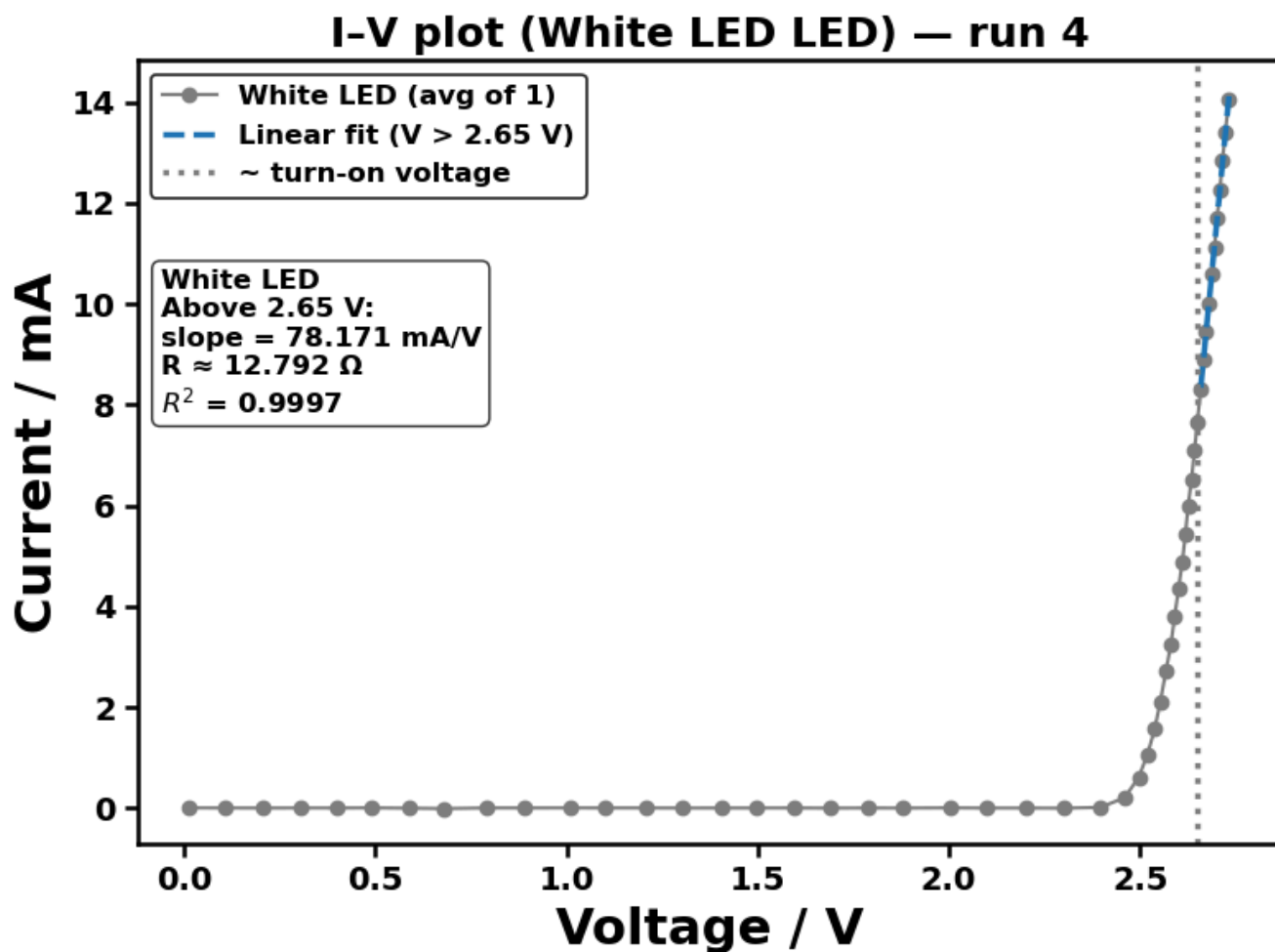
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/White_LED/White_LED_run2_20260123_104738.png



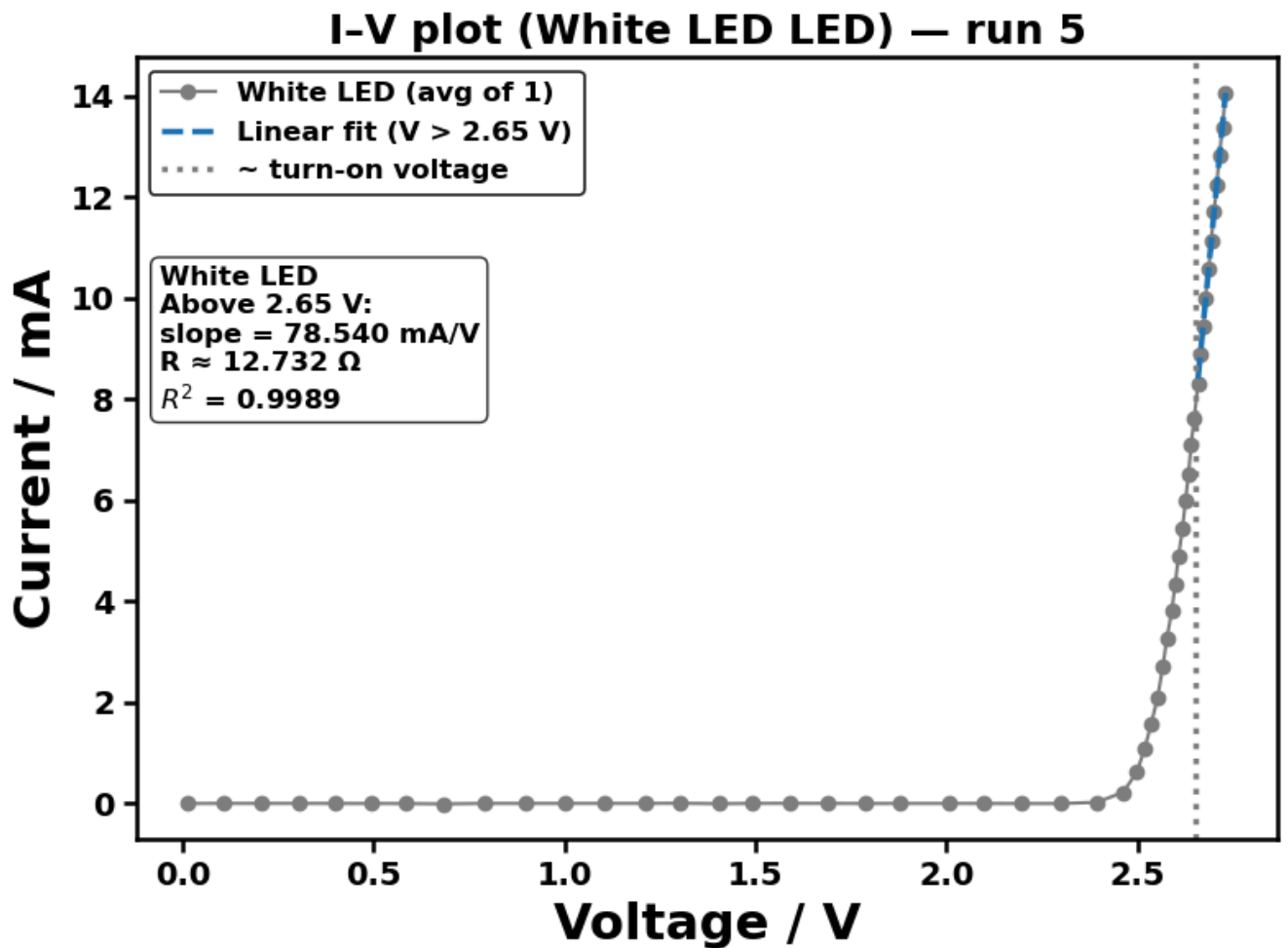
Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/White_LED/White_LED_run3_20260123_104738.png



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/White_LED/White_LED_run4_20260123_104738.png



Saved figure → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/Figures/5_LED_Colours/White_LED/White_LED_run5_20260123_104738.png



Saved CSV → /Users/liangze/Desktop/26 Spring/Practical Chemistry/Introduction to computer controlled experiments/ICCE 20260123/raw data_finalised/5_LED_Colours/White_LED/White_LED_repeats5_20260123_104738.csv

Experiment 5 done.

3. Extension - Liquid Nitrogen Cooling

Aim

To investigate the effect of temperature on the turn-on voltage of LEDs.

Explanation

The LED I-V characteristics were cooled by liquid nitrogen to ~77 K during measurements while maintaining consistent voltage sweep parameters. Changes in turn-on voltage were analysed to assess thermal effects on carrier mobility and band gap energy.

An increase in turn-on voltage with decreasing temperature is expected due to band gap widening and reduced thermal carrier excitation, providing further evidence that LED behaviour is governed by intrinsic semiconductor physics rather than simple resistive conduction.

```
In [21]: # note: This function was defined earlier and not updated,
# as the experiment could not be repeated due to safety considerations and limited
# The function was called in the same cell multiple times and only one plot is shown
# However, the full raw data and figures were saved successfully.

def save_iv_csv(filename, V, I):
    """Save (V, I) arrays to CSV."""
    with open(filename, mode='w', newline='') as f:
```



```

        w = csv.writer(f)
        w.writerow(["Voltage (V)", "Current (mA)"])
        w.writerows(zip(V, I))
    print(f"Saved: {filename}")

def run_led_iv_sweep(
    ser,
    V_split,
    filename="led_IV.csv",
    v_start=0.0,
    v_stop=5.0,
    v_step=0.1,
    delay=0.1,
    Pn=200,
):
    """
    Sweep applied voltage, measure (V_LED, I), do 2 linear fits split by V_split,
    plot, and (optionally) save to CSV.
    Returns: V, I, (m_low, c_low), (m_high, c_high), R_high
    """

    ser.write(f'<P{Pn}>'.encode())
    ser.readline()

    voltages, currents = [], []
    v_set = v_start
    while v_set <= v_stop + 1e-12:
        ser.write(f'<S{v_set:.2f}>'.encode())
        ser.readline()
        time.sleep(delay)

        ser.write(b'<V2>')
        V_meas = float(ser.readline().decode().split()[1])

        ser.write(b'<I1>')
        I_meas = float(ser.readline().decode().split()[1])

        voltages.append(V_meas)
        currents.append(I_meas)
        v_set += v_step

    V = np.array(voltages)
    I = np.array(currents)

    low = V < V_split
    high = V > V_split

    if low.sum() < 2 or high.sum() < 2:
        raise ValueError(
            f"Not enough points for fitting. "
            f"Below split: {low.sum()}, above split: {high.sum()}. "
            f"Try a different V_split."
        )

    m_low, c_low = np.polyfit(V[low], I[low], 1)
    m_high, c_high = np.polyfit(V[high], I[high], 1)

    R_high = 1000 / m_high

    V_fit_low = np.linspace(V[low].min(), V[low].max(), 100)
    V_fit_high = np.linspace(V[high].min(), V[high].max(), 100)

    plt.figure(figsize=(8, 6))

```

```

ax = plt.gca()
for spine in ax.spines.values():
    spine.set_linewidth(1.8)
plt.plot(V, I, 'o', label='Data')
plt.plot(V_fit_low, m_low * V_fit_low + c_low, '--', label=f'Fit: V < {V_split}')
plt.plot(V_fit_high, m_high * V_fit_high + c_high, '--', label=f'Fit: V > {V_sp')
plt.axvline(V_split, linestyle=':', label='Split', color='grey', linewidth=2.5)

plt.text(
    0.02, 0.6,
    f"Above {V_split:.2f} V:\n"
    f"slope = {m_high:.3f} mA/V\n"
    f" $R \approx \{R\_high:.3f\} \Omega$ ",
    transform=plt.gca().transAxes,
    fontsize=14,
    fontweight='bold',
    bbox=dict(boxstyle='round', facecolor='white', alpha=0.8)
)

plt.xlabel('Voltage / V', fontsize=22, fontweight="bold")
plt.ylabel('Current / mA', fontsize=22, fontweight="bold")
plt.xticks(fontsize=14, fontweight='bold')
plt.yticks(fontsize=14, fontweight='bold')
plt.title('LED I-V characteristic', fontsize=20, fontweight="bold")
leg = plt.legend(
    prop={'size': 12, 'weight': 'bold'},
    loc='upper left',
    frameon=True,
    fancybox=True
)

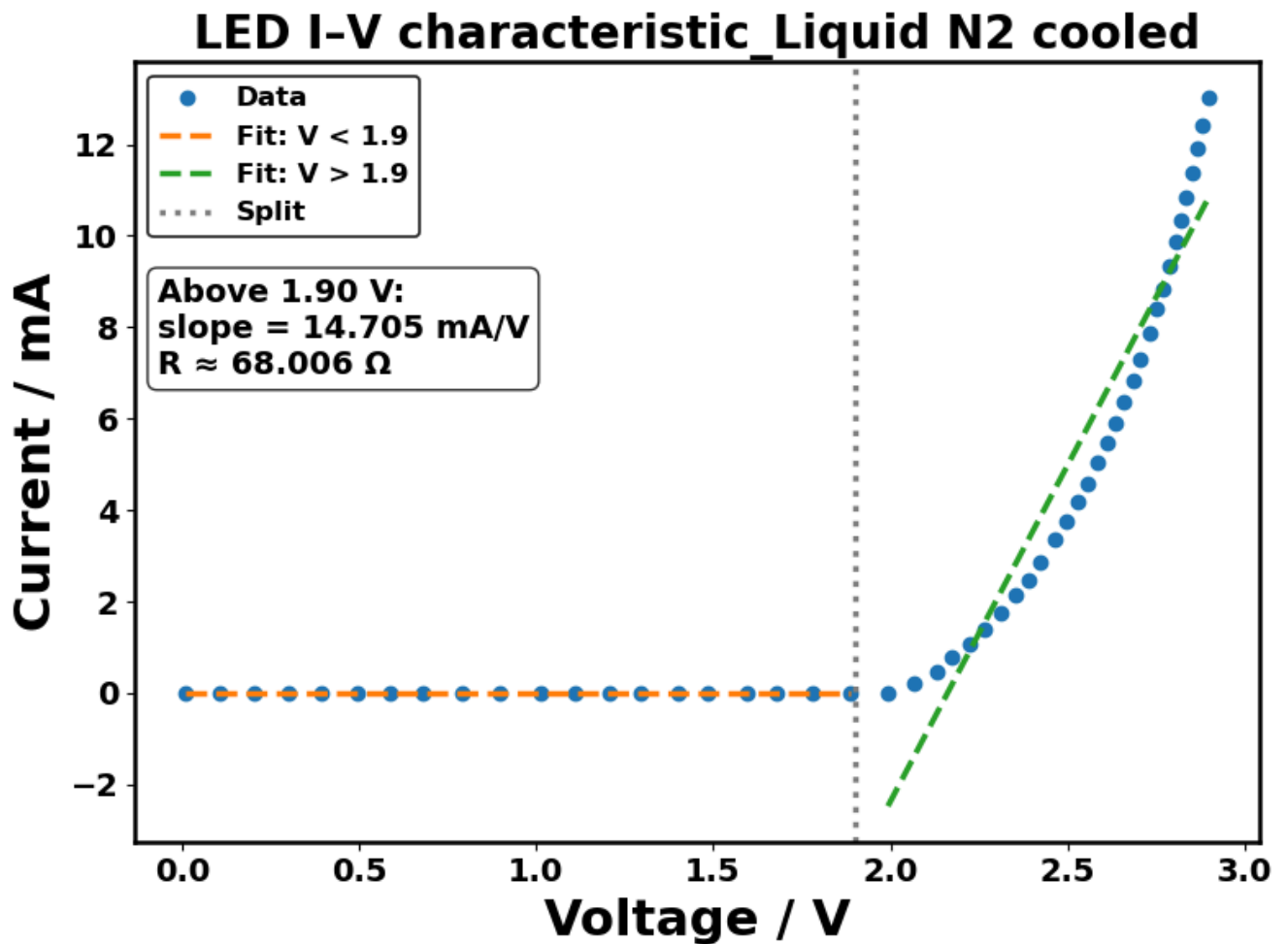
frame = leg.get_frame()
frame.set_facecolor('white')
frame.set_edgecolor('black')
frame.set_linewidth(1.2)
frame.set_alpha(0.85)

plt.tight_layout()
plt.show()

save_iv_csv(filename, V, I)
return

```

```
In [36]: run_led_iv_sweep(ser, 1.9, filename="green_LiquidN2_cooled.csv")
```



Saved: green_LiquidN2_cooled.csv

4. Additional observation

Observation 1.1: When two LEDs were connected simultaneously, the circuit was assumed to behave as a parallel configuration, since both LEDs were connected across the same applied voltage. Under this condition, an uneven brightness distribution was consistently observed between the two LEDs, as shown in Figure 1-3.

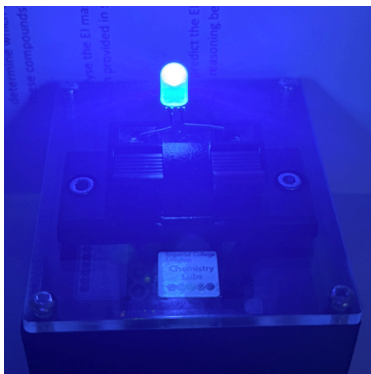


Figure 2(a): Single blue LED

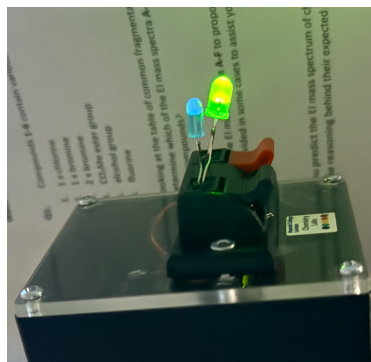


Figure 2(b): Blue and green LED

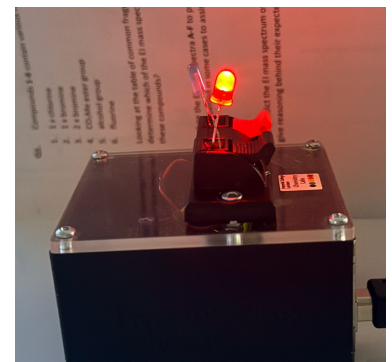


Figure 2(c): Blue and red LED

Figure 2. LED brightness comparison for different colour combinations under the same applied voltage.

Observation 1.2: When two LEDs of the same colour were connected in parallel across the same applied voltage, they exhibited equal brightness, as shown in Figure 4.

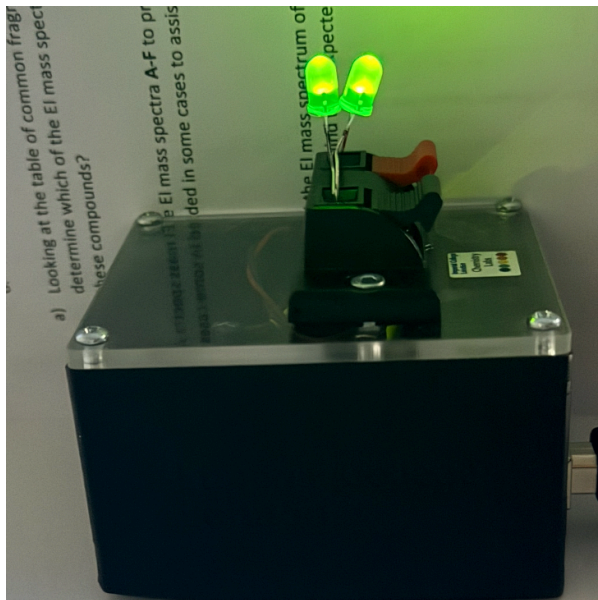


Figure 3. Two green LEDs connected under the same applied voltage.

Observation 1.3: When a LED was connected with resistors, the brightness of LED would decrease as the numbers of the resistors increased. When two resistors were added, the red LED was not emitting light anymore.

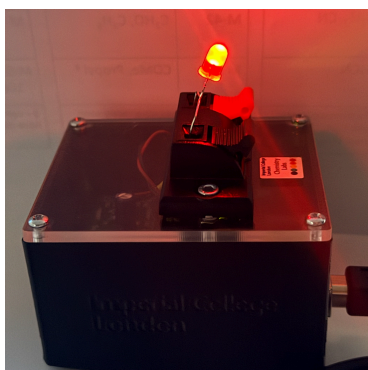


Figure 4(a): Single red LED

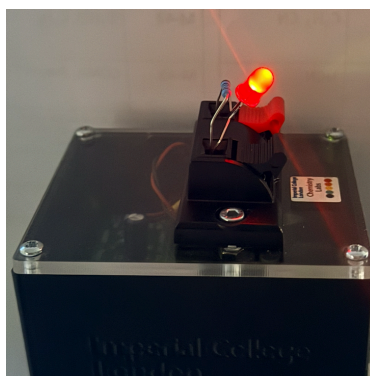


Figure 4(b): Red LED with one resistor ($100\ \Omega$)



Figure 4(c): Red LED with two resistors ($100\ \Omega$ each)

Figure 4. Red LED brightness comparison with the addition of resistors under the same applied voltage.

Explanation 1: The brightness of an LED can be influenced by the presence of other LEDs or resistors connected in parallel within the circuit.

In Observation 1.1 (Figure 1-3), when multiple LEDs were connected in parallel, the LED with the lower turn-on voltage tended to dominate conduction and therefore appeared significantly brighter, while LEDs with higher turn-on voltages exhibited reduced brightness. This behaviour is consistent with the relative turn-on voltage of colored LEDs, which follows the order: red (1.681 V), yellow (1.779 V), green (1.898 V), blue (2.495 V), and white (2.585 V).

If the turn-on voltages are the same, as in the case of LEDs of the same colour observed in Observation 1.2 (Figure 4), the two LEDs display identical brightness.

For Observation 1.3, a similar trend was observed when standard resistors were connected in parallel. For example, as shown in Figure 6 and 7, the brightness of the LED decreased when one resistor was added, and when an additional resistor was introduced, the LED no longer emitted visible light.

Observation 2: fun fact: traffic light can be made in this way!

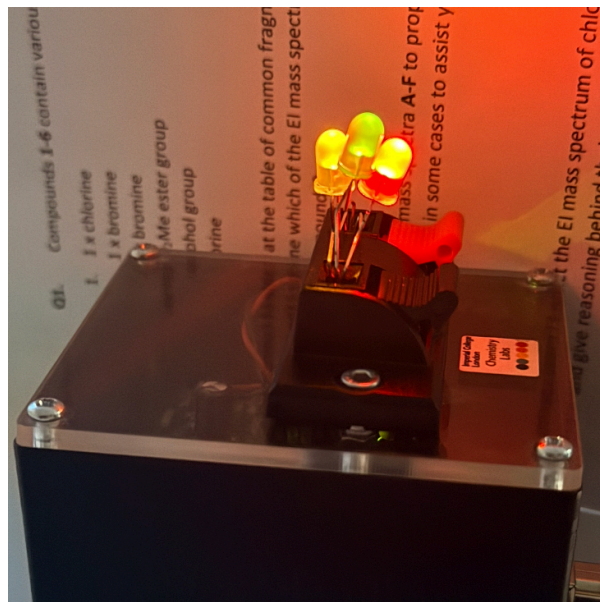


Figure 5. Our nice-looking traffic light by combining red, yellow and green LEDs in the circuit.

```
In [ ]: ser.close()
```